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Number Theory in the Spirit of Ramanujan

Bruce C. Berndt

$$1 + \frac{q^{1/5}}{1 + \frac{q}{1 + \frac{q^2}{1 + \frac{q^3}{1 + \dots}}}}$$

Ramanujan is recognized as one of the great number theorists of the twentieth century. Here now is the first book to provide an introduction to his work in number theory. Most of Ramanujan's work in number theory arose out of q -series and theta functions. This book provides an introduction to these two important subjects and to some of the topics in number theory that are inextricably intertwined with them, including the theory of partitions, sums of squares and triangular numbers, and the Ramanujan tau function. The majority of the results discussed here are originally due to Ramanujan or were rediscovered by him. Ramanujan did not leave us proofs of the thousands of theorems he recorded in his notebooks, and so it cannot be claimed that many of the proofs given in this book are those found by Ramanujan. However, they are all in the spirit of his mathematics.



The subjects examined in this book have a rich history dating back to Euler and Jacobi, and they continue to be focal points of contemporary mathematical research. Therefore, at the end of each of the seven chapters, Berndt discusses the results established in the chapter and places them in both historical and contemporary contexts. The book is suitable for advanced undergraduates and beginning graduate students interested in number theory.

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Preface

Generally acknowledged as India's greatest mathematician, Srinivasa Ramanujan is most often thought of as a number theorist, although he made substantial contributions to analysis and several other areas of mathematics. For most number theorists, when Ramanujan's name is mentioned, the partition and tau functions immediately come to mind. His interest in these arithmetic functions was inextricably intertwined with his primary interests of theta functions and other *q-series*. In fact, most of Ramanujan's research in number theory arose out of *q-series* and theta functions. Theta functions are the fundamental building blocks in the theory of elliptic functions, and Ramanujan independently developed his own theory of elliptic functions, which is quite unlike the classical theory. We do not formally define an elliptic function, but, roughly, elliptic functions are meromorphic functions with two linearly independent periods over the real numbers. The concept of double periodicity is not used in this book, and, to the best of our knowledge, Ramanujan never utilized this idea.

The purpose of this book is to provide an introduction to this large expanse of Ramanujan's work in number theory. Needless to say, we shall be able to cover only a very small fraction of Ramanujan's work on theta functions and *q-series* and their connections with number theory. However, after developing only a few facts about

q -series and theta functions, we will be equipped to prove many interesting theorems. The arithmetic functions on which we focus are the partition function $p(n)$, Ramanujan's tau function $\tau(n)$, the number of representations of a positive integer n as a sum of $2k$ squares denoted by $r_{2k}(n)$, and other arithmetic functions closely allied to $r_{2k}(n)$. Most of the material upon which we draw can be found in Ramanujan's published papers on $p(n)$ and $\tau(n)$, the later chapters in his second notebook, his lost notebook, and his handwritten manuscript on $p(n)$ and $\tau(n)$ published with his lost notebook. We emphasize that Ramanujan left behind few of his proofs, especially for his claims in his notebooks and lost notebook. Thus, for many of the theorems that we discuss, we do not know Ramanujan's proofs. This is particularly true for the theorems on sums of squares and similar arithmetic functions that we prove in Chapter 3.

The requirements for reading and understanding the material in this book are relatively modest. An undergraduate course in elementary number theory is advisable. For some of the analytic arguments, a solid undergraduate course in complex analysis is essential. However, the occasions when deep analytical rigor is needed are few, and so readers who do not have a strong background in analysis can simply verify formally the needed manipulations. Our intent here is not to give a rigorous course in analysis but to emphasize the most important ideas about q -series and theta functions and how they interplay with number theory. This book should be suitable for junior and senior undergraduates and beginning graduate students.

Since many readers may not be familiar with Ramanujan's life, we begin with a short account of his life where readers learn about the notebooks and lost notebook in which he recorded his theorems over several years. We provide brief histories, first of the "ordinary" notebooks, and second of the lost notebook. After these biographical and historical narrations, we provide short summaries of the book's seven chapters.

Ramanujan was born on December 22, 1887 in the home of his maternal grandmother in Erode, located in the southern Indian state of Tamil Nadu. After a few months, his mother, Komalatammal, returned with her son to her home in Kumbakonam approximately 160

miles south-southwest of Madras, where her husband was a clerk in the office of a cloth merchant. At the age of twelve, Ramanujan borrowed a copy of the second part of Loney's *Plane Trigonometry* [149] from an older student and worked all the problems in it. This longtime popular textbook in India has much more in it than its title suggests. For example, infinite series and elementary functions of a complex variable are two of its topics. At the age of about fifteen, he borrowed from the Kumbakonam College library a copy of G. S. Carr's *A Synopsis of Elementary Results in Pure Mathematics* [64], which served as his primary source for learning mathematics. Carr was a tutor in London and compiled this compendium of 4417 results (with very few proofs) to facilitate his tutoring. At the age of sixteen, Ramanujan entered the Government College in Kumbakonam. By that time, Ramanujan was completely devoted to mathematics and consequently failed his examinations at the end of his first year, because he would not study any other subject. He therefore lost his scholarship and, because his family was poor, was forced to terminate his formal education. He later twice tried to obtain an education at Pachaiyappa's College in Madras, but each time he failed his examinations.

After leaving the Government College in Kumbakonam, Ramanujan devoted all of his time to mathematics, recording his results without proofs in notebooks. It was probably around the age of sixteen that Ramanujan began to record his mathematical discoveries in notebooks, although the entries on magic squares in Chapter 1 in both his first and second notebooks likely emanate from his school days. Living in poverty with no means of financial support, suffering at times from serious illnesses, and working in isolation, Ramanujan devoted all of his efforts to mathematics and continued to record his discoveries without proofs in notebooks for the next five years. In 1909, he married Janaki, who was only nine or ten years old. With mounting pressure to find a job, Ramanujan visited V. Ramaswami Ayyar, the founder of the Indian Mathematical Society. Ayyar contacted R. Ramachandra Rao, who agreed to give Ramanujan, who now had moved to Madras, a monthly stipend so that Ramanujan could continue his mathematical research unabated.

After being supported for about fifteen months, for reasons that are unclear, Ramanujan refused further financial assistance and became a clerk in the Madras Port Trust Office. This turned out to be a watershed in Ramanujan's career. Several people, including S. Narayana Iyer, the Chief Accountant, and Sir Francis Spring, the Chairman, offered support, and Ramanujan was persuaded to write English mathematicians about his mathematical discoveries. Two of them, H. F. Baker and E. W. Hobson, evidently did not reply. M. J. M. Hill replied but was not encouraging. But on January 16, 1913, Ramanujan wrote G. H. Hardy, who responded immediately and encouragingly, inviting Ramanujan to come to Cambridge to develop his mathematical gifts. Ramanujan and his family were Iyengars, a conservative branch in the Brahmin tradition. Travelling to a distant land would make a person unclean, and so Ramanujan's mother was particularly adamant about her son's not accepting Hardy's invitation. After a pilgrimage to Namakal with S. N. Iyer and after Goddess Namagiri appeared in a dream to Komalatammal, Ramanujan received permission to travel. So on March 17, 1914, Ramanujan boarded a passenger ship for England.

At about this time, Ramanujan evidently stopped recording his theorems in notebooks, although a few entries in his third notebook were undoubtedly recorded in England. That Ramanujan no longer concentrated on logging entries in his notebooks is evident from two letters that he wrote to friends in Madras during his first year in England. In a letter of November 13, 1914 to his friend R. Krishna Rao [51, pp. 112–113], Ramanujan confided, "I have changed my plan of publishing my results. I am not going to publish any of the old results in my notebooks till the war is over." And in a letter of January 7, 1915 to S. M. Subramanian [51, pp. 123–125], Ramanujan admitted, "I am doing my work very slowly. My notebook is sleeping in a corner for these four or five months. I am publishing only my present researches as I have not yet proved the results in my notebooks rigorously."

Ramanujan soon became famous for the papers he published in England, some of them coauthored with Hardy. One of his most important papers is [186], [192, pp. 136–162], in which he introduced

his famous tau function $\tau(n)$, discussed in Chapter 2 of this book, and the Eisenstein series, P , Q , and R , which are introduced here in Chapter 4 and which were key players in so much of his research. In their paper [109], [192, pp. 262–275], Hardy and Ramanujan launched the new field of probabilistic number theory, which became an important branch in number theory. In another paper [110], [192, pp. 276–309], in the course of obtaining an asymptotic series for the partition function $p(n)$, Hardy and Ramanujan introduced the *circle method*, which still today is the primary tool for analytically attacking problems in additive number theory. The genesis of the circle method can be found in Ramanujan's notebooks [193], but unfortunately it is most frequently called the Hardy–Littlewood circle method today. In the latter part of his stay in England, Ramanujan wrote his famous papers on congruences for $p(n)$ [188], [192, pp. 210–213] and [190], [192, p. 230], about which much is written in this book.

Although Ramanujan never doubted his decision to accept Hardy's invitation to Cambridge, not all was well with Ramanujan. World War I began shortly after his arrival, and being a strict vegetarian, he could not always obtain familiar food and spices from India. On March 24, 1915, near the end of his first winter in Cambridge, Ramanujan wrote his friend E. Vinayaka Row in Madras [51, pp. 116–117], "I was not well till the beginning of this term owing to the weather and consequently I couldn't publish any thing for about 5 months." By the end of his third year in England, Ramanujan was critically ill, and, for the next two years, he was confined to sanitariums and nursing homes.

Ramanujan's health turned slightly upward when in 1918 he became the second Indian to be elected as a Fellow of the Royal Society and the first Indian to be chosen as a Fellow of Trinity College. After World War I ended, in 1919, Ramanujan returned home, but his health continued to deteriorate, and on April 26, 1920 Ramanujan died at the age of 32.

Doctors in both England and India had difficulty diagnosing Ramanujan's illness. He was treated for tuberculosis, but a severe vitamin deficiency, liver cancer, lead poisoning purportedly from not properly cleaning his cooking vessels, and a rare tropical disease were

other diagnoses. However, D. A. B. Young [52, pp. 65–75] made a careful examination of all extant records and recorded symptoms of Ramanujan's illness and convincingly concluded that Ramanujan suffered from hepatic amoebiasis (a parasitic infection of the liver). Not only do all of Ramanujan's symptoms suggest this disease, but Ramanujan's medical history in India also favors this diagnosis. Amoebiasis is a protozoal infection of the large intestine that gives rise to dysentery. In 1906 Ramanujan left home to attend Pachaiyappa's College in Madras, where he contracted a severe case of dysentery and had to return home for three months. Unless adequately treated, the infection is permanent, although the patient may go for long periods without exhibiting any symptoms. Relapses occur when the host-parasite relationship is disturbed, which likely happened when he endured a colder climate and perhaps inadequate nutrition after his arrival in England. The illness is difficult to diagnose.

Our description of Ramanujan's life has been necessarily brief. For several years, the standard sources about Ramanujan's life have been the obituaries of P. V. Seshu Aiyar, R. Ramachandra Rao, and Hardy, found in Ramanujan's *Collected Papers* [192] and Chapter 1 of Hardy's book [107]. By far, the most comprehensive biography of Ramanujan has been written by R. Kanigel [134]. The letters from and to Ramanujan are also a source of both mathematical and personal information about Ramanujan, and most of the extant letters have been compiled with commentary by R. A. Rankin and the author [51].

After Ramanujan died, Hardy strongly urged that Ramanujan's notebooks be edited and published. By "editing," Hardy meant that each claim made by Ramanujan in his notebooks should be examined and proved, if it cannot be found in the literature. Ramanujan, in fact, had left his first notebook with Hardy when he returned to India in 1919, and in 1923 Hardy wrote a paper [106], [108, pp. 505–516] about a chapter on hypergeometric series found in the first notebook. In this paper, Hardy pointed out that Ramanujan had independently discovered most of the important classical results in the subject while also discovering several new theorems as well. For the definition of a hypergeometric series, see Chapter 5 of this monograph.

Hardy sent the first notebook to the University of Madras where Ramanujan's other notebooks and papers were being preserved. Plans were undertaken to publish Ramanujan's collected papers and, possibly, his notebooks and other manuscripts. Handwritten copies of the notebooks were sent to Hardy along with other manuscripts and papers in 1923, but the papers were never returned to the University of Madras. It transpired that Ramanujan's *Collected Papers* [192] were published in 1927, but his notebooks and other manuscripts were not published.

Sometime in the late 1920s, G. N. Watson and B. M. Wilson began the task of editing Ramanujan's notebooks. The second notebook, being a revised, enlarged edition of the first, was their primary focus. Wilson was assigned Chapters 2–14, and Watson was to examine Chapters 15–21. Wilson devoted his efforts to this task until 1935, when he died from an infection at the early age of 38. Watson wrote over 30 papers inspired by the notebooks before his interest evidently waned in the late 1930s. Thus, the project was never completed.

It was not until 1957 that the notebooks were made available to the public when the Tata Institute of Fundamental Research in Bombay published a photocopy edition [193], but no editing was undertaken. The first notebook was published in volume 1, and volume 2 comprises the second and third notebooks. The present author undertook the task of editing Ramanujan's notebooks in 1977. With the help of several mathematicians, the author completed his work with the publication of his fifth volume [38] on the notebooks in 1998.

In the spring of 1976, George Andrews of Pennsylvania State University visited Trinity College, Cambridge, to examine the papers left by Watson. Among Watson's papers, he found a manuscript containing 138 pages in the handwriting of Ramanujan. In view of the fame of Ramanujan's notebooks [193], it was natural for Andrews to call this newly found manuscript "Ramanujan's lost notebook." How did this manuscript reach Trinity College?

Watson died in 1965 at the age of 79. Shortly thereafter, on separate occasions, J. M. Whittaker and R. A. Rankin visited Mrs. Watson. Whittaker was a son of E. T. Whittaker, who coauthored with

Watson probably the most popular and frequently used text on analysis in the 20th century [221]. Rankin had succeeded Watson as Mason Professor of Mathematics at the University of Birmingham, where Watson served for most of his career, but was now Professor of Mathematics at the University of Glasgow. Both Whittaker and Rankin went to Watson's attic office to examine the papers left by him, and Whittaker found the aforementioned manuscript by Ramanujan. Rankin suggested to Mrs. Watson that he might sort her late husband's papers and send those worth preserving to Trinity College Library, Cambridge. During the next three years, Rankin sorted through Watson's papers sending them in batches to Trinity College Library, with Ramanujan's manuscript being sent on December 26, 1968. Not realizing the importance of Ramanujan's papers, neither Rankin nor Whittaker mentioned them in their obituaries of Watson [195], [222]. The next question is: How did Watson come into possession of this sheaf of 138 pages of Ramanujan's work?

We mentioned above that in 1923 the University of Madras had sent a package of Ramanujan's papers to Hardy. Most likely, this shipment contained the "lost notebook." Of the over 30 papers that Watson wrote on Ramanujan's work, two of his last papers were devoted to Ramanujan's mock theta functions, which Ramanujan discovered in the last year of his life, which he described in a letter to Hardy only about three months before he died [51, pp. 220–223], and which are also found in the lost notebook. In these two papers, Watson made some conjectures about the existence of certain mock theta functions. If he had the lost notebook at that time, he would have seen that his conjectures were correct. Thus, probably sometime after Watson's interest in Ramanujan's work declined in the late 1930s, Hardy passed Ramanujan's papers to Watson.

In early 1988, just after the centenary of Ramanujan's birth, Narosa Publishing House in New Delhi published a photocopy edition of the lost notebook [194]. Included in this publication are partial manuscripts, loose papers, and fragments by Ramanujan, as well as letters from Ramanujan to Hardy written from nursing homes during the last two years of Ramanujan's sojourn in England.

The first chapter of this book is devoted to basic facts about q -series and theta functions, including the q -binomial theorem, the Jacobi triple product identity, the pentagonal number theorem, Ramanujan's ${}_1\psi_1$ summation theorem, and the quintuple product identity. Many of the theorems proved in Chapter 1 can be found in Chapter 16 of Ramanujan's second notebook [193], [34].

Chapter 2 focuses on congruences for the partition function $p(n)$ and Ramanujan's tau function $\tau(n)$. Much of this material is taken from Ramanujan's handwritten manuscript on $p(n)$ and $\tau(n)$, which was first published in 1988 along with Ramanujan's lost notebook [194]. Adding details to many of Ramanujan's proofs and discussing Ramanujan's theorems in light of the literature written after Ramanujan's death, the present author and K. Ono [50] published an expanded version of this manuscript.

In his notebooks [193], Ramanujan recorded a large number of entries on Lambert series. These identities for Lambert series were used by Ramanujan to establish theta function identities and formulas for the number of representations of an integer as a sum of a certain numbers of squares or of triangular numbers. We introduce readers to Lambert series in Chapter 3 and establish many identities leading to formulas for sums of squares and triangular numbers. A manuscript with no proofs on precisely this subject is another of those manuscripts published with Ramanujan's lost notebook [194], [19, Chapter 18]. His second notebook also contains a large number of such theorems.

Eisenstein series permeate Ramanujan's notebooks [193] and lost notebook [194]. Much of our exposition on Eisenstein series in Chapter 4, however, is taken from Ramanujan's epic paper [186], [192, 136–162]. One of Ramanujan's approaches to congruences for $p(n)$ is based on Eisenstein series, which we demonstrate at the close of Chapter 4.

In Chapter 5, we introduce readers to hypergeometric functions and elliptic integrals. Our goal in this chapter is to prove one of the most fundamental theorems of elliptic functions relating hypergeometric functions and elliptic integrals to theta functions. This theorem enables us to express theta functions and Eisenstein series

at various arguments in terms of certain elliptic parameters. Our exposition is derived from Chapter 17 of Ramanujan's second notebook [193], [34, Chapter 17], where, through a series of preliminary lemmas, Ramanujan leads us to the aforementioned key theorem.

Applications of the aforementioned representations for Eisenstein series and theta functions form the content of Chapter 6. First, we return to the topic of sums of squares and demonstrate how the formulas for Eisenstein series lead to short proofs of some of the results from Chapter 3. However, most of Chapter 6 is devoted to modular equations, a topic to which Ramanujan made more contributions than any other mathematician. Chapters 19–21 in his second notebook are devoted to modular equations, and our short introduction to this topic is drawn from our previous account of Ramanujan's work in these chapters [34].

One of Ramanujan's favorite topics was the Rogers–Ramanujan continued fraction, the focus of Chapter 7. Because we wish to share so much about this continued fraction with readers and because the length of the chapter would be prohibitive if we proved all theorems offered in this chapter, we forego some of the proofs. However, we do prove two key theorems relating the continued fraction with its reciprocal. These theorems are then used to give an alternative, cleaner proof of an identity of Ramanujan in Chapter 2, yielding immediately the congruence $p(5n + 4) \equiv 0 \pmod{5}$. The famous Rogers–Ramanujan functions are also discussed, and, in particular, we prove that the Rogers–Ramanujan continued fraction can be represented as a quotient of the two Rogers–Ramanujan functions. Our fervent wish is that our sampling of the many beautiful properties satisfied by this continued fraction will motivate readers to turn to original sources to learn more about it.

Ubiquitous in this book are products of the form

$$(1 - a)(1 - aq)(1 - aq^2) \cdots (1 - aq^{n-1}) =: (a; q)_n$$

as well as their infinite versions

$$(1 - a)(1 - aq)(1 - aq^2) \cdots (1 - aq^n) \cdots =: (a; q)_\infty, \quad |q| < 1,$$

which are called q -products. Although we assume that readers of this book are familiar with infinite series, it may well be that some

are not familiar with infinite products. A reader desiring to learn a few basic facts about the convergence of infinite products may consult a good text on complex analysis, such as that of N. Levinson and R. Redheffer [142, pp. 382–385], for basic properties of infinite products. In particular, all the infinite products in the present text converge absolutely and uniformly on compact subsets of $|q| < 1$. In particular, taking logarithms of infinite products and differentiating the resulting series termwise is permitted. At first, you may find that working with the products $(a; q)_n$ and $(a; q)_\infty$ is somewhat tedious. In order to verify q -product identities or to manipulate q -products, it may be helpful to write out the first three or four terms of each q -product. This should provide the needed insight in order to justify a given step. After working with q -products for awhile, you will begin to handle them more quickly and adroitly, and no longer need to write out any of their terms longhand. When you reach this stage, you should feel quite comfortable in manipulating q -series. It is assumed throughout the entire book that $|q| < 1$. Over 50 exercises are interspersed within the exposition.

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Chapter 1

Introduction

1.1. Notation and Arithmetical Functions

In this first section, we introduce notation that will be used throughout the entire monograph. Secondly, we define the arithmetical functions on which we will focus for most of this book, and which can be studied by employing the theory of theta functions and q -series. A brief introduction to theta functions and q -series will be given in the next section, to be followed in Section 1.3 by a few of the most useful theorems about these functions.

Definition 1.1.1. *Define*

(1.1.1)

$$(a)_0 := (a; q)_0 := 1, \quad (a)_n := (a; q)_n := \prod_{k=0}^{n-1} (1 - aq^k), \quad n \geq 1,$$

$$(1.1.2) \quad (a)_\infty := (a; q)_\infty := \prod_{k=0}^{\infty} (1 - aq^k), \quad |q| < 1.$$

We call q the base, and if the identification of the base is clear, we often omit q from the notation.

Replacing a by q^a , where $a > 0$, and using L'Hôpital's rule while letting q tend to 1, we find that

$$(1.1.3) \quad \lim_{q \rightarrow 1} \frac{(q^a; q)_n}{(1 - q)^n} = \lim_{q \rightarrow 1} \frac{1 - q^a}{1 - q} \frac{1 - q^{a+1}}{1 - q} \cdots \frac{1 - q^{a+n-1}}{1 - q} \\ = a(a + 1) \cdots (a + n - 1).$$

The expression on the right side above is called a rising or shifted factorial, and so we see that $(a; q)_n$ may be considered as an analogue of the rising factorial. We next define the q -binomial coefficient, or Gaussian coefficient, $\begin{bmatrix} n \\ m \end{bmatrix}$, which is an analogue of the ordinary binomial coefficient $\binom{n}{m}$.

Definition 1.1.2. Let n and m denote integers. Then the Gaussian coefficient is defined by

$$(1.1.4) \quad \begin{bmatrix} n \\ m \end{bmatrix} := \begin{cases} \frac{(q)_n}{(q)_m (q)_{n-m}}, & \text{if } 0 \leq m \leq n, \\ 0, & \text{otherwise.} \end{cases}$$

Exercise 1.1.3. Using (1.1.3) three times, each time with $a = 1$, show that

$$\lim_{q \rightarrow 1} \begin{bmatrix} n \\ m \end{bmatrix} = \binom{n}{m}.$$

Thus, the q -binomial coefficient tends to the ordinary binomial coefficient when $q \rightarrow 1$.

From the definition (1.1.4) it is not obvious that the q -binomial coefficients are polynomials in q .

Exercise 1.1.4. Using the definition (1.1.4), readers should first prove the first q -analogue of Pascal's formula given below.

Lemma 1.1.5. For $n \geq 1$,

$$(1.1.5) \quad \begin{bmatrix} n \\ m \end{bmatrix} = \begin{bmatrix} n-1 \\ m-1 \end{bmatrix} + q^m \begin{bmatrix} n-1 \\ m \end{bmatrix}$$

Exercise 1.1.6. Second, employing Lemma 1.1.5 and induction on n , prove that $\begin{bmatrix} n \\ m \end{bmatrix}$ is a polynomial in q of degree $m(n - m)$.

Exercise 1.1.7. Establish a second q -analogue of Pascal's formula, which is given for $n > 1$ by

$$\begin{bmatrix} n \\ m \end{bmatrix} = q^{n-m} \begin{bmatrix} n-1 \\ m-1 \end{bmatrix} + \begin{bmatrix} n-1 \\ m \end{bmatrix}.$$

Exercise 1.1.8. Prove that, for each pair of nonnegative integers m, n ,

$$\sum_{j=0}^n \begin{bmatrix} m+j \\ j \end{bmatrix} q^j = \begin{bmatrix} m+n+1 \\ m+1 \end{bmatrix}.$$

We now define four primary arithmetical functions on which we focus throughout the monograph.

Definition 1.1.9. If n is a positive integer, let $p(n)$ denote the number of unrestricted representations of n as a sum of positive integers, where representations with different orders of the same summands are not regarded as distinct. We call $p(n)$ the partition function.

For example, $p(4) = 5$, because there are 5 ways to represent 4 as a sum of positive integers, namely, $4 = 3+1 = 2+2 = 2+1+1 = 1+1+1+1$. Readers should check that $p(5) = 7$ and $p(6) = 11$, but readers should not check (at least by hand) that $p(200) = 3,972,999,029,388$. A few moments of reflection convinces one that $p(n)$ grows rapidly. More precisely, G. H. Hardy and Ramanujan [110], [192, pp. 276–309] showed that, as $n \rightarrow \infty$,

$$(1.1.6) \quad p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right),$$

i.e., the ratio of the left and right sides of (1.1.6) tends to 1 as n tends to ∞ .

The generating function for $p(n)$, due to Euler, is given by

$$(1.1.7) \quad \sum_{n=0}^{\infty} p(n)q^n = \frac{1}{(q; q)_{\infty}} = \frac{1}{1-q} \frac{1}{1-q^2} \cdots \frac{1}{1-q^k} \cdots,$$

where we define $p(0) = 1$. To see this, observe that the factor

$$\frac{1}{1-q^k} = \sum_{j=0}^{\infty} q^{jk}$$

generates the number of k 's that appear in a particular partition of n . Each partition of n appears once and only once on the right side of (1.1.7) when interpreted in this manner.

The generating function for the number of partitions of n into distinct parts, denoted by $p_d(n)$, is given by

$$(1.1.8) \quad \sum_{n=0}^{\infty} p_d(n)q^n = (-q; q)_{\infty} = (1+q)(1+q^2)\cdots(1+q^k)\cdots,$$

because any particular integer can occur at most once in any given representation of n . We now show that, by elementary product manipulations, we can deduce the following famous theorem of Euler.

Theorem 1.1.10 (Euler). *The number of partitions of a positive integer n into distinct parts equals the number of partitions of n into odd parts, denoted by $p_o(n)$.*

Proof. From (1.1.8),

$$(1.1.9) \quad \sum_{n=0}^{\infty} p_d(n)q^n = (-q; q)_{\infty} = \frac{(q^2; q^2)_{\infty}}{(q; q)_{\infty}} = \frac{1}{(q; q^2)_{\infty}} = \sum_{n=0}^{\infty} p_o(n)q^n$$

Equating coefficients of q^n , $n \geq 1$, on both sides of (1.1.9), we complete the proof of Euler's identity. $\square$

For example, $p_d(6) = 4$, because 6 has 4 representations into distinct parts, namely, $6 = 5 + 1 = 4 + 2 = 3 + 2 + 1$. On the other hand, $p_o(6) = 4$, because 6 has 4 representations into odd parts, namely, $5 + 1 = 3 + 3 = 3 + 1 + 1 + 1 = 1 + 1 + 1 + 1 + 1 + 1$.

Exercise 1.1.11. *Euler's identity can be slightly refined. Prove that the number of partitions of n into an even (odd) number of odd parts equals the number of partitions of n into distinct parts, where the number of odd parts is even (odd).*

Exercise 1.1.12. *Prove that the number of partitions of the positive integer n into parts that are not divisible by 3 is equal to the number of partitions of n in which no part appears more than twice.*

Exercise 1.1.13. *Euler's identity can be refined in yet another way. Let k and n be positive integers with $k \geq 2$. Prove that the number of*

partitions of n in which no part is divisible by k equals the number of partitions of n such that there are strictly less than k copies of each part. Note that when $k = 2$, we obtain Euler's identity. The case $k = 3$ is the previous exercise.

Using elementary product manipulations as in the proof of Theorem 1.1.10, readers are encouraged to prove the theorems in the following exercises.

Exercise 1.1.14. Let $A(n)$ denote the number of partitions of the positive integer n into parts congruent to 2, 5, or 11 modulo 12. Let $B(n)$ denote the number of partitions of n into distinct parts congruent to either 2, 4, or 5 modulo 6. Prove that

$$A(n) = B(n).$$

Exercise 1.1.15. Prove that the number of partitions of n in which only odd parts may be repeated is equal to the number of partitions of n in which no part appears more than 3 times.

Exercise 1.1.16. Prove that the number of partitions of n in which each part appears exactly 2, 3, or 5 times equals the number of partitions of n into parts congruent to ± 2 , ± 3 , or 6 modulo 12.

Exercise 1.1.17. Prove that the number of partitions of n in which no part appears exactly once equals the number of partitions of n into parts not congruent to ± 1 modulo 6.

Definition 1.1.18. Define the coefficients $\tau(n)$, $n \geq 1$, by

$$(1.1.10) \quad q(q; q)_{\infty}^{24} = \sum_{n=1}^{\infty} \tau(n) q^n, \quad |q| < 1.$$

Then $\tau(n)$ is called Ramanujan's tau function.

For example, $\tau(2) = -24$ and $\tau(15) = 1,217,160$. A glance at a table of $\tau(n)$ shows that $|\tau(n)|$ grows quickly. It has long been conjectured, but never proved, that $\tau(n) \neq 0$ for all $n > 0$. It may seem at first sight that the definition of $\tau(n)$ is artificial. However, it is one of the most important arithmetical functions in number theory and arises in many contexts, in particular, in the theory of modular forms. We study this arithmetical function in Chapter 2.

Definition 1.1.19. For positive integers n and k , let $r_k(n)$ denote the number of representations of n as a sum of k squares, where representations with different orders and different signs are counted as distinct. By convention, $r_k(0) = 1$.

For example, $r_2(2) = 4$, because $2 = 1^2 + 1^2 = 1^2 + (-1)^2 = (-1)^2 + 1^2 = (-1)^2 + (-1)^2$; $r_2(9) = 4$, because $9 = 3^2 + 0^2 = 0^2 + 3^2 = (-3)^2 + 0^2 = 0^2 + (-3)^2$; and $r_2(7) = 0$, because there are not any ways we can write 7 as a sum of 2 squares.

Definition 1.1.20. For $n \geq 0$, the triangular numbers are the numbers $n(n+1)/2$. If n and k are positive integers, let $t_k(n)$ denote the number of representations of n as a sum of k triangular numbers, where representations with different orders are counted as distinct. Set $t_k(0) = 1$.

For example, $t_2(7) = 2$, because $7 = 1 + 6 = 6 + 1$, while $t_2(16) = 4$, because $16 = 1 + 15 = 15 + 1 = 6 + 10 = 10 + 6$.

1.2. What are q -Series and Theta Functions?

Generally, a q -series has expressions of the type $(a; q)_n$ in the summands. In Section 1.3, we consider a few of the most important examples. However, many q -series do not have such products in their summands, but they may arise as limiting cases of [series](#) containing products (1.1.1), or their theory may be intimately intertwined with such [series](#). Theta functions are such examples.

Definition 1.2.1. Ramanujan's general theta function $f(a, b)$ is defined by

$$(1.2.1) \quad f(a, b) := \sum_{n=-\infty}^{\infty} a^{n(n+1)/2} b^{n(n-1)/2}, \quad |ab| < 1.$$

Three special cases are defined by, in Ramanujan's notation,

$$(1.2.2) \quad \varphi(q) := f(q, q) = \sum_{n=-\infty}^{\infty} q^{n^2},$$

$$(1.2.3) \quad \psi(q) := f(q, q^3) = \sum_{n=0}^{\infty} q^{n(n+1)/2},$$

$$(1.2.4) \quad f(-q) := f(-q, -q^2) = \sum_{n=-\infty}^{\infty} (-1)^n q^{n(3n-1)/2}.$$

Exercise 1.2.2. Justify the second equality of (1.2.3). See also (1.2.7) of Lemma 1.2.4 below.

The numbers $n(3n-1)/2$, $n \geq 0$, are the *pentagonal numbers*, and the numbers $n(3n \pm 1)/2$, $n \geq 0$, are the *generalized pentagonal numbers*.

Note that

$$(1.2.5) \quad \varphi^k(q) = \sum_{n=0}^{\infty} r_k(n)q^n \quad \text{and} \quad \psi^k(q) = \sum_{n=0}^{\infty} t_k(n)q^n.$$

These generating functions will be used in the sequel to find formulas for $r_k(n)$ and $t_k(n)$, for certain positive integers k .

Readers may wonder why the functions defined above are called theta functions, since no theta appears in the notations. In the classical definitions, e.g., see Whittaker and Watson's text [221, p. 464], the notation ϑ is employed. We emphasize that Ramanujan's definition is different from the usual definitions, but the generality is identical to that in [221], i.e., all of Ramanujan's theta functions fall under the purview in [221], and conversely.

Exercise 1.2.3. Prove the following basic and enormously useful properties of theta functions.

Lemma 1.2.4. We have

$$(1.2.6) \quad f(a, b) = f(b, a),$$

$$(1.2.7) \quad f(1, a) = 2f(a, a^3),$$

$$(1.2.8) \quad f(-1, a) = 0,$$

and, if n is any integer,

$$(1.2.9) \quad f(a, b) = a^{n(n+1)/2} b^{n(n-1)/2} f(a(ab)^n, b(ab)^{-n}).$$

1.3. Fundamental Theorems about q -Series and Theta Functions

Perhaps the most important theorem in the subject of q -series is the q -binomial theorem. *Copyrighted Material*

Theorem 1.3.1 (*q*-analogue of the binomial theorem). For $|q|, |z| < 1$,

$$(1.3.1) \quad \sum_{n=0}^{\infty} \frac{(a)_n}{(q)_n} z^n = \frac{(az)_{\infty}}{(z)_{\infty}}.$$

Proof. Note that the product on the right side of (1.3.1) converges uniformly on compact subsets of $|z| < 1$ and so represents an analytic function on $|z| < 1$. Thus, we may write

$$(1.3.2) \quad F(z) := \frac{(az)_{\infty}}{(z)_{\infty}} = \sum_{n=0}^{\infty} A_n z^n, \quad |z| < 1.$$

From the product representation in (1.3.2), we can readily verify that

$$(1.3.3) \quad (1 - z)F(z) = (1 - az)F(qz).$$

Equating coefficients of z^n , $n \geq 1$, on both sides of (1.3.3), we find that

$$A_n - A_{n-1} = q^n A_n - a q^{n-1} A_{n-1},$$

or

$$(1.3.4) \quad A_n = \frac{1 - a q^{n-1}}{1 - q^n} A_{n-1}, \quad n \geq 1.$$

Iterating (1.3.4) and using the value $A_0 = 1$, which is readily apparent from (1.3.2), we deduce that

$$(1.3.5) \quad A_n = \frac{(a)_n}{(q)_n}, \quad n \geq 0.$$

Using (1.3.5) in (1.3.2), we complete the proof of (1.3.1). $\square$

To understand why Theorem 1.3.1 is called the *q*-analogue of the binomial theorem, replace a by q^a in Theorem 1.3.1. Arguing as we did in (1.1.3), we find that

$$\lim_{q \rightarrow 1} \frac{(q^a)_n}{(q)_n} = \frac{a(a+1) \cdots (a+n-1)}{n!}.$$

Assuming now that a is a positive integer, and remembering that a has been replaced by q^a , we find that formally letting $q \rightarrow 1$ in (1.3.1) yields

$$\sum_{n=0}^{\infty} \frac{a(a+1) \cdots (a+n-1)}{n!} z^n = \prod_{n=0}^{a-1} \frac{1}{1-z} = (1-z)^{-a},$$

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which in calculus is often called the generalized binomial theorem.

Corollary 1.3.2 (Euler). For $|q| < 1$,

$$(1.3.6) \quad \sum_{n=0}^{\infty} \frac{z^n}{(q)_n} = \frac{1}{(z)_{\infty}}, \quad |z| < 1,$$

and

$$(1.3.7) \quad \sum_{n=0}^{\infty} \frac{(-z)^n q^{n(n-1)/2}}{(q)_n} = (z)_{\infty}, \quad |z| < \infty.$$

Proof. Equality (1.3.6) follows immediately from (1.3.1) by setting $a = 0$.

Replace a by a/b and z by bz in the q -binomial theorem. Thus, for $|bz| < 1$,

$$(1.3.8) \quad \sum_{n=0}^{\infty} \frac{(a/b)_n}{(q)_n} (bz)^n = \frac{(az)_{\infty}}{(bz)_{\infty}}.$$

We let $b \rightarrow 0$. Now,

$$(1.3.9) \quad \begin{aligned} \lim_{b \rightarrow 0} (a/b)_n b^n &= \lim_{b \rightarrow 0} \left(1 - \frac{a}{b}\right) \left(1 - \frac{aq}{b}\right) \cdots \left(1 - \frac{aq^{n-1}}{b}\right) b^n \\ &= (-a)^n q^{n(n-1)/2}. \end{aligned}$$

Equality (1.3.7) now follows upon setting $a = 1$. □

Alert readers will have immediately noticed that we have taken the limit on b under the summation sign without justifying it. In the theory of q -series, taking limits under the summation sign, as we have done, is always assumed to be justified, but hardly anyone ever does it. Nonetheless, a rigorous argument almost always runs along the same lines. In the case at hand, choose a number M such that $|q| \leq M < 1$. Let $\epsilon > 0$ be given such that $0 < 2\epsilon < 1 - M$. With $|b| \leq \epsilon$ temporarily fixed, let N_0 be that unique positive integer such that

$$|b| + |a|M^k \geq 2\epsilon, \quad 0 \leq k < N_0,$$

and

$$|b| + |a|M^{N_0} < 2\epsilon.$$

Then, for $|b| \leq \epsilon$ and $n \geq N_0$,

$$\begin{aligned} \left| \frac{(a/b)_n}{(q)_n} b^n \right| &\leq \frac{\prod_{k=0}^{n-1} (|b| + |a|M^k)}{(1-M)^n} \\ &\leq \frac{(\epsilon + |a|)^{N_0} (2\epsilon)^{n-N_0}}{(1-M)^n} \\ &= \left(\frac{\epsilon + |a|}{2\epsilon} \right)^{N_0} \left(\frac{2\epsilon}{1-M} \right)^n \end{aligned}$$

Since $2\epsilon < 1 - M$,

$$\sum_{n=N_0}^{\infty} \left(\frac{2\epsilon}{1-M} \right)^n < \infty.$$

Thus,

$$\sum_{n=0}^{\infty} \frac{(a/b)_n}{(q)_n} (bz)^n$$

converges uniformly for $|b| \leq \epsilon$ by the Weierstrass M -test, and so letting $b \rightarrow 0$ under the summation sign is justified.

As the q -analogue of the binomial theorem is perhaps the most fundamental result in the theory of q -series, so the Jacobi triple product identity in the next theorem is likely the most important and useful result in the theory of theta functions.

Theorem 1.3.3 (Jacobi Triple Product Identity). *For $z \neq 0$ and $|q| < 1$,*

$$(1.3.10) \quad \sum_{n=-\infty}^{\infty} z^n q^{n^2} = (-zq; q^2)_{\infty} (-q/z; q^2)_{\infty} (q^2; q^2)_{\infty}.$$

In Ramanujan's notation (1.2.1), the Jacobi triple product identity takes the shape

$$(1.3.11) \quad f(a, b) = (-a; ab)_{\infty} (-b; ab)_{\infty} (ab; ab)_{\infty}.$$

Proof. In (1.3.7), replace q by q^2 and z by $-zq$ to deduce that

$$\begin{aligned} (-zq; q^2)_{\infty} &= \sum_{n=0}^{\infty} \frac{z^n q^{n^2}}{(q^2; q^2)_n} = \frac{1}{(q^2; q^2)_{\infty}} \sum_{n=0}^{\infty} z^n q^{n^2} (q^{2n+2}; q^2)_{\infty} \\ (1.3.12) \quad &= \frac{1}{(q^2; q^2)_{\infty}} \sum_{n=-\infty}^{\infty} z^n q^{n^2} (q^{2n+2}; q^2)_{\infty}, \end{aligned}$$

since $(q^{2n+2}; q^2)_\infty = 0$ when n is a negative integer. Now apply (1.3.7) again with q replaced by q^2 but this time with $z = q^{2n+2}$. Thus, from (1.3.12),

$$\begin{aligned}
 (-zq; q^2)_\infty &= \frac{1}{(q^2; q^2)_\infty} \sum_{n=-\infty}^{\infty} z^n q^{n^2} \sum_{r=0}^{\infty} \frac{(-1)^r q^{(2n+2)r+r^2-r}}{(q^2; q^2)_r} \\
 &= \frac{1}{(q^2; q^2)_\infty} \sum_{r=0}^{\infty} \frac{(-1)^r z^{-r} q^r}{(q^2; q^2)_r} \sum_{n=-\infty}^{\infty} z^{n+r} q^{(n+r)^2} \\
 &= \frac{1}{(q^2; q^2)_\infty} \sum_{r=0}^{\infty} \frac{(-q/z)^r}{(q^2; q^2)_r} \sum_{m=-\infty}^{\infty} z^m q^{m^2} \\
 &= \frac{1}{(q^2; q^2)_\infty (-q/z; q^2)_\infty} \sum_{m=-\infty}^{\infty} z^m q^{m^2},
 \end{aligned}$$

by (1.3.6) with z replaced by $-q/z$ and q replaced by q^2 , and therefore with the restriction $|q/z| < 1$. Rearranging above, we complete the proof of (1.3.10) for $|q/z| < 1$. However, by analytic continuation, (1.3.10) holds for all complex $z \neq 0$, and so the proof is complete. $\square$

Corollary 1.3.4. Recall that Ramanujan's theta functions $\varphi(q)$, $\psi(q)$, and $f(-q)$ are defined by (1.2.2), (1.2.3), and (1.2.4), respectively. Then,

$$(1.3.13) \quad \varphi(q) = (-q; q^2)_\infty^2 (q^2; q^2)_\infty,$$

$$(1.3.14) \quad \psi(q) = \frac{(q^2; q^2)_\infty}{(q; q^2)_\infty},$$

$$(1.3.15) \quad f(-q) = (q; q)_\infty.$$

Proof. The product (1.3.13) follows from (1.2.2) and (1.3.11) by setting $a = b = q$.

Letting $a = q$ and $b = q^3$ in (1.3.11) and using (1.2.3) and (1.1.9), we find that

$$(1.3.16)$$

$$\begin{aligned}
 \psi(q) &= (-q; q^4)_\infty (-q^3; q^4)_\infty (q^4; q^4)_\infty = (-q; q^2)_\infty (-q^2; q^2)_\infty (q^2; q^2)_\infty \\
 &= (-q; q)_\infty (q^2; q^2)_\infty = \frac{(q^2; q^2)_\infty}{(q; q^2)_\infty}.
 \end{aligned}$$

Lastly, set $a = -q, b = -q^2$ in (1.3.11). Then, from (1.2.4),

$$(1.3.17) \quad f(-q) = (q; q^3)_\infty (q^2; q^3)_\infty (q^3; q^3)_\infty = (q; q)_\infty.$$

□

Combining (1.2.4) and (1.3.15), we obtain Euler's pentagonal number theorem, which, because of its importance, we state in a separate corollary.

Corollary 1.3.5 (Euler's Pentagonal Number Theorem).

$$(1.3.18) \quad \sum_{n=-\infty}^{\infty} (-1)^n q^{n(3n-1)/2} = \sum_{n=-\infty}^{\infty} (-1)^n q^{n(3n+1)/2} = (q; q)_\infty.$$

Note that the second representation in (1.3.18) arises from the former by replacing n by $-n$ in the first sum. In the sequel, we shall use these representations interchangeably.

From Corollary 1.3.5, we can easily derive a recurrence formula for calculating values of $p(n)$.

Corollary 1.3.6. *For brevity, set $\omega_j = j(3j-1)/2$, $-\infty < j < \infty$. Then*

$$(1.3.19) \quad p(n) = \sum_{0 < \omega_j \leq n} (-1)^{j+1} p(n - \omega_j).$$

Proof. By Corollary 1.3.5,

$$(1.3.20) \quad 1 = \sum_{j=-\infty}^{\infty} (-1)^j q^{j(3j-1)/2} \sum_{m=0}^{\infty} p(m) q^m.$$

Now equate coefficients of q^n , $n \geq 1$, on both sides of (1.3.20). The recurrence relation (1.3.19) then immediately follows. □

Exercise 1.3.7. *There is another simple recurrence relation for $p(n)$ that is due to Euler. Consider the generating function*

$$(1.3.21) \quad F(q) := \sum_{n=0}^{\infty} p(n) q^n = \frac{1}{(q; q)_\infty}.$$

Logarithmically differentiate both sides of (1.3.21). Then expand the resulting denominators on the right-hand side into geometric series.

If $\sigma(n) := \sum_{d|n} d$, deduce that, for each integer $n > 1$,

$$np(n) = \sum_{j=0}^{n-1} p(j)\sigma(n-j).$$

Euler's pentagonal number theorem has a beautiful combinatorial interpretation given in the next corollary.

Corollary 1.3.8 (Combinatorial Version of Euler's Pentagonal Number Theorem). *Let $D_e(n)$ denote the number of partitions of n into an even number of distinct parts, and let $D_o(n)$ denote the number of partitions of n into an odd number of distinct parts. Then*

$$(1.3.22) \quad D_e(n) - D_o(n) = \begin{cases} (-1)^j, & \text{if } n = j(3j \pm 1)/2, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. Let us begin by closely examining the product

$$(1.3.23) \quad (q; q)_\infty = (1 - q)(1 - q^2)(1 - q^3) \cdots,$$

which we see is similar to the generating function (1.1.8) for $p_d(n)$, except that in (1.3.23) there are minus signs in the product. When we multiply the terms of (1.3.23) together, we obtain a plus sign if we multiply an even number of powers of q , and we obtain a minus sign if we multiply an odd number of powers of q . Thus, partitions of an integer n into an even number of distinct parts are weighted by a plus sign, while partitions of n into an odd number of distinct parts are weighted by a minus sign. Thus, from the pentagonal number theorem (1.3.18),

$$1 + \sum_{n=1}^{\infty} \{D_e(n) - D_o(n)\} q^n = (q; q)_\infty = \sum_{j=-\infty}^{\infty} (-1)^j q^{j(3j-1)/2}.$$

The result now follows. □

For example, $D_e(5) - D_o(5) = 1$, since $5 = 2(3 \cdot 2 - 1)/2$, i.e., $j = 2$. Indeed, the partitions of 5 into distinct parts are 5, 4+1, and 3+2. Second, $D_e(6) - D_o(6) = 0$, since 6 is not a generalized pentagonal number. Indeed, the partitions of 6 into distinct parts are 6, 3+2+1, 4+2, and 5+1.

The next theorem is fundamental, and we shall use it several times in the sequel.

Theorem 1.3.9 (Jacobi's Identity). *We have*

$$(1.3.24) \quad \sum_{n=0}^{\infty} (-1)^n (2n+1) q^{n(n+1)/2} = (q; q)_{\infty}^3.$$

Proof. In (1.3.10), replace z by $z^2 q$ to deduce that

$$(1.3.25) \quad \sum_{n=-\infty}^{\infty} z^{2n} q^{n^2+n} = (-z^2 q^2; q^2)_{\infty} (-1/z^2; q^2)_{\infty} (q^2; q^2)_{\infty}.$$

Divide both sides of (1.3.25) by $1 + 1/z^2$ to find that

$$(1.3.26) \quad \frac{\sum_{n=-\infty}^{\infty} z^{2n+1} q^{n^2+n}}{z + 1/z} = (-z^2 q^2; q^2)_{\infty} (-q^2/z^2; q^2)_{\infty} (q^2; q^2)_{\infty}.$$

Observe that, by (1.2.8),

$$(1.3.27) \quad \sum_{n=-\infty}^{\infty} (-1)^n q^{n^2+n} = 0.$$

Thus, letting $z \rightarrow i$ in (1.3.26), employing (1.3.27), and applying L'Hôpital's rule, we find that

$$(1.3.28) \quad \frac{1}{2} \sum_{n=-\infty}^{\infty} (-1)^n (2n+1) q^{n(n+1)} = (q^2; q^2)_{\infty}^3.$$

We now divide the sum above into two parts, $-\infty < n \leq -1$, $0 \leq n < \infty$. In the former sum, replace n by $-n-1$ and simplify. Lastly, replace q^2 by q in (1.3.28). We thus arrive at (1.3.24) to complete the proof. $\square$

Besides the functions $\varphi(q)$, $\psi(q)$, and $f(-q)$, Ramanujan defines one further function

$$(1.3.29) \quad \chi(q) := (-q; q^2)_{\infty},$$

which is not a theta function but which plays a prominent role in the theory of theta functions. These four functions satisfy a myriad of relations. We offer here most of those recorded by Ramanujan in Entry 24 of Chapter 16 of his second notebook [193]; see [34, pp. 39–40] for proofs.

Theorem 1.3.10.

$$(1.3.30) \quad \frac{f(q)}{f(-q)} = \frac{\psi(q)}{\psi(-q)} = \frac{\chi(q)}{\chi(-q)} = \sqrt{\frac{\varphi(q)}{\varphi(-q)}},$$

$$(1.3.31) \quad \chi(q) = \frac{f(q)}{f(-q^2)} = \frac{\varphi(q)}{f(q)} = \frac{f(-q^2)}{\psi(-q)},$$

$$(1.3.32) \quad \varphi^2(-q^2) = \varphi(-q)\varphi(q),$$

$$(1.3.33) \quad \chi(q)\chi(-q) = \chi(-q^2),$$

$$(1.3.34) \quad f^3(-q^2) = \varphi^2(-q^2)\psi(q^2) = \varphi(-q)\psi^2(q).$$

Exercise 1.3.11. Use the functions' product representations to prove some of the identities of Theorem 1.3.10.

The q -binomial theorem is perhaps the most widely used summation formula in the theory of q -series. Ramanujan's ${}_1\psi_1$ summation theorem is a bilateral (i.e., the summation index runs from $-\infty$ to ∞ instead of from a finite integer to ∞) analogue and is undoubtedly the most useful bilateral summation in q -series. In both (1.3.1) and (1.3.36) below, the **series** on the left sides have exactly one q -product in the numerator and one q -product in the denominator, and both theorems give product representations for the **series**. The prefix 1 on ψ indicates that there is one q -product in the numerator, and the suffix 1 on ψ indicates that there is one q -product in the denominator. Before we state and prove Ramanujan's ${}_1\psi_1$ summation theorem, it will be convenient to define an analogue of $(a; q)_n$ for negative integers n . So, for $-\infty < n < \infty$, define

$$(1.3.35) \quad (a; q)_n = \frac{(a; q)_\infty}{(aq^n; q)_\infty}.$$

Observe that (1.3.35) agrees with Definition 1.1.1 when n is a non-negative integer.

Theorem 1.3.12 (Ramanujan's ${}_1\psi_1$ Summation). For $|b/a| < |z| < 1$ and $|q| < 1$,

$$(1.3.36) \quad \sum_{n=-\infty}^{\infty} \frac{(a)_n}{(b)_n} z^n = \frac{(az)_\infty (q/(az))_\infty (q)_\infty (b/a)_\infty}{(z)_\infty (b/(az))_\infty (b)_\infty (q/a)_\infty}.$$

Proof. Define

$$(1.3.37) \quad f(z) := \frac{(az)_\infty (q/(az))_\infty}{(z)_\infty (b/(az))_\infty}.$$

In the annulus, $|b/a| < |z| < 1$, $f(z)$ is analytic, and so therefore has a Laurent expansion

$$(1.3.38) \quad f(z) = \sum_{n=-\infty}^{\infty} c_n z^n.$$

From the definition (1.3.37), we can deduce that

$$(1.3.39) \quad (b - aqz)f(qz) = q(1 - z)f(z).$$

Observe that the Laurent expansion of $f(qz)$ is valid in the annulus, $|b/(aq)| < |z| < 1/|q|$, and so the Laurent expansions of both $f(z)$ and $f(qz)$ are valid in the intersection of the two annuli, i.e., for $|b/(aq)| < |z| < 1$. Thus, we need to assume that $|q| > |b/a|$, a restriction that we shall later remove by analytic continuation. Hence, by (1.3.38) and (1.3.39), for $|b/(aq)| < |z| < 1$,

$$(1.3.40) \quad \sum_{n=-\infty}^{\infty} (bc_n - ac_{n-1})q^n z^n = \sum_{n=-\infty}^{\infty} q(c_n - c_{n-1})z^n.$$

Equating coefficients of z^n on both sides of (1.3.40), we find that

$$(bc_n - ac_{n-1})q^n = q(c_n - c_{n-1}).$$

Solving for c_n , we find that

$$(1.3.41) \quad c_n = \frac{1 - aq^{n-1}}{1 - bq^{n-1}} c_{n-1}.$$

If $n > 0$, iterate (1.3.41) to conclude that

$$(1.3.42) \quad c_n = \frac{(a)_n}{(b)_n} c_0.$$

If $n < 0$, rewrite (1.3.41) in the form

$$(1.3.43) \quad c_{n-1} = \frac{1 - bq^{n-1}}{1 - aq^{n-1}} c_n.$$

Iterate (1.3.43), use (1.3.35), and replace n by $n + 1$ to arrive at

$$(1.3.44) \quad c_n = \frac{(a)_n}{(b)_n} c_0.$$

Putting (1.3.42) and (1.3.44) in (1.3.38) and using analytic continuation, we conclude that

$$(1.3.45) \quad f(z) = c_0 \sum_{n=-\infty}^{\infty} \frac{(a)_n}{(b)_n} z^n, \quad |b/a| < |z| < 1.$$

It remains to evaluate c_0 .

Write

$$(1.3.46) \quad f(z) = c_0 \sum_{n=-\infty}^{-1} \frac{(a)_n}{(b)_n} z^n + c_0 \sum_{n=0}^{\infty} \frac{(a)_n}{(b)_n} z^n =: c_0 (g(z) + h(z)).$$

From the definition (1.3.37) of $f(z)$, we see that $f(z)$ has a simple pole at $z = 1$. Since $g(z)$ is a power series about $z = \infty$ converging for $|z| > |b/a|$, it follows that $g(z)$ is analytic at $z = 1$. Hence, by (1.3.46), if we let $G(z) := c_0(1 - z)h(z)$,

$$(1.3.47) \quad \lim_{z \rightarrow 1} (1 - z)f(z) = c_0 \lim_{z \rightarrow 1} (1 - z)h(z) =: \lim_{z \rightarrow 1} G(z).$$

The function $G(z)$ is analytic on $0 \leq |z| < 1/|q|$, because $G(z)$ converges in a neighborhood about $z = 0$, its simple pole at $z = 1$ is removable, and the next largest pole of both $f(z)$ and $h(z)$ is at $z = 1/q$. Thus, from the definition of $h(z)$ given in (1.3.46) and the definition of $G(z)$,

$$G(z) = \lim_{N \rightarrow \infty} c_0 \sum_{n=0}^N \left\{ \frac{(a)_n}{(b)_n} - \frac{(a)_{n-1}}{(b)_{n-1}} \right\} z^n,$$

with the understanding that $(a)_{-1}/(b)_{-1} = 0$, and so, by (1.3.47),

$$\begin{aligned} G(1) &= \lim_{N \rightarrow \infty} c_0 \sum_{n=0}^N \left\{ \frac{(a)_n}{(b)_n} - \frac{(a)_{n-1}}{(b)_{n-1}} \right\} \\ (1.3.48) \quad &= c_0 \lim_{N \rightarrow \infty} \frac{(a)_N}{(b)_N} = c_0 \frac{(a)_{\infty}}{(b)_{\infty}}. \end{aligned}$$

On the other hand, by (1.3.37),

$$(1.3.49) \quad \lim_{z \rightarrow 1} (1 - z)f(z) = \frac{(a)_{\infty}(q/a)_{\infty}}{(q)_{\infty}(b/a)_{\infty}}.$$

Combining together (1.3.48) and (1.3.49), we conclude that

$$(1.3.50) \quad c_0 = \frac{(b)_{\infty}(q/a)_{\infty}}{(a)_{\infty}(b/a)_{\infty}}.$$

Putting (1.3.50) in (1.3.45), we complete the proof of (1.3.36). $\square$

Exercise 1.3.13. *As a corollary, convert Theorem 1.3.12 into the following symmetric form not involving a bilateral series.*

Corollary 1.3.14. *If $|\beta q| < |z| < 1/|\alpha q|$, then*

$$(1.3.51) \quad 1 + \sum_{n=1}^{\infty} \frac{(1/\alpha; q^2)_n (-\alpha q)^n}{(\beta q^2; q^2)_n} z^n + \sum_{n=1}^{\infty} \frac{(1/\beta; q^2)_n (-\beta q)^n}{(\alpha q^2; q^2)_n} z^{-n} \\ = \frac{(q^2; q^2)_{\infty} (\alpha \beta q^2; q^2)_{\infty} (-qz; q^2)_{\infty} (-q/z; q^2)_{\infty}}{(\alpha q^2; q^2)_{\infty} (\beta q^2; q^2)_{\infty} (-\alpha qz; q^2)_{\infty} (-\beta q/z; q^2)_{\infty}}.$$

Exercise 1.3.15. *Prove that Jacobi's triple product identity, (1.3.11), can be deduced from Ramanujan's ${}_1\psi_1$ summation theorem. Hint: first let $b = 0$ and replace a by c . Now let $z = -b/c$ and $q = ab$, and then let $c \rightarrow 0$.*

Exercise 1.3.16. *Prove that the q -binomial theorem, Theorem 1.3.1, is a special case of Ramanujan's ${}_1\psi_1$ summation theorem.*

We close this chapter with one further major identity, the quintuple product identity, which is enormously useful in the theory of theta functions. We follow our proof of the quintuple product identity with proofs of two beautiful corollaries that are analogues of Jacobi's identity in (1.3.24). We use these two corollaries in Chapter 6 to help derive certain modular equations of degree 3.

The quintuple product identity can be formulated in several ways, and we give three of them below. We provide a proof of the identity in the first setting.

Theorem 1.3.17 (The Quintuple Product Identity; First Version). *For $z \neq 0$,*

$$(1.3.52) \quad \sum_{n=-\infty}^{\infty} q^{3n^2+n} (z^{3n} q^{-3n} - z^{-3n-1} q^{3n+1}) \\ = (q^2; q^2)_{\infty} (qz; q^2)_{\infty} (q/z; q^2)_{\infty} (z^2; q^4)_{\infty} (q^4/z^2; q^4)_{\infty}.$$

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Theorem 1.3.18 (The Quintuple Product Identity; Second Version).
For $a \neq 0$,

$$\begin{aligned}
 & (-aq; q)_{\infty} (-1/a; q)_{\infty} (a^2q; q^2)_{\infty} (q/a^2; q^2)_{\infty} (q; q)_{\infty} \\
 & = a^{-1} (a^3q; q^3)_{\infty} (q^2/a^3; q^3)_{\infty} (q^3; q^3)_{\infty} \\
 (1.3.53) \quad & + (a^3q^2; q^3)_{\infty} (q/a^3; q^3)_{\infty} (q^3; q^3)_{\infty}.
 \end{aligned}$$

Theorem 1.3.19 (The Quintuple Product Identity; Third Version).
We have

$$(1.3.54) \quad \frac{f(-x^2, -\lambda x)f(-\lambda x^3)}{f(-x, -\lambda x^2)} = f(-\lambda^2 x^3, -\lambda x^6) + x f(-\lambda, -\lambda^2 x^9).$$

To deduce (1.3.53) from (1.3.52), replace q^2 by q , set $z = -a\sqrt{q}$, and employ the Jacobi triple product identity (1.3.10). Setting $a = 1/x$ and $a^3 = \lambda/q$ in (1.3.53) and utilizing the Jacobi triple product identity again, we deduce (1.3.54).

Proof. Let $f(z)$ denote the right side of (1.3.52). Then, for $0 < |z| < \infty$, we can express $f(z)$ as a Laurent series

$$(1.3.55) \quad f(z) = \sum_{n=-\infty}^{\infty} a_n z^n.$$

From the definition of f , we find that

$$f(1/z) = -z^{-2}f(z),$$

or, from (1.3.55),

$$\sum_{n=-\infty}^{\infty} a_{-n} z^n = - \sum_{n=-\infty}^{\infty} a_n z^{n-2} = - \sum_{n=-\infty}^{\infty} a_{n+2} z^n.$$

Equating constant terms, we deduce that $a_0 = -a_2$, and equating coefficients of z^{-1} , we find that $a_1 = 0$. From the definition of f , we also find that

$$f(q^2 z) = \frac{f(z)}{qz^3}.$$

Thus, from (1.3.55),

$$\sum_{n=-\infty}^{\infty} a_n z^n = \sum_{n=-\infty}^{\infty} a_n q^{2n+1} z^{n+3}.$$

Equating coefficients of z^n on both sides, we find that, for each integer n ,

$$a_n = q^{2n-5} a_{n-3}.$$

Exercise 1.3.20. By iteration, prove that, for every integer n ,

$$(1.3.56) \quad a_{3n} = q^{6n-5} a_{3n-3} = \cdots = q^{3n^2-2n} a_0,$$

$$(1.3.57) \quad a_{3n+1} = q^{6n-3} a_{3n-2} = \cdots = q^{3n^2} a_1,$$

$$(1.3.58) \quad a_{3n+2} = q^{6n-1} a_{3n-1} = \cdots = q^{3n^2+2n} a_2.$$

By the Jacobi triple product identity, Theorem 1.3.3,

$$(1.3.59) \quad \begin{aligned} (q^4; q^4)_\infty f(z) &= f(-z^2, -q^4/z^2) f(-qz, -q/z) \\ &= \sum_{j,k=-\infty}^{\infty} (-1)^{j+k} q^{2j(j-1)+k^2} z^{2j+k}. \end{aligned}$$

Equating constant coefficients on both sides of (1.3.59), we find that

$$(q^4; q^4)_\infty a_0 = \sum_{j=-\infty}^{\infty} (-1)^j q^{6j^2-2j} = (q^4; q^4)_\infty,$$

by the pentagonal number theorem, Corollary 1.3.5. Hence, $a_0 = 1$, and since $a_2 = -a_0$, we also deduce that $a_2 = -1$. Recalling also that $a_1 = 0$, we conclude from (1.3.56)–(1.3.58) that, for each integer n ,

$$a_{3n} = q^{3n^2-2n}, \quad a_{3n+1} = 0, \quad a_{3n+2} = -q^{3n^2+2n}.$$

Putting these values in (1.3.55), we complete the proof of (1.3.52), after replacing n by $-n-1$ in the second sum arising from (1.3.55). $\square$

Corollary 1.3.21. Recall that $\varphi(q)$ and $f(-q)$ are defined in (1.2.2) and (1.2.4), respectively. Then

$$(1.3.60) \quad \sum_{n=-\infty}^{\infty} (6n+1) q^{3n^2+n} = \varphi^2(-q^2) f(-q^2).$$

Proof. Divide both sides of (1.3.52) by $1 - q/z$. By L'Hôpital's rule,

$$\lim_{z \rightarrow q} \frac{z^{3n} q^{-3n} - z^{-3n-1} q^{3n+1}}{1 - q/z} = 6n + 1.$$

Hence, letting z tend to q in (1.3.52) and employing (1.3.13) and (1.3.15), we find that

$$\sum_{n=-\infty}^{\infty} (6n+1)q^{3n^2+n} = (q^2; q^2)_{\infty}^3 (q^2; q^4)_{\infty}^2 = \varphi^2(-q^2)f(-q^2),$$

and this completes the proof. $\square$

Corollary 1.3.22. *If $\psi(q)$ and $f(-q)$ are defined by (1.2.3) and (1.2.4), respectively, then*

$$(1.3.61) \quad \sum_{n=-\infty}^{\infty} (3n+1)q^{3n^2+2n} = \psi(q^2)f^2(-q).$$

Proof. We again use the quintuple product identity. Divide both sides of (1.3.52) by $1-z^2$ and let z tend to 1. First note that as $z \rightarrow 1$, the left side of (1.3.52) tends to

$$\begin{aligned} \sum_{n=-\infty}^{\infty} q^{3n^2-2n} - \sum_{n=-\infty}^{\infty} q^{3n^2+4n+1} \\ = f(q, q^5) - qf(q^7, q^{-1}) = f(q^5, q) - f(q^5, q) = 0, \end{aligned}$$

by (1.2.9). Thus, we apply L'Hôpital's rule to deduce that

$$\begin{aligned} \lim_{z \rightarrow 1} \frac{\sum_{n=-\infty}^{\infty} q^{3n^2+n} (z^{3n} q^{-3n} - z^{-3n-1} q^{3n+1})}{1-z^2} \\ = -\frac{1}{2} \left(\sum_{n=-\infty}^{\infty} 3nq^{3n^2-2n} + \sum_{n=-\infty}^{\infty} (3n+1)q^{3n^2+4n+1} \right) \\ = \frac{1}{2} \left(\sum_{m=-\infty}^{\infty} 3mq^{3m^2+2m} + \sum_{m=-\infty}^{\infty} (3m+2)q^{3m^2+2m} \right) \\ = \sum_{m=-\infty}^{\infty} (3m+1)q^{3m^2+2m}, \end{aligned}$$

where, in the antepenultimate line, we set $n = -m$ in the former sum and $n = -m - 1$ in the latter sum. Remembering that we have divided both sides of (1.3.52) by $1-z^2$ and have let $z \rightarrow 1$, we find

from the calculation above, (1.3.14), and (1.3.15) that

$$\sum_{m=-\infty}^{\infty} (3m+1)q^{3m^2+2m} = (q^2; q^2)_{\infty} (q; q^2)_{\infty}^2 (q^4; q^4)_{\infty}^2 = \psi(q^2) f^2(-q),$$

which completes the proof. $\square$

Although we have established only a few theorems about q -series and theta functions, we have developed enough machinery to prove some beautiful, significant theorems about the arithmetical functions introduced at the beginning of this chapter.

1.4. Notes

For an introduction to the elementary theory of partitions that is eminently accessible to undergraduates, read *Integer Partitions* [23], by G. E. Andrews and K. Eriksson. The most authoritative account on the theory of partitions is also by Andrews [14], several portions of which are suitable for undergraduates.

Hardy and Ramanujan [110], [192, pp. 276–309] actually proved more than (1.1.6); namely, they derived an infinite asymptotic **series** for $p(n)$ in decreasing exponential functions. They suspected that their **series** did not converge, but it was not until 1937 that D. H. Lehmer [140] showed that it diverges. In 1936, while preparing his lectures for a graduate course in analytic number theory at the University of Pennsylvania, H. Rademacher [181], [182, pp. 108–121] found an exact infinite **series** representation for $p(n)$. At about the same time, A. Selberg [209, pp. 695–706], [52, pp. 203–213] also discovered this same exact formula, but he never published his work.

In their classic paper [110], [192, pp. 276–309], Hardy and Ramanujan introduced their famous “circle method,” which remains today as the primary tool of number theorists using analytic techniques in studying problems in additive number theory. The principal idea behind the “circle method” can be found in Ramanujan’s notebooks [193, pp. 362–363], although he did not rigorously develop his ideas. See Berndt’s book [37, pp. 60–66] for a discussion of Ramanujan’s first attempt at formulating the circle method. Despite its genesis in Ramanujan’s work, today it is often called the Hardy–Littlewood

circle method, because Hardy and J. E. Littlewood extensively developed the method in a [series](#) of papers.

In 1947, Lehmer [141] wrote, "...it is natural to ask whether $\tau(n) = 0$ for any $n > 0$." Although not stated as a conjecture, since that time the nonvanishing of $\tau(n)$ has been known as Lehmer's conjecture.

Theorem 1.3.1, the q -binomial theorem, is due to A. Cauchy [66, p. 45], while Corollary 1.3.2 was first proved by L. Euler [91, Chap. 16]. The Jacobi triple product identity was proved by C. G. J. Jacobi in his *Fundamenta Nova* [131], the paper in which the theory of elliptic functions was founded and one of the most important papers in the history of mathematics. However, the Jacobi triple product identity was first proved by C. F. Gauss [97, p. 464]. The proof that we have given here is independently due to Andrews [12] and P. K. Menon [155]. F. Franklin devised a beautiful combinatorial proof of Corollary 1.3.5 (or Corollary 1.3.8). See Hardy and Wright's book [112, pp. 286–287] or Andrews's text [14, pp. 10–11] for Franklin's proof. Jacobi's identity, Theorem 1.3.9, has an elegant generalization found by S. Bhargava, C. Adiga, and D. D. Somashekara [60].

Ramanujan's ${}_1\psi_1$ summation theorem was first stated by Ramanujan in his notebooks [193, Chap. 16, Entry 17]. In fact, he stated it in precisely the form given in Corollary 1.3.14. It was found by Hardy, who called it, "a remarkable formula with many parameters" and intimated that it could be established by employing the q -binomial theorem [107, pp. 222–223]. However, it was not until 1949 and 1950 that the first published proofs were given by W. Hahn [103] and M. Jackson [130], respectively. The proof that we have given here is due to K. Venkatachaliengar and was presented for the first time in the monograph by Adiga, [Berndt](#), Bhargava, and G. N. Watson [5]; see also Berndt's book [34, pp. 32–34]. As with the other theorems in this chapter, there are now many proofs of the ${}_1\psi_1$ theorem. Almost all proofs employ the q -binomial theorem at some stage. W. P. Johnson [132] has written an interesting semi-expository paper showing how Cauchy could have discovered Ramanujan's ${}_1\psi_1$ summation formula (but did not). Ramanujan left no clues about his proof

or proofs. However, it is quite possible that Ramanujan used the theory of partial fractions, which is a generalization of the method of partial fractions all calculus students learn to evaluate integrals of rational functions. For a proof of Ramanujan's ${}_1\psi_1$ summation theorem employing partial fractions, see a paper by S. H. Chan [73], but this proof also utilizes the q -binomial theorem.

Readers might consider the following instructive exercise.

Exercise 1.4.1. Consider the Laurent expansion of

$$f(z) := \frac{(az)_\infty (b/z)_\infty}{(dz)_\infty (q/(dz))_\infty} =: \sum_{n=-\infty}^{\infty} c_n z^n.$$

Proceeding like we did in the proof of Ramanujan's ${}_1\psi_1$ summation theorem, deduce that

$$(1.4.1) \quad c_n = \frac{(a/d)_n}{(q^2/(bd))_n} \left(\frac{q}{b}\right)^n c_0, \quad -\infty < n < \infty.$$

It follows that

$$(1.4.2) \quad f(z) = c_0 \sum_{n=-\infty}^{\infty} \frac{(a/d)_n}{(q^2/(bd))_n} \left(\frac{qz}{b}\right)^n.$$

Applying Ramanujan's ${}_1\psi_1$ summation theorem to the right side of (1.4.2), we find that

$$(1.4.3) \quad \frac{(az)_\infty (b/z)_\infty}{(dz)_\infty (q/(dz))_\infty} = c_0 \frac{(aqz/(bd))_\infty (bd/(az))_\infty (q)_\infty (q^2/(ab))_\infty}{(qz/b)_\infty (q/(az))_\infty (q^2/(bd))_\infty (qd/a)_\infty}.$$

However, examining the zeros and poles on each side of (1.4.3), we see that (1.4.3) is not true. Thus, (1.4.1) must be wrong. What is wrong with this argument? If you need help, see [26].

The q -binomial theorem is a special case of a product representation of a basic hypergeometric series, where here *basic* is not meant to be a synonym for *fundamental* but instead refers to the base q . Roughly, in general, a basic hypergeometric series ${}_j\phi_k$ has j q -products in the numerator and $k+1$ q -products in the denominator, where one of the q -products in the denominator is $(q; q)_n$, if n is the index of summation. Clear and more extensive introductions to q -series can be found in the books by Andrews [14] and Andrews,

R. Askey, and R. Roy [18, Chap. 10]. By far, the most comprehensive treatise on q -series is by G. Gasper and M. Rahman [96]. An introduction to Ramanujan's extensive contributions to theta functions can be found in Berndt's lecture notes [36]. The interaction of q -series with number theory is best demonstrated in N. J. Fine's beautiful but succinctly written monograph [94]. Some illuminating historical remarks on q -series are given in Roy's review [200].

The quintuple product identity has been discovered many times in the past century, and an account of all proofs known to the author in 1991 can be found in [34, p. 83]. However, a more comprehensive survey discussing more than 25 known proofs has been written by S. Cooper [79]. The formulation given in (1.3.54) is that given by Ramanujan in his lost notebook [194, p. 207]. The proof of the quintuple product identity that we have given is a variant of a proof due to Bhargava [57], which in turn is a combination of proofs of L. J. Mordell [163] and L. Carlitz and M. V. Subbarao [63]. A particularly simple and short proof was discovered Watson [218]. A similar proof was discovered by R. J. Evans [92] and reproduced in the author's lecture notes on theta functions [36, pp. 11–12]. For systematic approaches and several new identities relating sums and products of theta functions, see the comprehensive paper by S. Kongsiriwong and Z.-G. Liu [136].

To the best of our knowledge, both (1.3.60) and (1.3.61) are due to Ramanujan; they can be found in Entry 8(ix), (x) of Chapter 17 in his second notebook [193], [34, pp. 114–115], and also in his paper [186, eq. (65)], [192, p. 147]. The first published proofs are, however, due to B. Gordon [100]. A proof of (1.3.60) is also given in Fine's book [94, p. 83].

Chapter 2

Congruences for $p(n)$ and $\tau(n)$

2.1. Historical Background

This chapter is primarily devoted to proving some of Ramanujan's congruences for $p(n)$ and $\tau(n)$. We begin with a remark on notation. Throughout the chapter, we shall see congruences of the sort

$$(2.1.1) \quad f(q) \equiv g(q) \pmod{m},$$

where $f(q) = \sum a_n q^n$ and $g(q) = \sum b_n q^n$ are power series in q . The congruence (2.1.1) is equivalent to the condition $a_n \equiv b_n \pmod{m}$ for every integer n appearing as an index in either power series.

In 1919, Ramanujan [188], [192, pp. 210–213] announced that he had found three simple congruences satisfied by $p(n)$, namely,

$$(2.1.2) \quad p(5n + 4) \equiv 0 \pmod{5},$$

$$(2.1.3) \quad p(7n + 5) \equiv 0 \pmod{7},$$

$$(2.1.4) \quad p(11n + 6) \equiv 0 \pmod{11}.$$

He gave proofs of (2.1.2) and (2.1.3) in [188] and later in a short one page note [190], [192, p. 230] announced that he had also found a proof of (2.1.4). He also remarks in [190] that “It appears that there are no equally simple properties for any moduli involving primes other than these three.” In a posthumously published paper [191], [192,

pp. 232–238], Hardy extracted different proofs of (2.1.2)–(2.1.4) from an unpublished manuscript of Ramanujan on $p(n)$ and $\tau(n)$ [194, pp. 133–177], [50].

In [188], Ramanujan offered a more general conjecture. Let $\delta = 5^a 7^b 11^c$ and let λ be an integer such that $24\lambda \equiv 1 \pmod{\delta}$. Then

$$(2.1.5) \quad p(n\delta + \lambda) \equiv 0 \pmod{\delta}.$$

In his unpublished manuscript [194, pp. 133–177], [50], Ramanujan gave a proof of (2.1.5) for arbitrary a and $b = c = 0$. He also began a proof of his conjecture for arbitrary b and $a = c = 0$, but he did not complete it. If he had completed his proof, he would have noticed that his conjecture in this case needed to be modified. Ramanujan had formulated his conjectures after studying a table of values of $p(n)$, $0 \leq n \leq 200$, made by P. MacMahon. After Ramanujan died, H. Gupta extended MacMahon's table up to $n = 300$. Upon examining Gupta's table in 1934, S. Chowla [75] found that $p(243)$ is not divisible by 7^3 , despite the fact that $24 \cdot 243 \equiv 1 \pmod{7^3}$. To correct Ramanujan's conjecture, define $\delta' = 5^a 7^{b'} 11^c$, where $b' = b$, if $b = 0, 1, 2$, and $b' = [(b+2)/2]$, if $b > 2$. Then

$$(2.1.6) \quad p(n\delta + \lambda) \equiv 0 \pmod{\delta'}.$$

In 1938, G. N. Watson [218] published a proof of (2.1.6) for $a = c = 0$ and gave a more detailed version of Ramanujan's proof of (2.1.6) in the case $b = c = 0$. It was not until 1967 that A. O. L. Atkin [28] proved (2.1.6) for arbitrary c and $a = b = 0$.

The tau function $\tau(n)$ was introduced by Ramanujan in his famous paper [186], [192, pp. 136–162]. Although he proved little about $\tau(n)$ in this paper, he did formulate some fundamental conjectures about $\tau(n)$. In [190], Ramanujan stated without proof congruences for $\tau(n)$ modulo 5, 7, and 23. In his unpublished manuscript on $p(n)$ and $\tau(n)$ [194, pp. 133–177], [50], he proved these congruences and several further results on $\tau(n)$.

2.2. Elementary Congruences for $\tau(n)$

First, we show that $\tau(n)$ is seldom odd.

Theorem 2.2.1. *The number of values of $n \leq x$ for which $\tau(n)$ is odd equals*

$$(2.2.1) \quad \left[\frac{1 + \sqrt{x}}{2} \right],$$

where $[x]$ denotes the greatest integer less than or equal to x .

Proof. Observe that, for any positive integer j , by the binomial theorem,

$$(1 - q^j)^8 = 1 - 8q^j + 28q^{2j} - \cdots + q^{8j} \equiv 1 + q^{8j} \equiv 1 - q^{8j} \pmod{2}.$$

Hence,

$$(q; q)_\infty^8 \equiv (q^8; q^8)_\infty \pmod{2}.$$

Therefore, using the definition of $\tau(n)$ in (1.1.10), the congruence above, and Jacobi's identity, Theorem 1.3.9, we find that

$$\begin{aligned} \sum_{n=1}^{\infty} \tau(n)q^n &= q(q; q)_\infty^{24} \equiv q(q^8; q^8)_\infty^3 \\ &= \sum_{n=0}^{\infty} (-1)^n (2n+1) q^{(2n+1)^2} \pmod{2}. \end{aligned}$$

Thus, $\tau(n)$ is odd or even according as n is an odd square or not.

Exercise 2.2.2. *As an elementary exercise, show that the number of odd squares less than or equal to x is precisely (2.2.1).*

So the proof of Theorem 2.2.1 is complete. $\square$

Theorem 2.2.3. *For each nonnegative integer n ,*

$$(2.2.2) \quad \tau(7n), \tau(7n+3), \tau(7n+5), \tau(7n+6) \equiv 0 \pmod{7}.$$

Proof. Applying the binomial theorem, as we did in the previous proof, we easily deduce that

$$(q; q)_\infty^7 \equiv (q^7; q^7)_\infty \pmod{7},$$

and so

$$(2.2.3) \quad \sum_{n=1}^{\infty} \tau(n)q^n = q(q; q)_\infty^{24} \equiv q(q; q)_\infty^3 (q^7; q^7)_\infty^3 \pmod{7}.$$

Since the powers in $(q^7; q^7)_\infty^3$ are all multiples of 7, we need only consider

$$(2.2.4) \quad q(q; q)_\infty^3 = \sum_{n=0}^{\infty} (-1)^n (2n+1) q^{1+n(n+1)/2},$$

by Jacobi's identity, Theorem 1.3.9. Observe that $1 + n(n+1)/2 \equiv 0, 1, 2, 4 \pmod{7}$. Moreover, $1 + n(n+1)/2 \equiv 0 \pmod{7}$ if and only if $n \equiv 3 \pmod{7}$ or $2n+1 \equiv 0 \pmod{7}$. It now follows from (2.2.3) and (2.2.4) that $\tau(7n) \equiv 0 \pmod{7}$. It also follows that there are no powers of q on the right side of (2.2.4) that are congruent to either 3, 5, or 6 modulo 7. The latter three congruences in (2.2.2) thus follow from (2.2.3) and (2.2.4). $\square$

For example, $\tau(3) = 252 \equiv 0 \pmod{7}$, $\tau(5) = 4830 \equiv 0 \pmod{7}$, $\tau(6) = -6048 \equiv 0 \pmod{7}$, and $\tau(7) = -16744 \equiv 0 \pmod{7}$. The reader will observe that 3, 5, and 6 are the quadratic nonresidues modulo 7 and that indeed the proof of Theorem 2.2.3 demonstrates this.

Theorem 2.2.4. *Let r , $0 \leq r < 23$, denote any quadratic residue modulo 23. Then, for each positive integer n ,*

$$(2.2.5) \quad \tau(23n - r) \equiv 0 \pmod{23}.$$

Proof. By the binomial theorem,

$$\sum_{n=1}^{\infty} \tau(n) q^n = q(q; q)_\infty^{24} \equiv q(q; q)_\infty (q^{23}; q^{23})_\infty \pmod{23}.$$

Since the powers in $(q^{23}; q^{23})_\infty$ are all multiples of 23, we need only consider

$$(2.2.6) \quad q(q; q)_\infty = \sum_{n=-\infty}^{\infty} (-1)^n q^{1+n(3n+1)/2},$$

by Euler's pentagonal number theorem (1.3.18). Observe that

$$(2.2.7) \quad 1 + \frac{n(3n+1)}{2} = (6n+1)^2 - \frac{23n(3n+1)}{2} \equiv (6n+1)^2 \pmod{23}.$$

Thus, $\tau(m)$ will be a multiple of 23 when m is not congruent to a square modulo 23. In other words, if we set $m =: 23n + \ell = k^2$, $0 \leq \ell < 23$, for some integers n , ℓ , and k , then ℓ must be a quadratic

nonresidue modulo 23. Recall that r is a quadratic residue modulo 23. Since $23 \equiv -1 \pmod{4}$, $-r, r \neq 0$, is a quadratic nonresidue modulo 23. It follows that

$$(2.2.8) \quad \tau(23n - r) \equiv 0 \pmod{23}, \quad 0 < r < 23.$$

□

For example, 3 and 6 are quadratic residues modulo 23, and so $\tau(20) = -7109760 \equiv 0 \pmod{23}$ and $\tau(17) = -6905934 \equiv 0 \pmod{23}$.

We shall later examine congruences for $\tau(n)$ modulo 5. However, to establish these congruences, we first need to prove Ramanujan's famous congruence for $p(n)$ modulo 5.

2.3. Ramanujan's Congruence

$$p(5n + 4) \equiv 0 \pmod{5}$$

In this monograph, we present four proofs of Ramanujan's congruence for $p(n)$ modulo 5. We offer three proofs in this section. The first and the second are more elementary than the third, but the third gives more information.

Theorem 2.3.1. *For each nonnegative integer n ,*

$$(2.3.1) \quad p(5n + 4) \equiv 0 \pmod{5}.$$

First Proof of Theorem 2.3.1. Our first proof is taken from Ramanujan's paper [188], [192, pp. 210–213] and is reproduced in Hardy's book [107, pp. 87–88].

We begin by writing

$$(2.3.2) \quad q(q; q)_{\infty}^4 \frac{(q^5; q^5)_{\infty}}{(q; q)_{\infty}^5} = q \frac{(q^5; q^5)_{\infty}}{(q; q)_{\infty}} = (q^5; q^5)_{\infty} \sum_{m=0}^{\infty} p(m) q^{m+1}.$$

By the binomial theorem,

$$(2.3.3) \quad (q; q)_{\infty}^5 \equiv (q^5; q^5)_{\infty} \pmod{5} \quad \text{or} \quad \frac{(q^5; q^5)_{\infty}}{(q; q)_{\infty}^5} \equiv 1 \pmod{5}.$$

Hence, by (2.3.2) and (2.3.3),

$$(2.3.4) \quad q(q; q)_{\infty}^4 \equiv (q^5; q^5)_{\infty} \sum_{m=0}^{\infty} p(m) q^{m+1} \pmod{5}.$$

We now see from (2.3.4) that in order to show that $p(5n+4) \equiv 0 \pmod{5}$ we must show that the coefficients of q^{5n+5} on the left side of (2.3.4) are multiples of 5.

By the pentagonal number theorem, Corollary 1.3.5, and Jacobi's identity, Theorem 1.3.9,

$$\begin{aligned} q(q; q)_{\infty}^4 &= q(q; q)_{\infty} (q; q)_{\infty}^3 \\ &= q \sum_{j=-\infty}^{\infty} (-1)^j q^{j(3j+1)/2} \sum_{k=0}^{\infty} (-1)^k (2k+1) q^{k(k+1)/2} \\ (2.3.5) \quad &= \sum_{j=-\infty}^{\infty} \sum_{k=0}^{\infty} (-1)^{j+k} (2k+1) q^{1+j(3j+1)/2+k(k+1)/2} \end{aligned}$$

Our objective is to determine when the exponents on the right side are multiples of 5. Observe that

$$2(j+1)^2 + (2k+1)^2 = 8 \left\{ 1 + \frac{1}{2}j(3j+1) + \frac{1}{2}k(k+1) \right\} - 10j^2 - 5.$$

Thus, $1 + \frac{1}{2}j(3j+1) + \frac{1}{2}k(k+1)$ is a multiple of 5 if and only if

$$(2.3.6) \quad 2(j+1)^2 + (2k+1)^2 \equiv 0 \pmod{5}.$$

It is easily checked that $2(j+1)^2 \equiv 0, 2$, or 3 modulo 5 and that $(2k+1)^2 \equiv 0, 1$, or 4 modulo 5. We therefore see that (2.3.6) is true if and only if

$$2(j+1)^2 \equiv 0 \pmod{5} \quad \text{and} \quad (2k+1)^2 \equiv 0 \pmod{5}.$$

In particular, $2k+1 \equiv 0 \pmod{5}$, which, by (2.3.5), implies that the coefficient of q^{5n+5} , $n \geq 0$, in $q(q; q)_{\infty}^4$ is a multiple of 5. The coefficient of q^{5n+5} on the right side of (2.3.4) is therefore also a multiple of 5, i.e., $p(5n+4)$ is a multiple of 5. $\square$

We now give a second simple proof due to G. E. Andrews [15] and based on the simple lemma given below. See also an extensive generalization of this lemma by Andrews and R. Roy [24]. In particular, taking a special case of their general theorem, Andrews and Roy establish the congruence $p(7n+5) \equiv 0 \pmod{7}$.

Lemma 2.3.2. *Let $\{a_n\}$, $n \geq 0$, be any sequence of integers. Then the coefficient of q^{5n+3} , $n \geq 0$, in*

$$(2.3.7) \quad L(q) := \frac{1}{(q; q)_\infty^2} \sum_{n=0}^{\infty} a_n q^{n^2}$$

is divisible by 5.

Proof. Write (2.3.7) in the form

$$L(q) = (q; q)_\infty^3 \frac{\sum_{m=0}^{\infty} a_m q^{m^2}}{(q; q)_\infty^5} \equiv (q; q)_\infty^3 \frac{\sum_{m=0}^{\infty} a_m q^{m^2}}{(q^5; q^5)_\infty} \pmod{5},$$

by the binomial theorem. Using Jacobi's identity, Theorem 1.3.9, we thus see that it suffices to examine the coefficient of q^{5n+3} in

$$(2.3.8) \quad (q; q)_\infty^3 \sum_{m=0}^{\infty} a_m q^{m^2} = \sum_{j=0}^{\infty} (-1)^j (2j+1) q^{j(j+1)/2} \sum_{m=0}^{\infty} a_m q^{m^2}.$$

We want those terms above for which $j(j+1)/2 + m^2 = 5n+3$, where $n \geq 0$. It is easy to see that this condition is equivalent to the congruence

$$(2.3.9) \quad (2j+1)^2 + 3m^2 \equiv 0 \pmod{5}.$$

Since $(2j+1)^2 \equiv 0, \pm 1 \pmod{5}$ and $3m^2 \equiv 0, 2, 3 \pmod{5}$, we see that (2.3.9) holds only when

$$(2.3.10) \quad m \equiv 2j+1 \equiv 0 \pmod{5}.$$

The coefficients of q^{5n+3} in (2.3.8) are then composed of terms of the sort $(-1)^j (2j+1) a_m$, which, by (2.3.10), are all multiples of 5. $\square$

Exercise 2.3.3. *Use (1.3.13) to show that*

$$(2.3.11) \quad \varphi(-q) = \frac{(q; q)_\infty}{(-q; q)_\infty}.$$

Second Proof of Theorem 2.3.1. Using (2.3.11), we find that

$$\begin{aligned} \sum_{k=0}^{\infty} p(k) q^{2k} &= \frac{1}{(q^2; q^2)_\infty} = \frac{1}{(q; q)_\infty (-q; q)_\infty} = \frac{1}{(q; q)_\infty^2} \frac{(q; q)_\infty}{(-q; q)_\infty} \\ &= \frac{1}{(q; q)_\infty^2} \left(1 + 2 \sum_{m=1}^{\infty} (-1)^m q^{m^2} \right) \end{aligned}$$

By Lemma 2.3.2, the coefficients $p(k)$ on the left side above are multiples of 5 whenever $2k \equiv 5j + 3 \pmod{5}$, i.e., whenever $k = 5n + 4$. This then completes our second proof. $\square$

Our third proof is also due to Ramanujan in [188], but it is only briefly indicated in that paper. In his unpublished manuscript on $p(n)$ and $\tau(n)$, [194], [50], Ramanujan gives a more detailed sketch. In this proof, the congruence $p(5n + 4) \equiv 0 \pmod{5}$ follows from a beautiful identity. Following Ramanujan, throughout the proof we let $J_k(q)$, $k = 1, 2$, denote power series with integral powers and integral coefficients, not necessarily the same at each appearance. The precise identities of $J_1(q)$ and $J_2(q)$ are not important for the proof.

Theorem 2.3.4. *We have*

$$(2.3.12) \quad \sum_{n=0}^{\infty} p(5n + 4)q^n = 5 \frac{(q^5; q^5)_{\infty}^5}{(q; q)_{\infty}^6}.$$

Proof. We begin by writing

$$(2.3.13) \quad \frac{(q^{1/5}; q^{1/5})_{\infty}}{(q^5; q^5)_{\infty}} = J_1(q) - q^{1/5} + J_2(q)q^{2/5}.$$

To see this, use the pentagonal number theorem (1.3.18) in the numerator on the left side of (2.3.13) and, if n is the index of summation, divide the terms into residue classes modulo 5. If $n \equiv 0, 2 \pmod{5}$, then the powers are integral, and these, when divided by $(q^5; q^5)_{\infty}$, account for the terms in $J_1(q)$. If $n \equiv 3, 4 \pmod{5}$, then the powers are of the form $k + 2/5$, where k is a positive integer, i.e., we obtain the terms $J_2(q)q^{2/5}$.

Exercise 2.3.5. *Lastly, when $n \equiv 1 \pmod{5}$, use the pentagonal number theorem, Corollary 1.3.5, to show that these terms in $(q^{1/5}; q^{1/5})_{\infty}$ are equal to $-q^{1/5}(q^5; q^5)_{\infty}$.*

Hence, we obtain the term $-q^{1/5}$ in (2.3.13). Now cube both sides of (2.3.13) to obtain

$$(2.3.14) \quad \frac{(q^{1/5}; q^{1/5})_{\infty}^3}{(q^5; q^5)_{\infty}^3} = (J_1^3 - 3J_2^2q) - q^{1/5}(3J_1^2 - J_2^3q) + 3J_1q^{2/5}(1 + J_1J_2) \\ - q^{3/5}(1 + 6J_1J_2) + 3J_2q^{4/5}(1 + J_1J_2),$$

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where for brevity we have deleted the argument q of J_1 and J_2 . Next, we use Jacobi's identity, Theorem 1.3.9, in the numerator on the left side of (2.3.14) and repeat the same kind of argument that we applied with the pentagonal number theorem in (2.3.13). We subdivide the indices n of the sum in the numerator into residue classes modulo 5. As an exercise, readers should use Jacobi's identity (1.3.24) to show that the contributions of the terms with $n \equiv 2 \pmod{5}$ are equal to $5(q^5; q^5)_\infty^3 q^{3/5}$. Thus, we obtain an equality of the sort

$$(2.3.15) \quad \frac{(q^{1/5}; q^{1/5})_\infty^3}{(q^5; q^5)_\infty^3} = G_1(q) + G_2(q)q^{1/5} + 5q^{3/5},$$

where $G_1(q)$ and $G_2(q)$ are power series with integral powers and integral coefficients. Hence, equating coefficients on the right sides of (2.3.14) and (2.3.15), we find that

$$J_1(1 + J_1 J_2) = 0, \quad 1 + 6J_1 J_2 = -5, \quad J_2(1 + J_1 J_2) = 0.$$

From any one of these equalities, we deduce that

$$(2.3.16) \quad J_1 J_2 = -1.$$

Our next task is to use (2.3.16), (2.3.13), and "rationalization" to show that

$$(2.3.17) \quad \begin{aligned} \frac{(q^5; q^5)_\infty}{(q^{1/5}; q^{1/5})_\infty} &= \frac{1}{J_1 - q^{1/5} + J_2 q^{2/5}} \\ &= \frac{(J_1^4 + 3J_2 q) + q^{1/5}(J_1^3 + 2J_2^2 q) + q^{2/5}(2J_1^2 + J_2^3 q)}{J_1^5 - 11q + q^2 J_2^5} \\ &\quad + \frac{q^{3/5}(3J_1 + J_2^4 q) + 5q^{4/5}}{J_1^5 - 11q + q^2 J_2^5}. \end{aligned}$$

We now demonstrate how to prove the second equality in (2.3.17).

Return to (2.3.13) and replace $q^{1/5}$ by $\omega q^{1/5}$, where ω is any fifth root of unity. Thus,

$$(2.3.18) \quad \frac{(\omega q^{1/5}; \omega q^{1/5})_\infty}{(q^5; q^5)_\infty} = J_1(q) - \omega q^{1/5} + J_2(q)\omega^2 q^{2/5}.$$

Let ω run through all five fifth roots of unity and multiply all five such equalities (2.3.18) to obtain

$$(2.3.19) \quad \prod_{\omega} \frac{(\omega q^{1/5}; \omega q^{1/5})_{\infty}}{(q^5; q^5)_{\infty}} = \prod_{\omega} \left\{ J_1(q) - \omega q^{1/5} + J_2(q) \omega^2 q^{2/5} \right\}.$$

First examine the product on the left side of (2.3.19). Using the fact that the sum of the five fifth roots of unity equals 0, we see that if n is not a multiple of 5, we obtain products of the form

$$(2.3.20) \quad (1 - q^{n/5})(1 - \omega q^{n/5})(1 - \omega^2 q^{n/5})(1 - \omega^3 q^{n/5})(1 - \omega^4 q^{n/5}) = 1 - q^n.$$

However, if $n = 5m$, then the corresponding terms are

$$(1 - q^m)(1 - q^m)(1 - q^m)(1 - q^m)(1 - q^m) = (1 - q^m)^5,$$

instead of $1 - q^n$ that we obtained in (2.3.20). Thus, we find that

$$(2.3.21) \quad \prod_{\omega} \frac{(\omega q^{1/5}; \omega q^{1/5})_{\infty}}{(q^5; q^5)_{\infty}} = \frac{(q; q)_{\infty}^6}{(q^5; q^5)_{\infty}^6}.$$

We now examine the product on the right side of (2.3.19). Since there are no fractional powers of q on the left side of (2.3.19) by (2.3.21), there are none on the right side as well. Thus, we only need to examine those terms in the product that give rise to integral powers of q . A brief inspection then convinces us that

$$(2.3.22) \quad \prod_{\omega} \left\{ J_1(q) - \omega q^{1/5} + J_2(q) \omega^2 q^{2/5} \right\} = J_1^5(q) - q + J_2^5(q) q^2 + C_1 q + C_2 q,$$

where C_1 is the sum of the $\binom{5}{1,3,1} = 20$ terms of the form $-J_1 J_2 \omega_1 \omega_2 \omega_3 \omega_4^2 = \omega_1 \omega_2 \omega_3 \omega_4^2$, and C_2 is the sum of the $\binom{5}{2,1,2} = 30$ terms of the form $-J_1^2 J_2^2 \omega_1 \omega_2^2 \omega_3^2 = -\omega_1 \omega_2^2 \omega_3^2$, since $J_1 J_2 = -1$. Here, for each j , ω_j is a fifth root of unity, and in each product $\omega_j \neq \omega_k$ if $j \neq k$. First, examine the terms $\omega_1 \omega_2 \omega_3 \omega_4^2 = \omega_4 \omega_5^{-1}$. Now,

$$\sum_{\omega_4 \neq \omega_5} \omega_4 \omega_5^{-1} = -\omega_5 \omega_5^{-1} = -1,$$

where the sum is over the four fifth roots of unity ω_4 , except ω_5 . Since there are five possibilities for ω_5 , we conclude that $C_1 = -5$. Second,

examine the terms $-\omega_1\omega_2^2\omega_3^2$. Now,

$$-\sum_{\omega_1 \neq \omega_2, \omega_3} \omega_1\omega_2^2\omega_3^2 = (\omega_2 + \omega_3)\omega_2^2\omega_3^2 - \omega_2^3\omega_3^2 + \omega_2^2\omega_3^3,$$

and

$$\sum_{\omega_2 \neq \omega_3} (\omega_2^3\omega_3^2 + \omega_2^2\omega_3^3) = -\omega_3^3\omega_3^2 - \omega_3^2\omega_3^3 = -2.$$

There are five possibilities for ω_3 , and so it would seem that we obtain a contribution of -10 to the value of C_2 . However, because of the symmetry of ω_2 and ω_3 , we have counted each contribution to C_2 twice, i.e., in fact, $C_2 = -5$. Using our values $C_1 = -5 = C_2$ in (2.3.22), we deduce that

$$(2.3.23) \quad \prod_{\omega} \{J_1(q) - \omega q^{1/5} + J_2(q)\omega^2 q^{2/5}\} = J_1^5(q) - 11q + J_2^5(q)q^2.$$

In summary, from (2.3.23), we have shown that

$$\begin{aligned} \frac{1}{J_1(q) - q^{1/5} + J_2(q)q^{2/5}} &= \frac{\prod_{\omega \neq 1} \{J_1(q) - \omega q^{1/5} + J_2(q)\omega^2 q^{2/5}\}}{\prod_{\omega} \{J_1(q) - \omega q^{1/5} + J_2(q)\omega^2 q^{2/5}\}} \\ &= \frac{\prod_{\omega \neq 1} \{J_1(q) - \omega q^{1/5} + J_2(q)\omega^2 q^{2/5}\}}{J_1^5(q) - 11q + J_2^5(q)q^2} \\ (2.3.24) \quad &= \frac{F(q)}{J_1^5(q) - 11q + J_2^5(q)q^2}, \end{aligned}$$

where

$$F(q) = \frac{J_1^5(q) - 11q + J_2^5(q)q^2}{J_1(q) - q^{1/5} + J_2(q)q^{2/5}}.$$

If we consider the numerator and denominator above as polynomials in $q^{1/5}$ and use long division, we find that $F(q)$ indeed is the numerator on the right side of (2.3.17). Thus, by (2.3.24), the proof of (2.3.17) is complete.

Recalling that on the left side of (2.3.17) $1/(q^{1/5}; q^{1/5})_{\infty}$ is the generating function for $p(n)$, we select those terms on both sides where the powers of q are congruent to $4/5$ modulo 1. We then divide both sides by $q^{4/5}$ to find that

$$(2.3.25) \quad (q^5; q^5)_{\infty} \sum_{n=0}^{\infty} p(5n+4)q^n = \frac{5}{J_1^5(q) - 11q + J_2^5(q)q^2}.$$

However, from (2.3.19), (2.3.21), and (2.3.23), we also know that

$$(2.3.26) \quad \frac{(q^5; q^5)_\infty^6}{(q; q)_\infty^6} = \frac{1}{J_1^5(q) - 11q + J_2^5(q)q^2}.$$

Utilizing (2.3.26) in (2.3.25), we complete the proof of (2.3.12). $\square$

Theorem 2.3.4 can be utilized to provide a proof of Ramanujan's congruence for $p(n)$ modulo 25.

Theorem 2.3.6. *For every nonnegative integer n ,*

$$(2.3.27) \quad p(25n + 24) \equiv 0 \pmod{25}.$$

Proof. Applying the binomial theorem on the right side of (2.3.12), we find that

$$(2.3.28) \quad \sum_{n=0}^{\infty} p(5n + 4)q^n \equiv 5 \frac{(q^5; q^5)_\infty^4}{(q; q)_\infty} = 5(q^5; q^5)_\infty^4 \sum_{n=0}^{\infty} p(n)q^n \pmod{25}.$$

From Theorem 2.3.1 we know that the coefficients of $q^4, q^9, q^{14}, \dots, q^{5n+4}, \dots$ on the far right side of (2.3.28) are all multiples of 25. It follows that the coefficients of $q^{5n+4}, n \geq 0$, on the far left side of (2.3.28) are also multiples of 25, i.e.,

$$p(25n + 24) \equiv 0 \pmod{25}.$$

This completes the proof. $\square$

Ramanujan's congruence in Theorem 2.3.1 yields a simple proof of Ramanujan's congruence for the tau function modulo 5, as we next demonstrate.

Theorem 2.3.7. *For each nonnegative integer n ,*

$$(2.3.29) \quad \tau(5n) \equiv 0 \pmod{5}.$$

Proof. By the definition of $\tau(n)$ and the binomial theorem,

$$(2.3.30) \quad \sum_{n=1}^{\infty} \tau(n)q^n = q(q; q)_\infty^{24} = q \frac{(q; q)_\infty^{25}}{(q; q)_\infty} \equiv q(q^5; q^5)_\infty^5 \sum_{n=0}^{\infty} p(n)q^n \pmod{5}.$$

By Theorem 2.3.1, the coefficient of q^{5n} , $n \geq 1$, on the far right side of (2.3.30) is a multiple of 5. Thus, the coefficient of q^{5n} on the far left side of (2.3.30) is a multiple of 5, i.e., $\tau(5n) \equiv 0 \pmod{5}$. $\square$

2.4. Ramanujan's Congruence

$$p(7n + 5) \equiv 0 \pmod{7}$$

We consider next Ramanujan's congruence for $p(n)$ modulo 7.

Theorem 2.4.1. *For each nonnegative integer n ,*

$$(2.4.1) \quad p(7n + 5) \equiv 0 \pmod{7}.$$

Proof. Our proof is again taken from Ramanujan's paper [188] and was sketched by Hardy [107, p. 88].

First, by the binomial theorem,

$$(2.4.2) \quad q^2(q^7; q^7)_\infty \sum_{n=0}^{\infty} p(n)q^n = q^2 \frac{(q^7; q^7)_\infty}{(q; q)_\infty} = q^2(q; q)_\infty^6 \frac{(q^7; q^7)_\infty}{(q; q)_\infty^7} \equiv q^2(q; q)_\infty^6 \pmod{7}.$$

Hence, if we can show that the coefficient of q^{7n+7} , $n \geq 0$, in $q^2(q; q)_\infty^6$ is a multiple of 7, it will follow from (2.4.2) that the coefficient of q^{7n+7} on the far left side is a multiple of 7, i.e., $p(7n + 5) \equiv 0 \pmod{7}$.

Applying Jacobi's identity, Theorem 1.3.9, we find that

$$(2.4.3) \quad q^2(q; q)_\infty^6 = q^2 \{(q; q)_\infty^3\}^2 = \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} (-1)^{j+k} (2j+1)(2k+1) q^{2+j(j+1)/2+k(k+1)/2}.$$

As we saw in the previous paragraph, we want to know when the exponents above are multiples of 7. Now observe that

$$(2j+1)^2 + (2k+1)^2 = 8\{2 + \frac{1}{2}j(j+1) + \frac{1}{2}k(k+1)\} - 14,$$

and so $2 + \frac{1}{2}j(j+1) + \frac{1}{2}k(k+1)$ is a multiple of 7 if and only if

$$(2.4.4) \quad (2j+1)^2 + (2k+1)^2 \equiv 0 \pmod{7}.$$

We easily see that $(2j+1)^2, (2k+1)^2 \equiv 0, 1, 2, 4 \pmod{7}$, and so the only way (2.4.4) can hold is if both $(2j+1)^2, (2k+1)^2 \equiv 0 \pmod{7}$. In such cases, we trivially see that the coefficients on the right side of

(2.4.3) are multiples of 7. Hence, the coefficient of q^{7n+7} , $n \geq 1$, on the left side of (2.4.3) is a multiple of 7. As we demonstrated in the foregoing paragraph, this implies that $p(7n+5) \equiv 0 \pmod{7}$. $\square$

We now consider the analogue of Theorem 2.3.4 for $p(7n+5)$, which was stated without proof by Ramanujan in his paper [188]. In his unpublished manuscript on $p(n)$ and $\tau(n)$, he gives a very brief sketch of its proof [194, pp. 133–177], [50]. There are now several proofs of Theorem 2.4.2, but the details of Ramanujan's proof were worked out only recently by Berndt, A. J. Yee, and J. Yi [54]. Because the details are cumbersome, we provide only the central ideas and refer readers to [54] for a complete proof. Readers should not attempt to complete missing details but merely try to grasp the ideas behind Ramanujan's proof.

Theorem 2.4.2. *We have*

$$(2.4.5) \quad \sum_{n=0}^{\infty} p(7n+5)q^n = 7 \frac{(q^7; q^7)_{\infty}^3}{(q; q)_{\infty}^4} + 49q \frac{(q^7; q^7)_{\infty}^7}{(q; q)_{\infty}^8}.$$

It is clear that Theorem 2.4.1 is an immediate corollary of Theorem 2.4.2.

Proof. Using (1.3.18) in both the numerator and denominator and then separating the indices of summation in the numerator into residue classes modulo 7, we readily find that

$$(2.4.6) \quad \frac{(q^{1/7}; q^{1/7})_{\infty}}{(q^7; q^7)_{\infty}} = J_1 + q^{1/7}J_2 - q^{2/7} + q^{5/7}J_3,$$

where J_1, J_2 , and J_3 are power series in q with integral coefficients, and where the pentagonal number theorem was used to calculate the coefficient of $q^{2/7}$. Cubing both sides of (2.4.6), we find that

$$\begin{aligned} & \frac{(q^{1/7}; q^{1/7})_{\infty}^3}{(q^7; q^7)_{\infty}^3} \\ &= (J_1^3 + 3J_2^2J_3q - 6J_1J_3q) + q^{1/7}(3J_1^2J_2 - 6J_2J_3q + J_3^2q^2) \\ & \quad + 3q^{2/7}(J_1J_2^2 - J_1^2 + J_3q) + q^{3/7}(J_2^3 - 6J_1J_2 + 3J_1J_3^2q) \\ & \quad + 3q^{4/7}(J_1 - J_2^2 + J_2J_3^2q) + 3q^{5/7}(J_2 + J_1^2J_3 - J_3^2q) \\ (2.4.7) \quad & + q^{6/7}(6J_1J_2J_3 - 1). \end{aligned}$$

On the other hand, using Jacobi's identity, Theorem 1.3.9, and separating the indices of summation in the numerator on the left side of (2.4.7) into residue classes modulo 7, we easily find that

$$(2.4.8) \quad \frac{(q^{1/7}; q^{1/7})_{\infty}^3}{(q^7; q^7)_{\infty}^3} = G_1 + q^{1/7}G_2 + q^{3/7}G_3 - 7q^{6/7},$$

where G_1, G_2 , and G_3 are power series in q with integral coefficients, and where Jacobi's identity, Theorem 1.3.9, was used to determine the coefficient of $q^{6/7}$. Comparing coefficients in (2.4.7) and (2.4.8), we conclude that

$$(2.4.9) \quad \begin{cases} J_1 J_2^2 - J_1^2 + J_3 q &= 0, \\ J_1 - J_2^2 + J_2 J_3^2 q &= 0, \\ J_2 + J_1^2 J_3 - J_3^2 q &= 0, \\ 6J_1 J_2 J_3 - 1 &= -7. \end{cases}$$

Replace $q^{1/7}$ by $\omega q^{1/7}$ in (2.4.6), where ω is any seventh root of unity. Therefore,

$$(2.4.10) \quad \frac{(\omega q^{1/7}; \omega q^{1/7})_{\infty}}{(q^7; q^7)_{\infty}} = J_1 + \omega q^{1/7} J_2 - \omega^2 q^{2/7} + \omega^5 q^{5/7} J_3.$$

Taking the products of both sides of (2.4.10) over all seven seventh roots of unity, we find that

$$(2.4.11) \quad \frac{(q; q)_{\infty}^8}{(q^7; q^7)_{\infty}^8} = \prod_{\omega} (J_1 + \omega q^{1/7} J_2 - \omega q^{2/7} + \omega q^{5/7} J_3).$$

Using the generating function for $p(n)$, (2.4.6), and (2.4.11), we find that

$$(2.4.12) \quad \begin{aligned} \sum_{n=0}^{\infty} p(n)q^n &= \frac{1}{(q; q)_{\infty}} = \frac{(q^{49}; q^{49})_{\infty}^7}{(q^7; q^7)_{\infty}^8} \frac{(q^7; q^7)_{\infty}^8}{(q^{49}; q^{49})_{\infty}^8} \frac{(q^{49}; q^{49})_{\infty}}{(q; q)_{\infty}} \\ &= \frac{(q^{49}; q^{49})_{\infty}^7}{(q^7; q^7)_{\infty}^8} \frac{\prod_{\omega} (J_1 + \omega q J_2 - \omega q^2 + \omega q^5 J_3)}{J_1 + q J_2 - q^2 + q^5 J_3} \\ &= \frac{(q^{49}; q^{49})_{\infty}^7}{(q^7; q^7)_{\infty}^8} \left\{ \prod_{\omega \neq 1} (J_1 + \omega q J_2 - \omega q^2 + \omega q^5 J_3) \right\}. \end{aligned}$$

We only need to compute the terms in $\prod_{\omega \neq 1} (J_1 + \omega q J_2 - \omega q^2 + \omega q^5 J_3)$ where the powers of q are of the form $7n+5$ to complete the proof. In

order to do this, we need to prove several identities using the identities of (2.4.9). We do not give the details.

Choosing only those terms on each side of (2.4.12) where the powers of q are of the form $7n + 5$ and using the omitted calculations, we find that

$$\sum_{\substack{n=0 \\ n \equiv 5 \pmod{7}}}^{\infty} p(n)q^n = q^5 \frac{(q^{49}; q^{49})_{\infty}^7}{(q^7; q^7)_{\infty}^8} \left(7 \frac{(q^7; q^7)_{\infty}^4}{(q^{49}; q^{49})_{\infty}^4} + 49q^7 \right),$$

or

$$(2.4.13) \quad \sum_{n=0}^{\infty} p(7n+5)q^{7n} = 7 \frac{(q^{49}; q^{49})_{\infty}^3}{(q^7; q^7)_{\infty}^4} + 49q^7 \frac{(q^{49}; q^{49})_{\infty}^7}{(q^7; q^7)_{\infty}^8}.$$

Replacing q^7 by q in (2.4.13), we complete the proof of (2.4.5). $\square$

Recall that the identity of Theorem 2.3.4 yielded in Theorem 2.3.6 a congruence for $p(n)$ modulo 5^2 . Similarly, the identity in Theorem 2.4.2 yields a congruence for $p(n)$ modulo 7^2 , as we now demonstrate.

Theorem 2.4.3. *For each nonnegative integer n ,*

$$(2.4.14) \quad p(49n+47) \equiv 0 \pmod{49}.$$

Proof. Write (2.4.5) in the form

$$(2.4.15) \quad \begin{aligned} \sum_{n=0}^{\infty} p(7n+5)q^n &= 7 \frac{(q^7; q^7)_{\infty}^3 (q; q)_{\infty}^3}{(q; q)_{\infty}^7} + 49q \frac{(q^7; q^7)_{\infty}^7}{(q; q)_{\infty}^8} \\ &\equiv 7(q^7; q^7)_{\infty}^2 \sum_{m=0}^{\infty} (-1)^m (2m+1) q^{m(m+1)/2} \pmod{49}, \end{aligned}$$

by the binomial theorem and Jacobi's identity (1.3.24). We now examine the terms on the right side of (2.4.15) where the powers of q are of the form $7n + 6$. Separating the summands into residue classes modulo 7, we see that the only terms yielding such exponents are when $m \equiv 3 \pmod{7}$. But then $2m + 1 \equiv 0 \pmod{7}$. Thus, the coefficient of the power q^{7n+6} , $n \geq 1$, on the right side of (2.4.15) is a multiple of 49. The same must be true, of course, on the left side of (2.4.15), i.e., the coefficient $p(49n+47)$ must be a multiple of 49, i.e., (2.4.14) has been established. $\square$

We shall return to congruences for $p(n)$ after we introduce Eisenstein series in Chapter 4.

2.5. The Parity of $p(n)$

In contrast to Theorem 2.2.1 which provides a criterion for determining when $\tau(n)$ is either even or odd, we know much less about the parity of $p(n)$. It has long been conjectured that $p(n)$ is even approximately half of the time, or, more precisely,

$$(2.5.1) \quad \#\{n \leq N : p(n) \text{ is even}\} \sim \frac{1}{2}N,$$

as $N \rightarrow \infty$. T. R. Parkin and D. Shanks [178] undertook the first extensive computations, providing strong evidence that indeed (2.5.1) is most likely true. Despite the venerability of the problem, it was not even known that $p(n)$ assumes either even or odd values infinitely often until 1959, when O. Kolberg [135] established these facts. Other proofs of Kolberg's theorem were later found by J. Fabrykowski and M. V. Subbarao [93] and by M. Newman [168]. In 1983, L. Mirsky [159] established the first quantitative result by showing that

$$(2.5.2) \quad \#\{n \leq N : p(n) \text{ is even (odd)}\} > \frac{\log \log N}{2 \log 2}.$$

An improvement was made by J.-L. Nicolas and A. Sárközy [171], who proved that

$$(2.5.3) \quad \#\{n \leq N : p(n) \text{ is even (odd)}\} > (\log N)^c,$$

for some positive constant c .

In the most recent investigations, the methods for finding lower bounds for the number of occurrences of even values of $p(n)$ have been somewhat different from those for odd values of $p(n)$. Greatly improving on previous results, Nicolas, I. Z. Ruzsa, and Sárközy [170] in 1998 proved that

$$(2.5.4) \quad \#\{n \leq N : p(n) \text{ is even}\} \gg \sqrt{N}$$

and, for each $\epsilon > 0$,

$$(2.5.5) \quad \#\{n \leq N : p(n) \text{ is odd}\} \gg \sqrt{N} e^{-(\log 2 + \epsilon) \frac{\log N}{\log \log N}}.$$

(We pause to explain the notation $\gg$. We write $F(N) \gg G(N)$, if and only if there exists a positive constant c such that $F(N) \geq cG(N)$,

for all N sufficiently large.) In an appendix to their paper [170], J.-P. Serre used modular forms to prove that

$$(2.5.6) \quad \lim_{N \rightarrow \infty} \frac{\#\{n \leq N : p(n) \text{ is even}\}}{\sqrt{N}} = \infty.$$

At present, this is the best known result for even values of $p(n)$. The lower bound (2.5.5) has been improved first by S. Ahlgren [6], who utilized modular forms, and second by Nicolas [169], who used more elementary methods, to prove that

$$(2.5.7) \quad \#\{n \leq N : p(n) \text{ is odd}\} \gg \frac{\sqrt{N}(\log \log N)^K}{\log N},$$

for some positive number K . Ahlgren proved (2.5.7) with $K = 0$. An elegant, elementary proof of (2.5.7) when $K = 0$ was established by D. Eichhorn [86]. The lower bound (2.5.7) is currently the best known result for odd values of $p(n)$.

Our goal in this section is to prove the results (2.5.5) and (2.5.4) of Nicolas, Ruzsa, and Sárközy [170] by relatively simple means. Our proof is a special instance of an argument devised by Berndt, Yee, and A. Zaharescu [55], who proved considerably more general theorems that are applicable to a wide variety of partition functions. Except for one step, when we must appeal to a theorem of S. Wigert [223] and Ramanujan [185], our proof is elementary and self-contained.

Theorem 2.5.1. *For each fixed c with $c > 2 \log 2$ and N sufficiently large,*

$$(2.5.8) \quad \#\{n \leq N : p(n) \text{ is odd}\} \geq N^{\frac{1}{2} - \frac{c}{\log \log N}}.$$

Theorem 2.5.2. *For each fixed constant c with $c < 1/\sqrt{6}$, and for N sufficiently large,*

$$(2.5.9) \quad \#\{n \leq N : p(n) \text{ is even}\} \geq c\sqrt{N}.$$

Before we begin our proofs of Theorems 2.5.1 and 2.5.2, we need to establish some terminology. Although we shall use language from modern algebra, readers need not know any theorems from the subject. In fact, some of the information conveyed in the next two paragraphs will not be used in the sequel, but we think these facts are interesting in themselves.

Let $A := \mathbb{F}_2[[X]]$ be the ring of formal power series in one variable X over the field with two elements $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$, i.e.,

$$(2.5.10) \quad A = \left\{ f(X) = \sum_{n=0}^{\infty} a_n X^n : a_n \in \mathbb{F}_2, \quad 0 \leq n < \infty \right\}.$$

The ring A is an integral domain; note that an element $f(X) = \sum_{n=0}^{\infty} a_n X^n \in A$ is invertible if and only if $a_0 = 1$. Since 0 and 1 are the only elements of $\mathbb{F}_2$, we may write any element $f(X) \in A$ in the form

$$(2.5.11) \quad f(X) = X^{n_1} + X^{n_2} + \dots,$$

where the sum may be finite or infinite and $0 \leq n_1 < n_2 < \dots$. For any $f(X) \in A$, observe that

$$(2.5.12) \quad f^2(X) = f(X^2).$$

In other words, if $f(X)$ is given by (2.5.11), then

$$(2.5.13) \quad f^2(X) = X^{2n_1} + X^{2n_2} + \dots.$$

On A there exists a natural derivation which sends $f(X) \in A$ to $f'(X) = \frac{df}{dX} \in A$, i.e., if

$$(2.5.14) \quad f(X) = \sum_{n=0}^{\infty} a_n X^n, \quad \text{then} \quad f'(X) = \sum_{n=1}^{\infty} n a_n X^{n-1}.$$

Note that for any $f(X) \in A$,

$$(2.5.15) \quad f''(X) = 0.$$

We also remark that for any $f(X)$ given in the form (2.5.11), the condition

$$(2.5.16) \quad f'(X) = 0$$

is equivalent to the condition that all the exponents n_j are even numbers.

In our proof of Theorem 2.5.2, we need to know the shape (2.5.11) of the series $f(X)/(1-X)$. For any integers $0 \leq a < b$, we see that in A

$$(2.5.17) \quad \frac{X^a + X^b}{1-X} = \frac{X^a(1-X^{b-a})}{1-X} = X^a + X^{a+1} + \dots + X^{b-1}.$$

We put together pairs of consecutive terms $X^{n_{2k+1}} + X^{n_{2k+2}}$ to obtain the equality

$$\begin{aligned}
 (2.5.18) \quad \frac{f(X)}{1-X} &= \frac{X^{n_1} + X^{n_2}}{1-X} + \frac{X^{n_3} + X^{n_4}}{1-X} + \cdots + \frac{X^{n_{2k+1}} + X^{n_{2k+2}}}{1-X} + \cdots \\
 &= (X^{n_1} + X^{n_1+1} + \cdots + X^{n_2-1}) + (X^{n_3} + \cdots + X^{n_4-1}) + \cdots \\
 &\quad + (X^{n_{2k+1}} + \cdots + X^{n_{2k+2}-1}) + \cdots.
 \end{aligned}$$

If the sum on the right side of (2.5.11) defining $f(X)$ is finite, say $f(X) = X^{n_1} + X^{n_2} + \cdots + X^{n_s}$, then

$$\begin{aligned}
 (2.5.19) \quad \frac{f(X)}{1-X} &= (X^{n_1} + X^{n_1+1} + \cdots + X^{n_2-1}) + \cdots \\
 &\quad + (X^{n_{s-1}} + X^{n_{s-1}+1} + \cdots + X^{n_s-1}),
 \end{aligned}$$

if s is even, and

$$\begin{aligned}
 (2.5.20) \quad \frac{f(X)}{1-X} &= (X^{n_1} + \cdots + X^{n_2-1}) + \cdots \\
 &\quad + (X^{n_{s-2}} + \cdots + X^{n_{s-1}-1}) + \sum_{n=n_s}^{\infty} X^n,
 \end{aligned}$$

if s is odd.

Before commencing our proofs of Theorems 2.5.1 and 2.5.2, we introduce some standard notation in analytic number theory. We say that $f(N) = O(g(N))$, as N tends to ∞ , if there exists a positive constant $A > 0$ and a number $N_0 > 0$ such that $|f(N)| \leq A|g(N)|$ for all $N \geq N_0$. To emphasize that this positive constant A above may depend upon another parameter c , we write $f(N) = O_c(g(N))$, as N tends to ∞ .

Proof of Theorem 2.5.1. We begin with the pentagonal number theorem

$$(2.5.21) \quad \sum_{n=0}^{\infty} (-1)^n q^{n(3n-1)/2} + \sum_{n=1}^{\infty} (-1)^n q^{n(3n+1)/2} = (q; q)_{\infty}.$$

By reducing the coefficients modulo 2 and replacing q by X in (2.5.21), we find that, if $1/F(X)$ is the image of the infinite series of (2.5.21)

in A , then

$$(2.5.22) \quad 1 = F(X) \left(1 + \sum_{n=1}^{\infty} \left(X^{n(3n-1)/2} + X^{n(3n+1)/2} \right) \right).$$

We write $F(X)$ in the form

$$(2.5.23) \quad F(X) = 1 + X^{n_1} + X^{n_2} + \cdots + X^{n_j} + \cdots,$$

where, of course, $n_1, n_2, \dots$ are positive integers. Clearly, from the generating function of the partition function $p(n)$ and (2.5.23),

$$(2.5.24) \quad \#\{1 \leq n \leq N : p(n) \text{ is odd}\} = \#\{n_j \leq N\}$$

and

$$(2.5.25) \quad \#\{1 \leq n \leq N : p(n) \text{ is even}\} = N - \#\{n_j \leq N\}.$$

We first establish a lower bound for $\#\{n_j \leq N\}$. Using (2.5.23), write (2.5.22) in the form

$$(2.5.26) \quad \left(\sum_{j=1}^{\infty} X^{n_j} \right) \left(1 + \sum_{n=1}^{\infty} \left(X^{n(3n-1)/2} + X^{n(3n+1)/2} \right) \right) = \sum_{m=1}^{\infty} \left(X^{m(3m-1)/2} + X^{m(3m+1)/2} \right).$$

Asymptotically, there are $\sqrt{2N/3}$ terms of the form $X^{m(3m-1)/2}$ less than X^N on the right side of (2.5.26). For a fixed positive integer n_j , we determine how many of these terms appear in a series of the form

$$(2.5.27) \quad X^{n_j} \left(1 + \sum_{n=1}^{\infty} \left(X^{n(3n-1)/2} + X^{n(3n+1)/2} \right) \right),$$

arising from the left side of (2.5.26). Thus, for fixed $n_j < N$, we estimate the number of integral pairs (m, n) of solutions of the equation

$$(2.5.28) \quad n_j + \frac{1}{2}n(3n-1) = \frac{1}{2}m(3m-1),$$

which we put in the form

$$(2.5.29) \quad 2n_j = (m-n)(3m+3n-1).$$

By a result of Wigert [223] and Ramanujan [185], [192, p. 80], the number of divisors of $2n_j$ is no more than $O_c \left(N^{\frac{c}{\log \log N}} \right)$ for any fixed $c > \log 2$. Thus, each of the numbers $m-n$ and $3m+3n-1$ can assume

at most $O_c \left(N^{\frac{c}{\log \log N}} \right)$ values. Since the pair $(m-n, 3m+3n-1)$ uniquely determines the pair (m, n) , it follows that the number of solutions to (2.5.29) is $O_c \left(N^{\frac{c}{\log \log N}} \right)$, where c is any constant such that $c > 2 \log 2$. A similar argument can be made for the terms in (2.5.26) of the form $X^{m(3m+1)/2}$.

Returning to (2.5.26) and (2.5.27), we see that each series of the form (2.5.27) has at most $O_c \left(N^{\frac{c}{\log \log N}} \right)$ terms $X^{m(3m-1)/2}$ up to X^N that appear on the right side of (2.5.26). It follows that there are at least $O_c \left(N^{\frac{1}{2} - \frac{c}{\log \log N}} \right)$ numbers $n_j \leq N$ that are needed to match all the (asymptotically $\sqrt{2N/3}$) terms $X^{m(3m-1)/2}$ up to X^N on the right side of (2.5.26). Again, an analogous argument holds for terms of the form $X^{m(3m+1)/2}$. We have therefore completed the proof of Theorem 2.5.1. $\square$

Proof of Theorem 2.5.2. Next, we provide a lower bound for $\#\{n \leq N : p(n) \text{ is even}\}$. Let $\{m_1, m_2, \dots\}$ be the complement of the set $\{0, n_1, n_2, \dots\}$ in the set of natural numbers $\{0, 1, 2, \dots\}$, and define

$$(2.5.30) \quad G(X) := X^{m_1} + X^{m_2} + \dots \in A.$$

Then

$$(2.5.31) \quad G(X) + F(X) = 1 + X + X^2 + \dots + X^k + \dots = \frac{1}{1-X}.$$

Since, by (2.5.25),

$$(2.5.32) \quad \#\{m_j \leq N\} = N - \#\{n_j \leq N\} = \#\{n \leq N : p(n) \text{ is even}\},$$

we need a lower bound for $\#\{m_j \leq N\}$. Using (2.5.31) in (2.5.22), we find that

$$\begin{aligned} & 1 + G(X) \left(1 + \sum_{n=1}^{\infty} \left(X^{n(3n-1)/2} + X^{n(3n+1)/2} \right) \right) \\ &= \frac{1}{1-X} \left(1 + \sum_{n=1}^{\infty} \left(X^{n(3n-1)/2} + X^{n(3n+1)/2} \right) \right) \\ &= \frac{1}{1-X} (1 + X + X^2 + X^5 + X^7 + \dots) \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{1-X} \left((1+X) + (X^2+X^5) + \dots \right. \\
&\quad + (X^{(n-1)(3(n-1)+1)/2} + X^{n(3n-1)/2}) \\
(2.5.33) \quad &\quad \left. + (X^{n(3n+1)/2} + X^{(n+1)(3(n+1)-1)/2}) + \dots \right).
\end{aligned}$$

By (2.5.18), we see that the right side of (2.5.33) equals

$$\begin{aligned}
(2.5.34) \quad &1 + (X^2 + X^3 + X^4) + \dots + (X^{(n-1)(3(n-1)+1)/2} + \dots + X^{n(3n-1)/2-1}) \\
&+ (X^{n(3n+1)/2} + \dots + X^{(n+1)(3(n+1)-1)/2-1}) + \dots.
\end{aligned}$$

Observe that the gap between $X^{n(3n-1)/2-1}$ and $X^{n(3n+1)/2}$ contains n terms that are missing from the series (2.5.34). This gap comes after a segment of

$$\frac{1}{2}n(3n-1) - 1 - \frac{1}{2}(n-1)(3(n-1)+1) + 1 = 2n-1$$

terms that do appear in (2.5.34). So we see that (2.5.34) contains asymptotically

$$\frac{2n-1}{n+(2n-1)}N = \frac{2n-1}{3n-1}N \sim \frac{2}{3}N$$

terms up to X^N . Now the sum in parentheses on the left side of (2.5.33) has asymptotically $2\sqrt{2N/3}$ nonzero terms up to X^N . Thus $G(X)$ must have at least $\sqrt{N/6}$ nonzero terms up to X^N in order for the left side of (2.5.33) to have at least $2N/3$ terms up to X^N to match those on the right side of (2.5.33). We have therefore completed the proof of Theorem 2.5.2. $\square$

2.6. Notes

Theorem 2.2.1 has been slightly refined by M. R. Murty, V. K. Murty, and T. N. Shorey [165], using a more sophisticated argument. They also obtain lower bounds for the values of $\tau(n)$ when $\tau(n)$ is odd.

Another proof of Theorem 2.3.1, rivalling Ramanujan's first proof in simplicity, has been given by J. Drost [84]. See M. D. Hirschhorn's paper [120] for still another elementary proof. Many references to further proofs of both Theorem 2.3.1 and Theorem 2.3.4 can be found

in the latest edition of Ramanujan's *Collected Papers* [192, pp. 372–375]. These pages also contain references to other proofs of Theorems 2.4.1 and 2.4.2. Ramanujan himself summarized the congruences he proved and the methods he utilized to prove them in a letter to Hardy written from the nursing home, Fitzroy House, in the summer of 1918. In particular, he wrote [51, pp. 192–193], “Thus the divisibility by $5^a 7^b 11^c$ when $a = 0, 1, 2, 3$; $b = 0, 1, 2, 3$; $c = 0, 1, 2$ amounting to $4 \times 4 \times 3 - 1$ or 47 cases of the conjectured theorem are proved.” This statement is interesting for several reasons. First, Ramanujan had evidently proved special cases of his general conjecture without leaving us proofs in these cases. Second, he claimed a proof for the modulus 7^3 , but we had noted in the introduction to this chapter that Ramanujan's conjecture was false in this case. Third, Ramanujan's proof of his conjecture for arbitrary powers of 5 was obviously established after this letter was written.

We have not given a proof of Ramanujan's congruence $p(11n+6) \equiv 0 \pmod{11}$. The most elementary proof is due to L. Winquist [227] and uses *Winquist's Identity*. Further proofs of Winquist's identity have been found by Hirschhorn [117] and S.-Y. Kang [133]. Another elementary approach to proving that $p(11n+6) \equiv 0 \pmod{11}$ has been devised by Berndt, S. H. Chan, Z.-G. Liu, and H. Yesilyurt [48], who established a new identity for $(q; q)_{\infty}^{10}$. Hirschhorn [118] has devised a common approach to proving all three congruences (2.1.2)–(2.1.4).

Hirschhorn and D. C. Hunt [126] gave an alternative proof along classical lines to that of Ramanujan and Watson for Ramanujan's congruence modulo 5^a . Those readers intrigued by our sketch of Ramanujan's Theorem 2.4.2 might consult F. G. Garvan's [95] proof of the more general congruence modulo 7^b .

Ramanujan appeared to conjecture that the only congruences of the form $p(\ell n + B) \equiv 0 \pmod{\ell}$, where ℓ is a prime, are those when $\ell = 5, 7$, or 11 . This was not proved until 2003, when Ahlgren and M. Boylan [8] proved that indeed these are the only three such congruences.

Suppose, however, that we drop the restriction that the moduli of the arithmetic progressions are the same as the moduli of the congruences. That is, are there congruences when the moduli are not the same or are not primes? The first theorems establishing lots of congruences for $p(n)$ were found by Atkin [29]. Then K. Ono [175] proved that, given any prime $\ell \geq 5$, there exist infinitely many congruences of the type $p(An + B) \equiv 0 \pmod{\ell}$. This was extended by Ahlgren [7] who established a similar result for any prime power ℓ^k . A consequence of their work is that if $\ell \geq 5$ is prime, then for a positive proportion of positive integers n , $p(n) \equiv 0 \pmod{\ell}$. Nonetheless, finding concrete examples illustrating their theorems is not easy. Atkin and J. N. O'Brien [30] had earlier found a couple of such congruences; one is

$$p(17303n + 237) \equiv 0 \pmod{13}.$$

R. Weaver [219] devised an algorithm based on Ono's work and found over 76,000 explicit examples, all with $\ell \leq 31$.

Define

$$\delta_\ell := \frac{\ell^2 - 1}{24}.$$

Except for a couple of sporadic examples, it turns out that for all of the congruences $p(An + B) \equiv 0 \pmod{\ell}$ found by Ramanujan, Ono, Ahlgren, and others, $B \equiv -\delta_\ell \pmod{\ell}$. In another breakthrough, Ahlgren and Ono [10], proved that this residue class is only one of $(\ell+1)/2$ residue classes where $p(n)$ possesses many such congruences. An informative historical description of the quest for congruences for the partition function has been given by Ahlgren and Ono [9].

Subbarao [211] first conjectured that in every arithmetic progression $n \equiv r \pmod{t}$ there are infinitely many values of n such that $p(n)$ is even and that there are infinitely many values of n for which $p(n)$ is odd. The most extensive results pointing toward the truth of this conjecture have been found by Ono [173], [174] and Ahlgren [6], with a summary of previous results provided in these papers. The best lower bounds for the number of even and odd values of $p(n)$ in arithmetic progressions are (2.5.6) [170] and (2.5.7) [6], respectively, while the most general theorems of this sort are found in [55] and [56].

In his unpublished manuscript on $p(n)$ and $\tau(n)$, Ramanujan asserted and, in some cases, proved further congruences for $\tau(n)$, most involving divisor functions. References to most of the many papers written on this subject can be found in Berndt and Ono's account of Ramanujan's manuscript [50]. We now have a complete understanding of congruences for $\tau(n)$ through the theory of ℓ -adic representations, and an account of this with many references can be found in H. P. F. Swinnerton-Dyer's survey paper [213]. There are, in essence, only six congruences for $\tau(n)$, with the largest modulus being 691.

Z.-H. Sun and K. S. Williams [212] have established a refinement of Theorem 2.2.4. To describe their theorem, we first need to define the order of an integer. Let p denote a prime. Then the unique nonnegative integer α such that $p^\alpha | n$ but $p^{\alpha+1} \nmid n$ is called the order of n modulo p and is denoted by $\text{ord}_p n$.

Theorem 2.6.1. *For any positive integer n ,*

$$\tau(n) \equiv \begin{cases} 0 \pmod{23}, & \text{if there exists a prime } p \text{ such that} \\ & \left(\frac{p}{23}\right) = -1 \text{ and } 2 \nmid \text{ord}_p n, \\ 0 \pmod{23}, & \text{if there exists a prime } p \text{ such that} \\ & p = 2x^2 + xy + 3y^2 \text{ and } \text{ord}_p n \equiv 2 \pmod{3}, \\ (-1)^\mu \prod_{p=2x^2+xy+3y^2 \neq 23} (1 + \text{ord}_p n) \pmod{23}, & \text{otherwise,} \end{cases}$$

where

$$\mu := \sum_{\substack{p=2x^2+xy+3y^2 \\ \text{ord}_p n \equiv 1 \pmod{3}}} 1,$$

where p runs over all primes represented by $2x^2 + xy + 3y^2$.

As mentioned in the Historical Background beginning this chapter, Ramanujan introduced his arithmetical function $\tau(n)$ in [186], [192, pp. 136–16], which was to become one of the most important papers in the history of number theory. In this paper, Ramanujan made three important conjectures about $\tau(n)$.

(a) $\tau(n)$ is multiplicative, i.e., $\tau(m)\tau(n) = \tau(mn)$, whenever $(m, n) = 1$.

(b) If p is any prime and n is any integer exceeding 1, then

$$\tau(p^{n+1}) = \tau(p)\tau(p^n) - p^{11}\tau(p^{n-1}).$$

(c) For each prime p ,

$$|\tau(p)| \leq 2p^{11/2}.$$

The first two conjectures were first proved by L. J. Mordell [161] in 1917 and then greatly generalized by E. Hecke [113], beginning one of the most important chapters in the theory of modular forms. Conjecture (c) was also considerably generalized and was one of the most famous unproved conjectures in number theory until it was proved by P. Deligne [81] in 1974.

Readers wanting to learn more about $\tau(n)$ might begin by reading two expository papers, one by R. A. Rankin [197], and the other by M. R. Murty [164]. See also a paper by V. K. Murty [166].

Chapter 3

Sums of Squares and Sums of Triangular Numbers

3.1. Lambert Series

In this chapter, we shall derive formulas for $r_{2k}(n)$, for $k = 1, 2, 3, 4$, and formulas for $t_{2k}(n)$, for $k = 1, 2$. Observe that $r_{2k}(n)$ and $t_{2k}(n)$ are generated by

$$\varphi^{2k}(q) = \sum_{n=0}^{\infty} r_{2k}(n)q^n \quad \text{and} \quad \psi^{2k}(q) = \sum_{n=0}^{\infty} t_{2k}(n)q^n,$$

respectively, where $\varphi(q)$ and $\psi(q)$ are defined in (1.2.2) and (1.2.3), respectively, and where, by convention, we define $r_{2k}(0) = 1 = t_{2k}(0)$. Appearing in our proofs are Lambert series. Strictly speaking, a Lambert series is a series of the type

$$(3.1.1) \quad \sum_{n=1}^{\infty} a_n \frac{q^n}{1 - q^n}.$$

A generalized Lambert series allows more general exponents in both the numerators and denominators of (3.1.1), with the minus sign in the denominator also possibly being replaced by a plus sign. Nonetheless, we use the appellation, Lambert series, for *all* such series.

We also prove three further theorems on the representations of numbers by certain quadratic forms. For our purpose, a quadratic form is a function of k variables, $n_1, n_2, \dots, n_k$, and has the shape $\sum a_{i,j} n_i n_j$, where $a_{i,j}$ are nonnegative integers and $n_1, n_2, \dots, n_k$ are integers. Thus, for example, $r_k(n)$ denotes the number of representations of n by the quadratic form $n_1^2 + n_2^2 + \dots + n_k^2$. In another example, we examine the number of ways n can be represented by the quadratic form $x^2 + xy + y^2$.

3.2. Sums of Two Squares

Theorem 3.2.1. *For each positive integer n ,*

$$(3.2.1) \quad r_2(n) = 4 \sum_{\substack{d|n \\ d \text{ odd}}} (-1)^{(d-1)/2}.$$

We can state Theorem 3.2.1 in the alternative formulation

$$(3.2.2) \quad r_2(n) = 4(d_{1,4}(n) - d_{3,4}(n)),$$

where

$$(3.2.3) \quad \begin{aligned} & d_{j,k}(n) \text{ denotes the number of positive divisors} \\ & d \text{ of } n \text{ such that } d \equiv j \pmod{k}. \end{aligned}$$

Immediately deducible from (3.2.2) is the well-known theorem that every prime p congruent to 1 modulo 4 can be represented as a sum of two squares [172, p. 54].

Exercise 3.2.2. *Using (3.2.2), prove that $r_2(n) > 0$ if and only if every prime p congruent to 3 modulo 4 in the canonical factorization of n appears with an even exponent [172, p. 55, Theorem 2.15].*

First Proof of Theorem 3.2.1. Using the Jacobi triple product identity, Theorem 1.3.3, we first deduce that

$$\begin{aligned} (a - 1/a)(a^2 q; q)_\infty (a^{-2} q; q)_\infty (q; q)_\infty &= a(a^2 q; q)_\infty (a^{-2}; q)_\infty (q; q)_\infty \\ &= \sum_{n=-\infty}^{\infty} (-1)^n a^{2n+1} q^{n(n+1)/2} \end{aligned}$$

$$\begin{aligned}
&= \left(\sum_{\substack{n=-\infty \\ n \text{ even}}}^{\infty} + \sum_{\substack{n=-\infty \\ n \text{ odd}}}^{\infty} \right) (-1)^n a^{2n+1} q^{n(n+1)/2} \\
&= \sum_{n=-\infty}^{\infty} a^{4n+1} q^{n(2n+1)} - \sum_{n=-\infty}^{\infty} a^{4n-1} q^{n(2n-1)} \\
&= a(-a^4 q^3; q^4)_{\infty} (-a^{-4} q; q^4)_{\infty} (q^4; q^4)_{\infty} \\
(3.2.4) \quad &- \frac{1}{a} (-a^4 q; q^4)_{\infty} (-a^{-4} q^3; q^4)_{\infty} (q^4; q^4)_{\infty},
\end{aligned}$$

where we applied the Jacobi triple product identity, Theorem 1.3.3, two additional times. We next use logarithmic differentiation to differentiate both sides of (3.2.4) with respect to a and then set $a = 1$. Note that on the far left side of (3.2.4) the differentiation of the infinite products is unnecessary, because when $a = 1$, the factor $a - 1/a = 0$. We therefore obtain

$$\begin{aligned}
(3.2.5) \quad 2(q; q)_{\infty}^3 &= 2(-q^3; q^4)_{\infty} (-q; q^4)_{\infty} (q^4; q^4)_{\infty} \\
&\quad + (-q^3; q^4)_{\infty} (-q; q^4)_{\infty} (q^4; q^4)_{\infty} \\
&\quad \times \sum_{n=1}^{\infty} \left(\frac{4q^{4n-1}}{1+q^{4n-1}} - \frac{4q^{4n-3}}{1+q^{4n-3}} - \frac{4q^{4n-3}}{1+q^{4n-3}} + \frac{4q^{4n-1}}{1+q^{4n-1}} \right) \\
&= 2(-q^3; q^4)_{\infty} (-q; q^4)_{\infty} (q^4; q^4)_{\infty} \\
&\quad \times \left\{ 1 - 4 \sum_{n=1}^{\infty} \left(\frac{q^{4n-3}}{1+q^{4n-3}} - \frac{q^{4n-1}}{1+q^{4n-1}} \right) \right\}.
\end{aligned}$$

Now divide both sides of (3.2.5) by 2 and by

$$\begin{aligned}
(-q; q)_{\infty}^2 (q; q)_{\infty} &= (-q; q)_{\infty} (q^2; q^2)_{\infty} \\
&= (-q; q^2)_{\infty} (-q^2; q^2)_{\infty} (q^2; q^2)_{\infty} \\
&= (-q; q^2)_{\infty} (q^4; q^4)_{\infty} \\
&= (-q^3; q^4)_{\infty} (-q; q^4)_{\infty} (q^4; q^4)_{\infty}
\end{aligned}$$

to deduce that

$$(3.2.6) \quad \frac{(q; q)_{\infty}^2}{(-q; q)_{\infty}^2} = 1 - 4 \sum_{n=1}^{\infty} \left(\frac{q^{4n-3}}{1+q^{4n-3}} - \frac{q^{4n-1}}{1+q^{4n-1}} \right).$$

By (1.3.13) and Euler's identity (1.1.9),

$$(3.2.7) \quad \varphi(-q) = (q; q^2)_\infty^2 (q^2; q^2)_\infty = \frac{(q; q)_\infty}{(-q; q)_\infty}.$$

Using (3.2.7) in (3.2.6) and replacing q by $-q$, we conclude that

$$\begin{aligned} \varphi^2(q) &= 1 + 4 \sum_{n=1}^{\infty} \left(\frac{q^{4n-3}}{1 - q^{4n-3}} - \frac{q^{4n-1}}{1 - q^{4n-1}} \right) \\ &= 1 + 4 \sum_{m=1}^{\infty} \left(\sum_{r=1}^{\infty} q^{(4m-3)r} - \sum_{r=1}^{\infty} q^{(4m-1)r} \right) \\ (3.2.8) \quad &= 1 + 4 \sum_{n=1}^{\infty} \left(\sum_{\substack{d|n \\ d \equiv 1 \pmod{4}}} 1 - \sum_{\substack{d|n \\ d \equiv 3 \pmod{4}}} 1 \right) q^n. \end{aligned}$$

Lastly, equate coefficients of q^n , $n \geq 1$, on both sides of (3.2.8) to deduce that

$$r_2(n) = 4(d_{1,4}(n) - d_{3,4}(n)),$$

which, as we have seen, is an alternative formulation of (3.2.1). $\square$

We now give a second proof of Theorem 3.2.1 that is quite different and somewhat shorter than the proof above. It requires, however, a deeper theorem, namely, Ramanujan's ${}_1\psi_1$ summation theorem.

Second Proof of Theorem 3.2.1. In Ramanujan's ${}_1\psi_1$ summation theorem, Theorem 1.3.12, set $\alpha = \beta = -1$ and $z = 1$. After simplifying the product on the right-hand side, we find that

$$(3.2.9) \quad 1 + 4 \sum_{n=1}^{\infty} \frac{q^n}{1 + q^{2n}} = (-q; q^2)_\infty^4 (q^2; q^2)_\infty^2 = \varphi^2(q),$$

where in the next to last step we used (1.1.9), and in the last step used (3.2.7).

On the other hand,

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{q^n}{1 + q^{2n}} &= \sum_{n=1}^{\infty} \sum_{m=0}^{\infty} (-1)^m q^{n+2mn} = \sum_{m=0}^{\infty} (-1)^m \sum_{n=1}^{\infty} q^{(2m+1)n} \\ (3.2.10) \quad &= \sum_{m=0}^{\infty} \frac{(-1)^m q^{(2m+1)}}{1 - q^{2m+1}}. \end{aligned}$$

Using (3.2.10) in (3.2.9), we see that we have reached the first equality of (3.2.8). The remainder of the proof then follows as before. $\square$

3.3. Sums of Four Squares

Theorem 3.3.1. *For each positive integer n ,*

$$(3.3.1) \quad r_4(n) = 8 \sum_{\substack{d|n \\ 4 \nmid d}} d.$$

Observe that the sum in (3.3.1) is clearly never equal to 0. Thus, as a corollary of Theorem 3.3.1, we deduce Lagrange's theorem that every positive integer n can be represented as a sum of 4 squares [172, p. 317].

First Proof of Theorem 3.3.1. We begin with Jacobi's identity in the form (1.3.28)

$$(3.3.2) \quad (q; q)_{\infty}^3 = \frac{1}{2} \sum_{n=-\infty}^{\infty} (-1)^n (2n+1) q^{n(n+1)/2}.$$

Squaring both sides of (3.3.2), we easily find that

$$\begin{aligned} (q; q)_{\infty}^6 &= \frac{1}{4} \sum_{m, n=-\infty}^{\infty} (-1)^{m+n} (2m+1)(2n+1) q^{(m^2+n^2+m+n)/2} \\ &= \frac{1}{4} \left(\sum_{\substack{m, n=-\infty \\ m+n \text{ even}}}^{\infty} + \sum_{\substack{m, n=-\infty \\ m+n \text{ odd}}}^{\infty} \right) (-1)^{m+n} (2m+1)(2n+1) \\ (3.3.3) \quad &\times q^{(m^2+n^2+m+n)/2}. \end{aligned}$$

In the first sum on the right side of (3.3.3), set

$$m = r + s, \quad n = r - s \quad \text{or} \quad r = \frac{1}{2}(m+n), \quad s = \frac{1}{2}(m-n),$$

and in the second on the right side of (3.3.3), set

$$m = r + s, \quad n = s - r - 1 \quad \text{or} \quad r = \frac{1}{2}(m-n-1), \quad s = \frac{1}{2}(m+n+1).$$

In both cases, as m and n run over their respective ranges, each of r and s runs through all of the integers. It follows from (3.3.3) that

(3.3.4)

$$\begin{aligned}
 (q; q)_{\infty}^6 &= \frac{1}{4} \left(\sum_{r,s=-\infty}^{\infty} (2r+2s+1)(2r-2s+1)q^{r^2+s^2+r} \right. \\
 &\quad \left. - \sum_{r,s=-\infty}^{\infty} (2r+2s+1)(2s-2r-1)q^{r^2+s^2+r} \right) \\
 &= \frac{1}{2} \sum_{r,s=-\infty}^{\infty} ((2r+1)^2 - (2s)^2) q^{r^2+s^2+r} \\
 &= \frac{1}{2} \left(\sum_{s=-\infty}^{\infty} q^{s^2} \sum_{r=-\infty}^{\infty} (2r+1)^2 q^{r^2+r} - \sum_{r=-\infty}^{\infty} q^{r^2+r} \sum_{s=-\infty}^{\infty} (2s)^2 q^{s^2} \right) \\
 &= \frac{1}{2} \left(\sum_{s=-\infty}^{\infty} q^{s^2} \left(1 + 4q \frac{d}{dq} \right) \sum_{r=-\infty}^{\infty} q^{r^2+r} \right. \\
 &\quad \left. - \sum_{r=-\infty}^{\infty} q^{r^2+r} \times 4q \frac{d}{dq} \sum_{s=-\infty}^{\infty} q^{s^2} \right) \\
 &= \frac{1}{2} \left((-q; q^2)_{\infty}^2 (q^2; q^2)_{\infty} \left(1 + 4q \frac{d}{dq} \right) 2(-q^2; q^2)_{\infty}^2 (q^2; q^2)_{\infty} \right. \\
 &\quad \left. - 2(-q^2; q^2)_{\infty}^2 (q^2; q^2)_{\infty} \times 4q \frac{d}{dq} (-q; q^2)_{\infty}^2 (q^2; q^2)_{\infty} \right),
 \end{aligned}$$

where we used the product representation (1.3.13) for $\varphi(q)$, and where we applied the Jacobi triple product identity (1.3.10) to

$$\begin{aligned}
 \sum_{r=-\infty}^{\infty} q^{r^2+r} &= f(1, q^2) = (-1; q^2)_{\infty} (-q^2; q^2)_{\infty} (q^2; q^2)_{\infty} \\
 &= 2(-q^2; q^2)_{\infty}^2 (q^2; q^2)_{\infty}.
 \end{aligned}$$

Now in (3.3.4), we logarithmically differentiate to deduce that

$$\begin{aligned}
 (q; q)_{\infty}^6 &= (-q; q^2)_{\infty}^2 (q^2; q^2)_{\infty}^2 (-q^2; q^2)_{\infty}^2 \\
 &\quad \times \left(1 + 8 \sum_{n=1}^{\infty} \frac{2nq^{2n}}{1+q^{2n}} - 4 \sum_{n=1}^{\infty} \frac{2nq^{2n}}{1-q^{2n}} \right) \\
 &\quad - (-q^2; q^2)_{\infty}^2 (q^2; q^2)_{\infty}^2 (-q; q^2)_{\infty}^2
 \end{aligned}$$

$$\begin{aligned}
 & \times \left(8 \sum_{n=1}^{\infty} \frac{(2n-1)q^{2n-1}}{1+q^{2n-1}} - 4 \sum_{n=1}^{\infty} \frac{2nq^{2n}}{1-q^{2n}} \right) \\
 & = (-q^2; q^2)_{\infty}^2 (q^2; q^2)_{\infty}^2 (-q; q^2)_{\infty}^2 \\
 (3.3.5) \quad & \times \left(1 - 8 \sum_{n=1}^{\infty} \left\{ \frac{(2n-1)q^{2n-1}}{1+q^{2n-1}} - \frac{2nq^{2n}}{1+q^{2n}} \right\} \right).
 \end{aligned}$$

Now divide both sides of (3.3.5) by

$$(-q; q)_{\infty}^4 (q; q)_{\infty}^2 = (-q; q)_{\infty}^2 (q^2; q^2)_{\infty}^2 = (-q; q^2)_{\infty}^2 (-q^2; q^2)_{\infty}^2 (q^2; q^2)_{\infty}^2$$

to deduce that

$$(3.3.6) \quad \frac{(q; q)_{\infty}^4}{(-q; q)_{\infty}^4} = 1 - 8 \sum_{n=1}^{\infty} \left\{ \frac{(2n-1)q^{2n-1}}{1+q^{2n-1}} - \frac{2nq^{2n}}{1+q^{2n}} \right\}.$$

Thus, substituting (3.2.7) into (3.3.6) and then replacing q by $-q$, we find that

$$\begin{aligned}
 \varphi^4(q) &= 1 + 8 \sum_{n=1}^{\infty} \left\{ \frac{(2n-1)q^{2n-1}}{1-q^{2n-1}} + \frac{2nq^{2n}}{1+q^{2n}} \right\} \\
 &= 1 + 8 \sum_{n=1}^{\infty} \left\{ \frac{(2n-1)q^{2n-1}}{1-q^{2n-1}} + \frac{2nq^{2n}}{1-q^{2n}} \right\} \\
 &\quad - 8 \sum_{n=1}^{\infty} \left\{ \frac{2nq^{2n}}{1-q^{2n}} - \frac{2nq^{2n}}{1+q^{2n}} \right\} \\
 &= 1 + 8 \sum_{n=1}^{\infty} \frac{nq^n}{1-q^n} - 8 \sum_{n=1}^{\infty} \frac{4nq^{4n}}{1-q^{4n}} \\
 &= 1 + 8 \sum_{\substack{n=1 \\ 4 \nmid n}}^{\infty} \frac{nq^n}{1-q^n} \\
 &= 1 + 8 \sum_{\substack{n=1 \\ 4 \nmid n}}^{\infty} nq^n \sum_{m=0}^{\infty} q^{nm} \\
 &= 1 + 8 \sum_{m=1}^{\infty} \sum_{\substack{d=1 \\ 4 \nmid d}}^{\infty} dq^{dm}
 \end{aligned}$$

$$(3.3.7) \quad = 1 + 8 \sum_{n=1}^{\infty} \sum_{\substack{d|n \\ 4 \nmid d}} dq^n,$$

where in the penultimate step we merely replaced n by d , and where in the last step we set $n = dm$ and collected all coefficients of q^n . Now equate coefficients of q^n , $n \geq 1$, on the extremal sides of (3.3.7) to deduce (3.3.1). $\square$

We now give a second proof of Theorem 3.3.1 that is based on our work in the second proof of Theorem 3.2.1.

Second Proof of Theorem 3.3.1. In Ramanujan's ${}_1\psi_1$ summation theorem, Theorem 1.3.12, first replace q by q^2 , and then set $z = qe^{i\theta}$, $a = -1$, and $b = -q^2$, where θ is real.

Exercise 3.3.2. Using (1.3.35), prove that for every integer n ,

$$(3.3.8) \quad \frac{(-1; q^2)_n}{(-q^2; q^2)_n} = \frac{2}{1 + q^{2n}}.$$

Using (3.3.8) and replacing n by $-n$ for each negative integer n , we find from (1.3.36) that

$$\begin{aligned} \sum_{n=-\infty}^{\infty} \frac{(-1; q^2)_n}{(-q^2; q^2)_n} q^n e^{in\theta} &= 1 + 2 \sum_{n=1}^{\infty} \frac{q^n e^{in\theta}}{1 + q^{2n}} + 2 \sum_{n=1}^{\infty} \frac{q^{-n} e^{-in\theta}}{1 + q^{-2n}} \\ &= 1 + 4 \sum_{n=1}^{\infty} \frac{q^n \cos(n\theta)}{1 + q^{2n}} \\ (3.3.9) \quad &= \frac{(-qe^{i\theta}; q^2)_{\infty} (-qe^{-i\theta}; q^2)_{\infty} (q^2; q^2)_{\infty}^2}{(qe^{i\theta}; q^2)_{\infty} (qe^{-i\theta}; q^2)_{\infty} (-q^2; q^2)_{\infty}^2}. \end{aligned}$$

Now replace θ by $\pi - \theta$ in (3.3.9). Multiply this new identity by (3.3.9) itself to deduce that

$$\begin{aligned} (3.3.10) \quad \left\{ 1 + 4 \sum_{n=1}^{\infty} \frac{q^n \cos(n\theta)}{1 + q^{2n}} \right\} \left\{ 1 + 4 \sum_{n=1}^{\infty} \frac{(-1)^n q^n \cos(n\theta)}{1 + q^{2n}} \right\} \\ = \frac{(q^2; q^2)_{\infty}^4}{(-q^2; q^2)_{\infty}^4} = \varphi^4(-q^2), \end{aligned}$$

by (3.2.7). Now integrate both sides of (3.3.10) with respect to θ over $[-\pi, \pi]$. Using the orthogonality of $\{\cos(n\theta)\}$, $1 \leq n < \infty$, on $[-\pi, \pi]$, we find that

$$(3.3.11) \quad 2\pi\varphi^4(-q^2) = 2\pi + 16\pi \sum_{n=1}^{\infty} \frac{(-1)^n q^{2n}}{(1+q^{2n})^2}.$$

Replacing $-q^2$ by q in (3.3.11), we deduce that

$$(3.3.12) \quad \varphi^4(q) = 1 + 8 \sum_{n=1}^{\infty} \frac{q^n}{(1+(-q)^n)^2}.$$

Now,

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{q^n}{(1+(-q)^n)^2} &= \sum_{n=1}^{\infty} q^n \sum_{m=1}^{\infty} m(-1)^{m-1}(-q)^{n(m-1)} \\ &= \sum_{m=1}^{\infty} (-1)^{m-1} m \sum_{n=1}^{\infty} (-1)^n (-q)^{mn} \\ (3.3.13) \quad &= \sum_{m=1}^{\infty} \frac{mq^m}{1+(-q)^m}. \end{aligned}$$

Substitute (3.3.13) in (3.3.12) and observe that we have now reached the first line of (3.3.7). Thus, the remainder of the proof is identical to that given above. $\square$

3.4. Sums of Six Squares

Of Jacobi's formulas for $r_{2k}(n)$, $1 \leq k \leq 4$, the most difficult to prove is perhaps that for $r_6(n)$. The proof of S. H. Chan [72] that we give is possibly the simplest. Arithmetic proofs of the formula for $r_6(n)$ are given in [167] and [154].

Theorem 3.4.1.

$$(3.4.1) \quad r_6(n) = 4 \left(\sum_{\substack{d|n \\ d \equiv 3 \pmod{4}}} d^2 - \sum_{\substack{d|n \\ d \equiv 1 \pmod{4}}} d^2 \right) + 16 \left(\sum_{\substack{d|n \\ \frac{k}{d} \equiv 1 \pmod{4}}} d^2 - \sum_{\substack{d|n \\ \frac{k}{d} \equiv 3 \pmod{4}}} d^2 \right).$$

Proof. Replacing a , b , and z by y , yq , and x , respectively, in Ramanujan's ${}_1\psi_1$ summation (1.3.36), we obtain the useful corollary

$$(3.4.2) \quad \sum_{n=-\infty}^{\infty} \frac{x^n}{1-yq^n} = \frac{(xy; q)_{\infty} (q/(xy); q)_{\infty} (q; q)_{\infty}^2}{(y; q)_{\infty} (q/y; q)_{\infty} (x; q)_{\infty} (q/x; q)_{\infty}},$$

where $|q| < |x| < 1$.

Exercise 3.4.2. Observe that the right-hand side of (3.4.2) is symmetric in x and y . Hence,

$$(3.4.3) \quad \sum_{n=-\infty}^{\infty} \frac{x^n}{1-yq^n} = \sum_{n=-\infty}^{\infty} \frac{y^n}{1-xq^n}.$$

Give a direct proof of (3.4.3), independent of (3.4.2).

Differentiating both sides of (3.4.2) with respect to x and multiplying both sides by x , we find that

$$(3.4.4) \quad \sum_{n=-\infty}^{\infty} \frac{nx^n}{1-yq^n} = \frac{(xy; q)_{\infty} (q/(xy); q)_{\infty} (q; q)_{\infty}^2}{(y; q)_{\infty} (q/y; q)_{\infty} (x; q)_{\infty} (q/x; q)_{\infty}} \\ \times \sum_{n=0}^{\infty} \left(-\frac{xyq^n}{1-xyq^n} + \frac{q^{n+1}/(xy)}{1-q^{n+1}/(xy)} + \frac{xq^n}{1-xq^n} - \frac{q^{n+1}/x}{1-q^{n+1}/x} \right).$$

Interchange x and y in (3.4.4) and subtract the resulting identity from (3.4.4) to deduce that

$$(3.4.5) \quad \sum_{n=-\infty}^{\infty} \frac{nx^n}{1-yq^n} - \sum_{n=-\infty}^{\infty} \frac{ny^n}{1-xq^n} = \frac{(xy; q)_{\infty} (q/(xy); q)_{\infty} (q; q)_{\infty}^2}{(y; q)_{\infty} (q/y; q)_{\infty} (x; q)_{\infty} (q/x; q)_{\infty}} \\ \times \sum_{n=0}^{\infty} \left(\frac{xq^n}{1-xq^n} - \frac{q^{n+1}/x}{1-q^{n+1}/x} - \frac{yq^n}{1-yq^n} + \frac{q^{n+1}/y}{1-q^{n+1}/y} \right),$$

where $|q| < |x|, |y| < 1$. Letting $y \rightarrow -x$ in (3.4.5) and simplifying, we find that

$$(3.4.6) \quad L(q) := \sum_{n=-\infty}^{\infty} \left(\frac{nx^n}{1+xq^n} - \frac{n(-x)^n}{1-xq^n} \right) = \frac{(-x^2; q)_{\infty} (-q/x^2; q)_{\infty} (q; q)_{\infty}^2}{(x^2; q^2)_{\infty} (q^2/x^2; q^2)_{\infty}} \\ \times 2 \sum_{n=0}^{\infty} \left(\frac{xq^n}{1-x^2q^{2n}} - \frac{q^{n+1}/x}{1-q^{2n+2}/x^2} \right) =: R(q).$$

Upon simplifying, we see that

$$(3.4.7) \quad R(q) = \frac{(-x^2; q)_{\infty} (-q/x^2; q)_{\infty} (q; q)_{\infty}^2}{(x^2; q^2)_{\infty} (q^2/x^2; q^2)_{\infty}} \sum_{n=-\infty}^{\infty} \frac{2xq^n}{1-x^2q^{2n}} \\ = 2x \frac{(-x^2; q)_{\infty} (-q/x^2; q)_{\infty} (q; q)_{\infty}^2}{(x^2; q^2)_{\infty} (q^2/x^2; q^2)_{\infty}} \frac{(x^2q; q^2)_{\infty} (q/x^2; q^2)_{\infty} (q^2; q^2)_{\infty}^2}{(x^2; q^2)_{\infty} (q^2/x^2; q^2)_{\infty} (q; q^2)_{\infty}^2} \\ = 2x(1+x^2) \frac{(-x^2q; q)_{\infty} (-q/x^2; q)_{\infty} (x^2q; q^2)_{\infty} (q/x^2; q^2)_{\infty} (q^2; q^2)_{\infty}^4}{(x^2; q^2)_{\infty}^2 (q^2/x^2; q^2)_{\infty}^2},$$

where in the second equality we applied (3.4.2), with x , y , and q replaced by q , x^2 , and q^2 , respectively.

Simplifying the left-hand side of (3.4.6) yields

$$L(q) = \sum_{n=1}^{\infty} \left(\frac{nx^n}{1+xq^n} - \frac{n(-x)^n}{1-xq^n} + \frac{-nx^{-n}}{1+xq^{-n}} - \frac{-n(-x)^{-n}}{1-xq^{-n}} \right) \\ = \sum_{n=1}^{\infty} \left(nx^n - \frac{nx^{n+1}q^n}{1+xq^n} - n(-x)^n + \frac{n(-x)^{n+1}q^n}{1-xq^n} \right. \\ \left. - \frac{nx^{-n-1}q^n}{1+q^n/x} + \frac{n(-x)^{-n-1}q^n}{1-q^n/x} \right) \\ = \sum_{n=1}^{\infty} nx^n (1 - (-1)^n) \\ + \sum_{n=1, m=0}^{\infty} ((-1)^{m+1} nx^{n+m+1} q^{mn+n} + (-1)^{n+1} nx^{n+m+1} q^{mn+n}) \\ + \sum_{n=1, m=0}^{\infty} ((-1)^{m+1} nx^{-n-m-1} q^{mn+n} + (-1)^{n+1} nx^{-n-m-1} q^{mn+n})$$

$$\begin{aligned}
&= 2 \sum_{n=1}^{\infty} (2n-1)x^{2n-1} \\
&\quad + \sum_{n,m=1}^{\infty} ((-1)^m - (-1)^n) n (x^{n+m} + x^{-n-m}) q^{mn}.
\end{aligned}$$

Noting that the summands disappear whenever m and n are of the same parity, we deduce that

$$\begin{aligned}
L(q) &= 2 \sum_{n=1}^{\infty} (2n-1)x^{2n-1} \\
&\quad + 2 \sum_{n,m=1}^{\infty} (2n-1) (x^{2n-1+2m} + x^{-2n+1-2m}) q^{(2n-1)2m} \\
&\quad - 2 \sum_{n,m=1}^{\infty} 2n (x^{2n+2m-1} + x^{-2n-2m+1}) q^{2n(2m-1)} \\
&= \frac{2x(1+x^2)}{(1-x^2)^2} + 2 \sum_{n,m=1}^{\infty} (2n-1-2m) \\
&\quad \times (x^{2n-1+2m} + x^{-2n+1-2m}) q^{(2n-1)2m} \\
&= \frac{2x(1+x^2)}{(1-x^2)^2} + 2x(1+x^2) \sum_{n,m=1}^{\infty} (2n-1-2m)x^{-2n-2m} \\
(3.4.8) \quad &\times (1-x^2+\dots+x^{4n+4m-4}) q^{(2n-1)2m}
\end{aligned}$$

We now multiply both $L(q)$ and $R(q)$ by $2/(x(1+x^2))$ and let $x \rightarrow i$. First, from (3.4.7) we find that

$$\begin{aligned}
\lim_{x \rightarrow i} \frac{2R(q)}{x(1+x^2)} &= \frac{(q; q)_{\infty}^2 (-q; q^2)_{\infty}^2 (q^2; q^2)_{\infty}^4}{(-q^2; q^2)_{\infty}^4} \\
(3.4.9) \quad &= \frac{(q^2; q^4)_{\infty}^2 (q^2; q^2)_{\infty}^6}{(-q^2; q^2)_{\infty}^4} = \frac{(q^2; q^2)_{\infty}^6}{(-q^2; q^2)_{\infty}^6},
\end{aligned}$$

where we have appealed to Euler's identity (1.1.9) in the last step. Second, from (3.4.8) we deduce that

$$\begin{aligned}
\lim_{x \rightarrow i} \frac{2L(q)}{x(1+x^2)} &= 1 + 4 \sum_{n,m=1}^{\infty} (2n-1-2m)(-1)^{n+m} \\
&\quad \times (2n-1+2m) q^{(2n-1)2m}
\end{aligned}$$

$$\begin{aligned}
&= 1 + 4 \sum_{n,m=1}^{\infty} ((2n-1)^2 - 4m^2) (-1)^n (-q^2)^{(2n-1)m} \\
&= 1 + 4 \sum_{n=1}^{\infty} (-1)^n (2n-1)^2 \sum_{m=1}^{\infty} (-q^2)^{(2n-1)m} \\
&\quad + 16 \sum_{m=1}^{\infty} m^2 \sum_{n=1}^{\infty} (-1)^{n+1} (-q^2)^{(2n-1)m} \\
&= 1 + 4 \sum_{k=1}^{\infty} (-q^2)^k \sum_{(2n-1)|k} (-1)^n (2n-1)^2 \\
&\quad + 16 \sum_{k=1}^{\infty} (-q^2)^k \sum_{(2n-1)m=k} (-1)^{n+1} m^2.
\end{aligned}
\tag{3.4.10}$$

Now substitute (3.4.9) and (3.4.10) into (3.4.6) to deduce that

$$\begin{aligned}
\frac{(q^2; q^2)_{\infty}^6}{(-q^2; q^2)_{\infty}^6} &= 1 + 4 \sum_{k=1}^{\infty} (-q^2)^k \sum_{(2n-1)|k} (-1)^n (2n-1)^2 \\
&\quad + 16 \sum_{k=1}^{\infty} (-q^2)^k \sum_{(2n-1)m=k} (-1)^{n+1} m^2.
\end{aligned}$$

Equating the coefficients of $(-q^2)^n$ on both sides above and using (3.2.7), we complete the proof of Jacobi's six squares theorem. $\square$

3.5. Sums of Eight Squares

Our proof of Jacobi's formula for $r_8(n)$ will focus on the function

$$(3.5.1) \quad F(z) = F(q; z) := \frac{(1+z)(-zq; q)_{\infty}(-q/z; q)_{\infty}(q; q)_{\infty}^2}{(1-z)(zq; q)_{\infty}(q/z; q)_{\infty}(-q; q)_{\infty}^2}.$$

Lemma 3.5.1. For $|q| < |z| < 1/|q|$ and $z \neq 1$,

$$(3.5.2) \quad F(z) = \frac{1+z}{1-z} + 2 \sum_{n=1}^{\infty} \frac{q^n(z^{-n} - z^n)}{1+q^n}.$$

Proof. In Ramanujan's ${}_1\psi_1$ summation formula, Corollary 1.3.14, let $\alpha = \beta = -1$, replace q^2 by q and lastly replace z by $\sqrt{q}z$. We then

readily find that, for $1 < |z| < 1/|q|$,

$$\begin{aligned}
 1 + 2 \sum_{n=1}^{\infty} \frac{q^n z^n}{1+q^n} + 2 \sum_{n=1}^{\infty} \frac{z^{-n}}{1+q^n} &= \frac{(q; q)_{\infty}^2 (-zq; q)_{\infty} (-1/z; q)_{\infty}}{(-q; q)_{\infty}^2 (zq; q)_{\infty} (1/z; q)_{\infty}} \\
 (3.5.3) \qquad &= -\frac{(1+z)(q; q)_{\infty}^2 (-zq; q)_{\infty} (-q/z; q)_{\infty}}{(1-z)(-q; q)_{\infty}^2 (zq; q)_{\infty} (q/z; q)_{\infty}} = -F(z).
 \end{aligned}$$

But, for $1 < |z|$,

$$\begin{aligned}
 1 + 2 \sum_{n=1}^{\infty} \frac{z^{-n}}{1+q^n} &= 1 + 2 \sum_{n=1}^{\infty} \left(z^{-n} - \frac{q^n z^{-n}}{1+q^n} \right) \\
 &= 1 + 2 \frac{z^{-1}}{1-z^{-1}} - 2 \sum_{n=1}^{\infty} \frac{q^n z^{-n}}{1+q^n} \\
 (3.5.4) \qquad &= \frac{z+1}{z-1} - 2 \sum_{n=1}^{\infty} \frac{q^n z^{-n}}{1+q^n}.
 \end{aligned}$$

Putting (3.5.4) in (3.5.3), multiplying both sides by -1 , and using analytic continuation, we complete the proof of (3.5.2) for $|q| < |z| < 1/|q|$ and $z \neq 1$. $\square$

Theorem 3.5.2. For $|q| < |z| < 1/|q|$ and $z \neq 1$,

$$(3.5.5) \qquad F^2(z) = \left(\frac{1+z}{1-z} \right)^2 + 8 \sum_{n=1}^{\infty} \frac{(-1)^{n-1} n q^n}{1-q^n} + 4 \sum_{n=1}^{\infty} \frac{n q^n}{1-q^n} (z^n + z^{-n}).$$

Proof. We begin by squaring both sides of (3.5.2), and in doing so, we use the elementary identities

$$(3.5.6) \qquad \left(\frac{1+z}{1-z} \right) (z^{-n} - z^n) = (z^n + z^{-n}) + 2 \sum_{k=1}^{n-1} (z^k + z^{-k}) + 2,$$

$$\begin{aligned}
 (3.5.7) \qquad (z^{-n} - z^n)(z^{-m} - z^m) &= (z^{n+m} + z^{-n-m}) - (z^{n-m} + z^{m-n}).
 \end{aligned}$$

It follows from (3.5.2), (3.5.6), and (3.5.7) that

$$(3.5.8) \quad \left(\frac{1+z}{1-z} + 2 \sum_{n=1}^{\infty} \frac{q^n(z^{-n} - z^n)}{1+q^n} \right)^2 \\ = \left(\frac{1+z}{1-z} \right)^2 + C_0 + \sum_{n=1}^{\infty} C_n(z^n + z^{-n}),$$

where the constants C_n , $n \geq 0$, are independent of z . Now applying explicitly (3.5.6) and (3.5.7) in the expansion of the left side of (3.5.8), we find that

$$(3.5.9) \quad \left(\frac{1+z}{1-z} + 2 \sum_{n=1}^{\infty} \frac{q^n(z^{-n} - z^n)}{1+q^n} \right)^2 \\ = \left(\frac{1+z}{1-z} \right)^2 + 4 \sum_{n=1}^{\infty} \frac{q^n}{1+q^n} \left\{ (z^n + z^{-n}) + 2 \sum_{k=1}^{n-1} (z^k + z^{-k}) + 2 \right\} \\ + 4 \sum_{m,n=1}^{\infty} \frac{q^n}{1+q^n} \frac{q^m}{1+q^m} \{ (z^{n+m} + z^{-n-m}) - (z^{n-m} + z^{m-n}) \}.$$

Equating coefficients of $(z^n + z^{-n})$ on the right sides of (3.5.8) and (3.5.9), we find that

$$(3.5.10) \quad C_n = 4 \frac{q^n}{1+q^n} + 8 \sum_{m=1}^{\infty} \frac{q^{n+m}}{1+q^{n+m}} + 4 \sum_{m=1}^{n-1} \frac{q^n}{(1+q^m)(1+q^{n-m})} \\ - 8 \sum_{m=1}^{\infty} \frac{q^m q^{n+m}}{(1+q^m)(1+q^{n+m})}.$$

For the purpose of simplification, rewrite each of the three sums in (3.5.10) and arrive at

$$(3.5.11) \quad C_n = 4 \frac{q^n}{1+q^n} + 8 \frac{q^n}{1-q^n} \sum_{m=1}^{\infty} \left\{ \frac{q^{n+m}}{q^n(1+q^{n+m})} - \frac{q^{n+m}}{1+q^{n+m}} \right\} \\ + 4 \frac{q^n}{1-q^n} \sum_{m=1}^{n-1} \left\{ 1 - \frac{q^m}{1+q^m} - \frac{q^{n-m}}{1+q^{n-m}} \right\} \\ - 8 \frac{q^n}{1-q^n} \sum_{m=1}^{\infty} \left\{ -\frac{q^m}{1+q^m} + \frac{q^{n+m}}{q^n(1+q^{n+m})} \right\}.$$

Observe that the first portion of the first sum on the right side cancels with the second portion of the third sum on the right side above. After still further cancellation, we find that (3.5.11) reduces simply to

$$\begin{aligned} C_n &= 4 \frac{q^n}{1+q^n} + 4(n-1) \frac{q^n}{1-q^n} + 8 \frac{q^n}{1-q^n} \frac{q^n}{1+q^n} \\ (3.5.12) \quad &= 4n \frac{q^n}{1-q^n}. \end{aligned}$$

Observe that the left side of (3.5.8) vanishes at $z = -1$. Thus, setting $z = -1$ in (3.5.8), we readily find that

$$(3.5.13) \quad C_0 = 8 \sum_{n=1}^{\infty} \frac{(-1)^{n-1} n q^n}{1-q^n}.$$

Putting (3.5.12) and (3.5.13) in (3.5.8), we complete the proof of (3.5.5). $\square$

Theorem 3.5.3. *We have*

$$(3.5.14) \quad \varphi^8(q) = 1 + 16 \sum_{n=1}^{\infty} \frac{n^3 q^n}{1 - (-q)^n}.$$

Proof. Rewrite (3.5.5) in the form

$$(3.5.15) \quad F^2(z) = \left(\frac{1+z}{1-z} \right)^2 + 4 \sum_{n=1}^{\infty} \frac{n q^n}{1-q^n} (z^n + z^{-n} - 2(-1)^n).$$

Divide both sides of (3.5.15) by $(1+z)^2$ and let z tend to -1 . Using L'Hôpital's rule, we find that

$$(3.5.16) \quad \lim_{z \rightarrow -1} \frac{z^n + z^{-n} - 2(-1)^n}{(1+z)^2} = (-1)^n n^2.$$

Letting $z \rightarrow -1$ in (3.5.15), using (3.5.16), using the definition (3.5.1) on the left side of (3.5.15), and lastly employing (3.2.7), we conclude that

$$(3.5.17) \quad \varphi^8(-q) = 1 + 16 \sum_{n=1}^{\infty} \frac{(-1)^n n^3 q^n}{1-q^n}.$$

Replacing q by $-q$ in (3.5.17), we deduce (3.5.14) to complete the proof. $\square$

Theorem 3.5.4. *For each positive integer n ,*

$$(3.5.18) \quad r_8(n) = 16(-1)^n \sum_{d|n} (-1)^d d^3.$$

Proof. From (3.5.14),

$$\begin{aligned} \varphi^8(q) &= 1 + 16 \sum_{d=1}^{\infty} \frac{d^3 q^d}{1 - (-q)^d} \\ &= 1 + 16 \sum_{d=1}^{\infty} \sum_{m=0}^{\infty} d^3 q^d (-q)^{dm} = 1 + 16 \sum_{n=1}^{\infty} (-1)^n \sum_{d|n} (-1)^d d^3 q^n, \end{aligned}$$

where we put $n = d(m+1)$. The desired result now follows by equating coefficients of q^n , $n \geq 1$, on both sides above. $\square$

3.6. Sums of Triangular Numbers

In this short section we provide easy proofs of formulas for $t_2(n)$ and $t_4(n)$. We begin with some elementary identities for theta functions.

Exercise 3.6.1. *Prove that*

$$(3.6.1) \quad \varphi(q) + \varphi(-q) = 2\varphi(q^4),$$

$$(3.6.2) \quad \varphi(q) - \varphi(-q) = 4q\psi(q^8),$$

$$(3.6.3) \quad \varphi(q)\psi(q^2) = \psi^2(q).$$

Theorem 3.6.2. *Recall that $d_{j,k}(n)$ is defined in (3.2.3). Then for each positive integer n ,*

$$(3.6.4) \quad t_2(n) = d_{1,4}(4n+1) - d_{3,4}(4n+1).$$

Proof. Using (3.6.1)–(3.6.3), we easily find that

$$(3.6.5) \quad \varphi^2(q) - \varphi^2(-q) = 8q\varphi(q^4)\psi(q^8) = 8q\psi^2(q^4).$$

Employing (3.2.2) and using the generating function (1.2.5) for $t_k(n)$, we see that

$$\begin{aligned} 4 \sum_{m=1}^{\infty} (d_{1,4}(m) - d_{3,4}(m)) q^m - 4 \sum_{m=1}^{\infty} (d_{1,4}(m) - d_{3,4}(m)) (-q)^m \\ = 8 \sum_{n=0}^{\infty} t_2(n) q^{4n+1}. \end{aligned}$$

Equating coefficients of q^{4n+1} , $n \geq 0$, on both sides above, we conclude that

$$d_{1,4}(4n+1) - d_{3,4}(4n+1) = t_2(n),$$

which is what we wanted to prove. $\square$

Theorem 3.6.3. *For each positive integer n ,*

$$(3.6.6) \quad t_4(n) = \sigma(2n+1),$$

where $\sigma(n) = \sum_{d|n} d$.

Exercise 3.6.4. *In analogy with (3.6.5), prove that*

$$(3.6.7) \quad \varphi^2(q) + \varphi^2(-q) = 2\varphi^2(q^2).$$

Proof. Multiply (3.6.5) and (3.6.7) together and use (3.6.3) to deduce that

$$(3.6.8) \quad \varphi^4(q) - \varphi^4(-q) = 16q\psi^4(q^2).$$

Now invoke Theorem 3.3.1 and employ the generating function (1.2.5) for $t_4(n)$ to arrive at

$$8 \sum_{n=0}^{\infty} \sum_{\substack{d|n \\ 4 \nmid d}} d q^n - 8 \sum_{n=0}^{\infty} \sum_{\substack{d|n \\ 4 \nmid d}} d (-q)^n = 16 \sum_{n=0}^{\infty} t_4(n) q^{2n+1}.$$

Equating coefficients of q^{2n+1} on both sides above, we conclude that

$$\sum_{\substack{d|(2n+1) \\ 4 \nmid d}} d = t_4(n).$$

Noting that the condition $4 \nmid d$ is superfluous for divisors of odd integers, we complete the proof. $\square$

3.7. Representations of Integers by $x^2 + 2y^2$, $x^2 + 3y^2$, and $x^2 + xy + y^2$

In our next theorems, we derive analogues of Theorem 3.2.1 for representations of positive integers by the quadratic forms $x^2 + 2y^2$ and $x^2 + 3y^2$. However, first we need a lemma which is a beautiful identity by itself.

Lemma 3.7.1. For $|a|, |b| < 1$,

$$(3.7.1) \quad 1 + 2 \sum_{n=1}^{\infty} \frac{a^n + b^n}{1 + (ab)^n} = \frac{(-a; ab)_{\infty} (-b; ab)_{\infty} (ab; ab)_{\infty}^2}{(a; ab)_{\infty} (b; ab)_{\infty} (-ab; ab)_{\infty}^2}$$

Proof. In Corollary 1.3.14, set $\alpha = \beta = -1$, $q = \sqrt{ab}$, and $z = \sqrt{a/b}$. Then (3.7.1) readily follows upon observing that

$$\frac{(-1; ab)_n}{(-ab; ab)_n} = \frac{2}{1 + (ab)^n},$$

for each positive integer n . □

Theorem 3.7.2. We have

$$(3.7.2) \quad \varphi(q)\varphi(q^2) = 1 - 2 \sum_{n=1}^{\infty} \frac{(-1)^{n(n+1)/2} q^{2n-1}}{1 - q^{2n-1}}.$$

Proof. Set $a = q$ and $b = q^3$ in Lemma 3.7.1. Using (3.2.7) or (1.3.13), (1.3.14), (1.3.32), and (1.3.30), we find that

$$\begin{aligned} 1 + 2 \sum_{n=1}^{\infty} \frac{q^n + q^{3n}}{1 + q^{4n}} &= \frac{(-q; q^4)_{\infty} (-q^3; q^4)_{\infty} (q^4; q^4)_{\infty}^2}{(q; q^4)_{\infty} (q^3; q^4)_{\infty} (-q^4; q^4)_{\infty}^2} \\ &= \frac{(-q; q^2)_{\infty}}{(q; q^2)_{\infty}} \varphi^2(-q^4) \\ &= \frac{\psi(q)}{\psi(-q)} \varphi(-q^2) \varphi(q^2) \\ &= \varphi(q^2) \sqrt{\frac{\varphi(q)}{\varphi(-q)}} \sqrt{\varphi(-q) \varphi(q)} \\ (3.7.3) \quad &= \varphi(q) \varphi(q^2). \end{aligned}$$

Expanding $1/(1 + q^{4n})$ in a geometric series on the left side of (3.7.3), we arrive at the following array of powers:

$$\begin{array}{ccccccc} q & q^3 & -q^5 & -q^7 & q^9 & q^{11} & \dots \\ q^2 & q^6 & -q^{10} & -q^{14} & q^{18} & q^{22} & \dots \\ q^3 & q^9 & -q^{15} & -q^{21} & q^{27} & q^{33} & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{array}$$

If we sum this array by successive columns and rows, we find that

$$(3.7.4) \quad \sum_{n=1}^{\infty} \frac{q^n + q^{3n}}{1 + q^{4n}} = \sum_{n=1}^{\infty} \frac{(-1)^{n(n+1)/2+1} q^{2n-1}}{1 - q^{2n-1}}.$$

If we use (3.7.4) in (3.7.3), we complete the proof of (3.7.2). $\square$

Theorem 3.7.3. *Let $r_{1,2}(n)$ denote the number of representations of the positive integer n as the sum of a square and twice a square. Then, for $n \geq 1$,*

$$r_{1,2}(n) = 2(d_{1,8}(n) + d_{3,8}(n) - d_{5,8}(n) - d_{7,8}(n)),$$

where $d_{j,8}(n)$, $j = 1, 3, 5, 7$, is defined by (3.2.3).

Proof. In (3.7.2), expand $1/(1 - q^{2n-1})$ in a geometric series, and the desired result follows. $\square$

Theorem 3.7.4. *We have*

$$(3.7.5) \quad \begin{aligned} \varphi(q)\varphi(q^3) &= 1 + 4 \sum_{n=0}^{\infty} \left(\frac{q^{4(3n+1)}}{1 - q^{4(3n+1)}} - \frac{q^{4(3n+2)}}{1 - q^{4(3n+2)}} \right) \\ &\quad + 2 \sum_{n=0}^{\infty} \left(\frac{q^{3n+1}}{1 - q^{3n+1}} - \frac{q^{3n+2}}{1 - q^{3n+2}} \right) \end{aligned}$$

Proof. In Lemma 3.7.1, set $a = -q$ and $b = -q^2$ to deduce that

$$(3.7.6) \quad \begin{aligned} 1 + 2 \sum_{n=1}^{\infty} \frac{(-q)^n + (-q^2)^n}{1 + q^{3n}} &= \frac{(q; q^3)_{\infty} (q^2; q^3)_{\infty} (q^3; q^3)_{\infty}^2}{(-q; q^3)_{\infty} (-q^2; q^3)_{\infty} (-q^3; q^3)_{\infty}^2} \\ &= \frac{(q; q)_{\infty} (q^3; q^3)_{\infty}}{(-q; q)_{\infty} (-q^3; q^3)_{\infty}} \\ &= \varphi(-q)\varphi(-q^3), \end{aligned}$$

by (3.2.7).

Next, on the left side of (3.7.6), expand $1/(1 + q^{3n})$ in a geometric series, invert the order of summation, and sum the two resulting geometric series. This elementary task then gives us

$$(3.7.7) \quad \sum_{n=1}^{\infty} \frac{(-q)^n + (-q^2)^n}{1 + q^{3n}} = \sum_{m=0}^{\infty} (-1)^{m+1} \left(\frac{q^{3m+1}}{1 + q^{3m+1}} + \frac{q^{3m+2}}{1 + q^{3m+2}} \right).$$

Inserting (3.7.7) in (3.7.6), replacing q by $-q$, and employing, for brevity, the Legendre symbol $\left(\frac{n}{3}\right)$, we find that

$$\begin{aligned}
 \varphi(q)\varphi(q^3) &= 1 + 2 \sum_{m=0}^{\infty} \left(\frac{q^{3m+1}}{1 + (-q)^{3m+1}} - \frac{q^{3m+2}}{1 + (-q)^{3m+2}} \right) \\
 (3.7.8) \quad &= 1 + 2 \sum_{n=1}^{\infty} \left(\frac{n}{3} \right) \frac{q^n}{1 + (-q)^n} \\
 &= 1 + 2 \sum_{n=1}^{\infty} \left(\frac{2n}{3} \right) \frac{q^{2n}}{1 + q^{2n}} + 2 \sum_{n=0}^{\infty} \left(\frac{2n+1}{3} \right) \frac{q^{2n+1}}{1 - q^{2n+1}} \\
 &= 1 + 2 \sum_{n=1}^{\infty} \left(\frac{2n}{3} \right) \left(\frac{q^{2n}}{1 - q^{2n}} - 2 \frac{q^{4n}}{1 - q^{4n}} \right) \\
 &\quad + 2 \sum_{n=0}^{\infty} \left(\frac{2n+1}{3} \right) \frac{q^{2n+1}}{1 - q^{2n+1}} \\
 &= 1 + 2 \sum_{n=1}^{\infty} \left(\frac{n}{3} \right) \frac{q^n}{1 - q^n} + 4 \sum_{n=1}^{\infty} \left(\frac{n}{3} \right) \frac{q^{4n}}{1 - q^{4n}} \\
 &= 1 + 2 \sum_{n=0}^{\infty} \left(\frac{q^{3n+1}}{1 - q^{3n+1}} - \frac{q^{3n+2}}{1 - q^{3n+2}} \right) \\
 &\quad + 4 \sum_{n=0}^{\infty} \left(\frac{q^{4(3n+1)}}{1 - q^{4(3n+1)}} - \frac{q^{4(3n+2)}}{1 - q^{4(3n+2)}} \right).
 \end{aligned}$$

□

Theorem 3.7.4 has the following arithmetical interpretation.

Theorem 3.7.5. Let $r_{1,3}(n)$ denote the number of representations of the positive integer n by $x^2 + 3y^2$, where x and y are integers, and where representations with x and y having different signs are counted as distinct. Recall that $d_{j,k}(n)$ is defined in (3.2.3). Then for each positive integer n ,

$$(3.7.9) \quad r_{1,3}(n) = 4(d_{4,12}(n) - d_{8,12}(n)) + 2(d_{1,3}(n) - d_{2,3}(n)).$$

Proof. If we expand each of the quotients $1/(1 - q^m)$ on the right side of (3.7.5) into geometric series, collect the coefficients of all common exponents in each of the four resulting double series, and then equate

coefficients of like powers of n on each side, we deduce (3.7.9) to complete the proof. $\square$

We prove one further theorem and its arithmetical consequence in this chapter, but it is closely related to Theorem 3.7.4. To effect the proof, we first need a beautiful analogue of Lemma 3.7.1.

Lemma 3.7.6. *For $|a|, |b| < 1$,*

$$(3.7.10) \quad \sum_{n=1}^{\infty} \frac{a^{2n-1} - b^{2n-1}}{1 - (ab)^{2n-1}} = a \frac{(b/a; a^2b^2)_{\infty} (a^3b; a^2b^2)_{\infty} (a^2b^2; a^2b^2)_{\infty}^2}{(a^2; a^2b^2)_{\infty} (b^2; a^2b^2)_{\infty} (ab; a^2b^2)_{\infty}^2}.$$

Proof. We apply Corollary 1.3.14 with $q = ab$, $z = -1/a^2$, $\alpha = ab$, and $\beta = 1/(ab)$ to deduce that

$$(3.7.11) \quad \begin{aligned} & \frac{(b/a; a^2b^2)_{\infty} (a^3b; a^2b^2)_{\infty} (a^2b^2; a^2b^2)_{\infty}^2}{(a^2; a^2b^2)_{\infty} (b^2; a^2b^2)_{\infty} (ab; a^2b^2)_{\infty} (a^3b^3; a^2b^2)_{\infty}} \\ &= 1 + \sum_{n=1}^{\infty} \frac{(1/(ab); a^2b^2)_n}{(ab; a^2b^2)_n} b^{2n} + \sum_{n=1}^{\infty} \frac{(ab; a^2b^2)_n}{(a^3b^3; a^2b^2)_n} a^{2n} \\ &= 1 + \sum_{n=1}^{\infty} \frac{1 - 1/(ab)}{1 - (ab)^{2n-1}} b^{2n} + \sum_{n=1}^{\infty} \frac{1 - ab}{1 - (ab)^{2n+1}} a^{2n} \\ &= (1 - ab) \left(\frac{1}{1 - ab} + \sum_{n=2}^{\infty} \frac{a^{2n-2}}{1 - (ab)^{2n-1}} - \sum_{n=1}^{\infty} \frac{b^{2n-1}/a}{1 - (ab)^{2n-1}} \right). \end{aligned}$$

Multiplying the extremal sides of (3.7.11) by $a/(1 - ab)$, we complete the proof of (3.7.10). $\square$

Theorem 3.7.7. *If $\psi(q)$ is defined by (1.2.3), then*

$$(3.7.12) \quad q\psi(q^2)\psi(q^6) = \sum_{n=0}^{\infty} \left(\frac{q^{6n+1}}{1 - q^{12n+2}} - \frac{q^{6n+5}}{1 - q^{12n+10}} \right).$$

Exercise 3.7.8. *Use Theorem 3.7.7 to find a formula for the number of representations of a positive integer by the sum of a triangular number and three times a triangular number.*

Proof. In Lemma 3.7.6, put $a = q$ and $b = q^5$ to find that

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{q^{2n-1} - q^{5(2n-1)}}{1 - q^{6(2n-1)}} &= q \frac{(q^4; q^{12})_{\infty} (q^8; q^{12})_{\infty} (q^{12}; q^{12})_{\infty}^2}{(q^2; q^{12})_{\infty} (q^{10}; q^{12})_{\infty} (q^6; q^{12})_{\infty}^2} \\ (3.7.13) \quad &= q \frac{(q^4; q^4)_{\infty} (q^{12}; q^{12})_{\infty}}{(q^2; q^4)_{\infty} (q^6; q^{12})_{\infty}} = q\psi(q^2)\psi(q^6). \end{aligned}$$

It remains to simplify the left side of (3.7.13). Expand $1/(1 - q^{6(2n-1)})$ in a geometric series and invert the order of summation to deduce that

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{q^{2n-1} - q^{5(2n-1)}}{1 - q^{6(2n-1)}} &= \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \left(q^{(6m+1)(2n-1)} - q^{(6m+5)(2n-1)} \right) \\ (3.7.14) \quad &= \sum_{m=0}^{\infty} \left(\frac{q^{6m+1}}{1 - q^{12m+2}} - \frac{q^{6m+5}}{1 - q^{12m+10}} \right). \end{aligned}$$

Substituting (3.7.14) in (3.7.13), we immediately deduce (3.7.12) to complete the proof. $\square$

Theorem 3.7.9. Recall that $\varphi(q)$ and $\psi(q)$ are defined by (1.2.2) and (1.2.3), respectively. Then

$$(3.7.15) \quad \varphi(q)\varphi(q^3) + 4q\psi(q^2)\psi(q^6) = 1 + 6 \sum_{n=0}^{\infty} \left(\frac{q^{3n+1}}{1 - q^{3n+1}} - \frac{q^{3n+2}}{1 - q^{3n+2}} \right).$$

Proof. Using (3.7.12) and the trivial identity

$$\frac{x}{1 - x^2} = \frac{x}{1 - x} - \frac{x^2}{1 - x^2},$$

we easily find that

$$\begin{aligned} q\psi(q^2)\psi(q^6) &= \sum_{n=0}^{\infty} \left(\frac{q^{6n+1}}{1 - q^{6n+1}} - \frac{q^{12n+2}}{1 - q^{12n+2}} \right. \\ &\quad \left. - \frac{q^{6n+5}}{1 - q^{6n+5}} + \frac{q^{12n+10}}{1 - q^{12n+10}} \right) \\ &= \sum_{n=0}^{\infty} \left(\frac{q^{3n+1}}{1 - q^{3n+1}} - \frac{q^{3n+2}}{1 - q^{3n+2}} \right) \\ &\quad - \sum_{n=0}^{\infty} \left(\frac{q^{4(3n+1)}}{1 - q^{4(3n+1)}} - \frac{q^{4(3n+2)}}{1 - q^{4(3n+2)}} \right). \end{aligned} \quad (3.7.16)$$

Multiply the extremal sides of (3.7.16) by 4 and add the resulting equality to (3.7.5) to complete the proof. $\square$

Theorem 3.7.10.

$$(3.7.17) \quad \sum_{j,k=-\infty}^{\infty} q^{j^2+jk+k^2} = 1 + 6 \sum_{n=0}^{\infty} \left(\frac{q^{3n+1}}{1-q^{3n+1}} - \frac{q^{3n+2}}{1-q^{3n+2}} \right).$$

Proof. By Theorem 3.7.9, it suffices to show that

$$(3.7.18) \quad \sum_{j,k=-\infty}^{\infty} q^{j^2+jk+k^2} = \varphi(q)\varphi(q^3) + 4q\psi(q^2)\psi(q^6).$$

We begin by observing that

$$j^2 + jk + k^2 = \left(\frac{1}{2}j + k\right)^2 + 3\left(\frac{1}{2}j\right)^2,$$

so that if $j = 2n + 1$,

$$\left(\frac{1}{2}j + k\right)^2 + 3\left(\frac{1}{2}j\right)^2 = (n + k)(n + k + 1) + 3n(n + 1) + 1.$$

Thus,

$$\begin{aligned} \sum_{j,k=-\infty}^{\infty} q^{j^2+jk+k^2} &= \left(\sum_{\substack{j,k=-\infty \\ j \text{ even}}}^{\infty} + \sum_{\substack{j,k=-\infty \\ j \text{ odd}}}^{\infty} \right) q^{\left(\frac{1}{2}j+k\right)^2+3\left(\frac{1}{2}j\right)^2} \\ &= \sum_{n,k=-\infty}^{\infty} q^{(n+k)^2+3n^2} + \sum_{n,k=-\infty}^{\infty} q^{(n+k)(n+k+1)+3n(n+1)+1} \\ &= \sum_{m,n=-\infty}^{\infty} q^{m^2+3n^2} + \sum_{m,n=-\infty}^{\infty} q^{m(m+1)+3n(n+1)+1} \\ &= \varphi(q)\varphi(q^3) + 4q\psi(q^2)\psi(q^6), \end{aligned}$$

where we have used the bilateral representation of $f(q, q^3) = 2\psi(q)$ in (1.2.3). Now employ Theorem 3.7.9 on the far right side above. This therefore completes the proof. $\square$

Theorem 3.7.10 has the following immediate and beautiful arithmetic interpretation.

Theorem 3.7.11. *Let $r(n)$ denote the number of representations of the positive integer n by the quadratic form $j^2 + jk + k^2$, where different*

signs of j and k yield distinct representations of n . Then for each positive integer n ,

$$(3.7.19) \quad r(n) = 6(d_{1,3}(n) - d_{2,3}(n)),$$

where $d_{j,k}(n)$ is defined by (3.2.3).

Proof. Return to (3.7.17) and expand the summands on the right side into geometric series. Hence,

$$(3.7.20) \quad \sum_{j,k=-\infty}^{\infty} q^{j^2+jk+k^2} = 1 + 6 \sum_{m=0}^{\infty} \sum_{r=1}^{\infty} (q^{(3m+1)r} - q^{(3m+2)r}).$$

Now equate coefficients of q^n , $n \geq 1$, on both sides of (3.7.20), and the desired formula (3.7.19) follows. $\square$

3.8. Notes

The formulas for $r_2(n)$, $r_4(n)$, $r_6(n)$, and $r_8(n)$ given in Theorems 3.2.1, 3.3.1, 3.4.1, and 3.5.4, respectively, are due to Jacobi [131]. All four theorems can be proved arithmetically. For example, see Hardy and Wright's book [112, pp. 241–242] for an elementary proof of Theorem 3.2.1. In Chapter 6, we return to these problems and give completely different proofs of Jacobi's formulas for $r_4(n)$ and $r_8(n)$ arising out of Ramanujan's theory of elliptic functions.

The first proof of Theorem 3.2.1 that we have given is due to M. D. Hirschhorn [115]. In a later paper [119], he gave another simple proof based on Jacobi's identity, Theorem 1.3.9. One year later, Hirschhorn [122] showed that one can derive further theorems about $r_2(n)$ from Theorem 3.2.1. The first proof of Theorem 3.3.1 that we have given is also due to Hirschhorn [116], who earlier [114] had given another proof. S. Bhargava and C. Adiga [58] have also utilized the ${}_1\psi_1$ summation formula to give simple proofs of Theorems 3.2.1 and 3.3.1. Starting with Theorem 3.2.1 for two squares, B. K. Spearman and K. S. Williams [210] provided an elementary arithmetic proof of Theorem 3.3.1 for four squares. For a proof of Theorem 3.3.1 based on a recurrence and computer algebra, see a paper by G. E. Andrews, S. B. Ekhad, and D. Zeilberger [22]. Our proof of Theorem 3.4.1 is

taken from a paper by S. H. Chan [72]. Chan used the same idea to also establish a formula for $t_6(n)$, namely,

$$(3.8.1) \quad t_6(n) = \frac{1}{8} \sum_{d|4n+3} (-1)^{(d+1)/2} d^2.$$

We are uncertain who first proved (3.8.1), but an equivalent formulation can be found in an unpublished manuscript of Ramanujan [194, p. 356], [19, pp. 398–401]. The first published proof known to us is by K. Ono, S. Robins, and P. T. Wahl [177, Theorem 4]; this proof uses the theory of modular forms. An arithmetic proof is given in [128, p. 262, Theorem 11]. S. Cooper and H. Y. Lam [80] also employed the ${}_1\psi_1$ summation formula to prove Theorem 3.4.1. The proof of Theorem 3.5.4 that we have given is due to J.-F. Lin [144], who in another paper [145] gave a variant of his proof. As we have seen, Lin's primary idea in [144] is based on squaring a particular Lambert series. This is the same idea used by Ramanujan in his proof of the fundamental identity (4.2.6) for Eisenstein series in Chapter 4. This idea was also employed and generalized by K. Venkatachaliengar [214, p. 37] and elaborated upon in more detail by Cooper [76, Theorem 2.2]. The theorem of Venkatachaliengar is beautiful, and we present Cooper's formulation. Define, for $|q| < 1$ and $a \neq q^{2k}$, where k is an integer,

$$F(a, t) := \sum_{n=-\infty}^{\infty} \frac{t^n}{1 - aq^{2n}}$$

and

$$\rho_1(z) := \frac{1}{2} + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{z^n}{1 - q^{2n}}.$$

Then

$$(3.8.2) \quad F(a, t)F(b, t) = t \frac{\partial}{\partial t} F(ab, t) + F(ab, t)(\rho_1(a) + \rho_1(b)).$$

Cooper [76] offers several formulas for the number of integers represented by an even number of squares or an even number of triangular numbers. Cooper and Lam in [80] use Ramanujan's ${}_1\psi_1$ summation formula and Venkatachaliengar's fundamental identity (3.8.2) to provide proofs of formulas for sums of k squares and sums of k triangular

numbers, for $k = 2, 4, 6, 8$. For completeness, the formula for $t_8(n)$ is given by

$$(3.8.3) \quad t_8(n) = \sum_{\substack{d|(n+1) \\ d \text{ odd}}} \left(\frac{n+1}{d} \right)^3.$$

A. M. Legendre [139, p. 133] was the first to prove (3.8.3). An analytic equivalent of (3.8.3) can be found in Ramanujan's lost notebook [194], [19, p. 401]. Ono, Robins, and Wahl [177, p. 82] have established (3.8.3) using the theory of modular forms. An arithmetic proof of (3.8.3) can be found in [128]. We provide a proof of (3.8.3) in Chapter 6; see Theorem 6.2.12.

The connection between formulas for sums of squares and sums of triangular numbers was further solidified by P. Barrucand, Cooper, and Hirschhorn [32], who proved that

$$r_k(8n+k) = c_k t_k(n), \quad \text{where} \quad c_k = 2^k \left(1 + \frac{1}{2} \binom{k}{4} \right), \quad 1 \leq k \leq 7.$$

Thus, the study of $t_k(n)$, for $1 \leq k \leq 7$, is reduced to the study of the subsequence $r_k(8n+k)$ of $r_k(n)$.

Another elegant approach was given by L. Carlitz [62]. He employed a beautiful formula due to W. N. Bailey [31], namely,

$$(3.8.4) \quad \sum_{n=-\infty}^{\infty} \left\{ \frac{xq^n}{(1-xq^n)^2} - \frac{yq^n}{(1-yq^n)^2} \right\} \\ = \frac{(xyq)_{\infty} (q/(xy))_{\infty} (xq/y)_{\infty} (yq/x)_{\infty} (q)_4^{\infty}}{(xq)_{\infty}^2 (q/x)_{\infty}^2 (yq)_{\infty}^2 (q/y)_{\infty}^2},$$

to give proofs of formulas for $r_4(n)$ and $r_6(n)$. An equivalent formulation of (3.8.4) can be found in N. J. Fine's book [94, p. 22, eq. (18.85)]. Bailey's proof [31] of (3.8.4) employs the Weierstrass $\wp$ -function from the theory of elliptic functions. Shortly thereafter, J. M. Dobbie [83] gave a shorter, more elementary proof of (3.8.4).

Williams gave an arithmetic proof of Theorem 3.5.4 based on an extension of an identity of J. Liouville [225].

Ramanujan's lost notebook [194, pp. 353–355] contains a fragment providing many of Ramanujan's theorems on Lambert series.

This fragment has been examined by Berndt [39], with the arithmetical consequences of Ramanujan's Lambert series identities also discussed by him. See also Berndt's book with Andrews [19, Chapter 17].

Andrews [13] used the theory of basic hypergeometric series to give a uniform approach to proving Jacobi's formulas for $r_{2k}(n)$, $1 \leq k \leq 4$.

Formulas for $r_k(n)$, when k is odd, have an entirely different flavor. Gauss found a formula for $r_3(n)$, which was put in a more concrete form by G. Eisenstein [88], [90, p. 505], who also gave an analogous formula for $r_5(n)$. Chapter 4 in E. Grosswald's book [101] is devoted to the study of $r_3(n)$, while pages 128 and 129 in the same text provide information about $r_5(n)$. For recent work on formulas for $r_k(n)$, when k is odd, see papers by Cooper [77], [78].

Theorem 3.6.2 is due to Jacobi. The first proof of Theorem 3.6.3 was found by Legendre [139, p. 133]. Further proofs were given by Cauchy [65, p. 572], [66, p. 64] and Plana [179, p. 147]. For an elementary proof based on an extension of an identity of Liouville, see a paper by J. G. Huard, Z. M. Ou, B. K. Spearman, and Williams [128]. Jacobi [131] claimed that V. Y. Bouniakowsky first proved Theorem 3.6.3, but he did not give a reference. Lin [143] derived a formula for the number of representations of a positive integer by 8 triangular numbers. The analogue of Lagrange's theorem is the theorem of Gauss [98, p. 497] stating that every positive integer can be represented as a sum of three triangular numbers. An elegant proof via q -series has been given by Andrews [16]. Williams [226] derived an elegant formula for the number of representations of a positive integer as a sum of two triangular numbers plus twice the sum of two triangular numbers. Liouville [146], [147] determined the positive integers a , b , and c , $1 \leq a \leq b \leq c$, such that the polynomial $a\Delta_1 + b\Delta_2 + c\Delta_3$, where Δ_1 , Δ_2 , and Δ_3 denote triangular numbers, represents all positive integers. They are

$$(a, b, c) = (1, 1, 1), (1, 1, 2), (1, 1, 4), (1, 1, 5), \\ (1, 2, 2), (1, 2, 3), (1, 2, 4).$$

For an exposition of Liouville's methods giving formulas for $r_{2k}(n)$, $1 \leq k \leq 5$, see Chapter 14 in M. B. Nathanson's text [167].

Although not explicitly stated by him, Theorem 3.7.5 is due to P. G. L. Dirichlet [82]. The first explicit statement of Theorem 3.7.5, however, is due to L. Lorenz in 1871 [152]. The proof of Theorem 3.7.5 that we have given was independently given by Berndt [35] and by Bhargava, Adiga, and D. D. Somashekara [59]. R. A. Askey [27] used Ramanujan's ${}_1\psi_1$ summation theorem to derive a formula for the number of ways a positive integer can be represented by a square and twice a square. Although not stated explicitly by him, Theorem 3.7.11 is also due to Dirichlet [82], who proved a general theorem for representations of integers by binary quadratic forms. Thus, every theorem in this chapter concerning representations of integers as sums of squares is, in fact, contained in Dirichlet's general theorem. Theorem 3.7.4 has been enormously generalized by Williams [224], who derived a representation for certain sums of the products $\varphi(q^a)\varphi(q^b)$ in terms of Lambert series. The analytic formulation of Theorem 3.7.11, that is, Theorem 3.7.10, can be found in a letter from Ramanujan to Hardy written on a Saturday, probably in 1918, from the nursing home, Fitzroy House [194, pp. 93–96], [51, pp. 196–198]. This letter is examined in detail in Berndt's paper [35], and indeed our proof of Theorem 3.7.10 has been taken from [35]. Hirschhorn [121] has used a general method involving partial fractions to give uniform proofs of Theorems 3.2.1, 3.3.1, and 3.7.5.

Equality (3.2.8), equality (3.3.12), Theorem 3.7.2, and Theorem 3.7.4 are given by Ramanujan in Entry 8(i)–(iv), respectively, in Chapter 17 of his second notebook [193], [34, p. 114]. Theorems 3.5.3, 3.6.2, and 3.6.3, or equivalent formulations, are given by Ramanujan as examples in Section 17 of Chapter 17 in his second notebook [193], [34, p. 139].

We have concentrated our remarks on $r_{2k}(n)$, for $k \leq 4$. Most readers by now have probably asked, "Are there formulas for $k > 4$?" Formulas for $k > 4$ are more complicated than those for $k \leq 4$. For those who have some familiarity with modular forms, we remark that the generating function $\varphi^{2k}(q)$ for $r_{2k}(n)$ is a modular form of weight k . For $k \leq 4$, the dimension of the space of modular forms in which

$\varphi^{2k}(q)$ lies is equal to 1. For $k > 4$, the dimension exceeds one, and so other modular forms play a role in formulas for $r_{2k}(n)$, making these formulas more complicated. In the latter part of the nineteenth century and the early part of the twentieth century, J. W. L. Glaisher wrote several papers on sums of squares. In particular, in [99], he derived formulas for $r_{2k}(n)$, $5 \leq k \leq 9$. In his paper [186], Ramanujan stated without proof a general formula for $r_{2k}(n)$, which was later proved by Mordell [162]. However, Ramanujan worked out the details for $1 \leq k \leq 12$ and recorded them in Table VI [186], [192, p. 159]. Using a modular transformation that connects $\varphi^{2k}(q)$ with the generating function $\psi^{2k}(q)$ of $t_{2k}(n)$, Ramanujan [187], [192, pp. 179–199] derived corresponding formulas for $t_{2k}(n)$, although he did not give them explicitly. In her doctoral thesis [183], V. Ramamani gave new proofs of Ramanujan's formulas for $r_{2k}(n)$, $k \leq 12$, and corrected two numerical errors in Ramanujan's formula for $r_{22}(n)$ given in his Table VI. A new class of formulas involving products of certain Eisenstein series has been conjectured by H. H. Chan and K. S. Chua [69].

The theorems of Jacobi have been enormously generalized by S. C. Milne [156]–[158], who also provides the most extensive literature survey known on formulas for sums of squares. K. Ono [176] and L. Long and Y. Yang [150] have given further proofs of some of Milne's results.

Elementary texts in number theory containing material related to that given in this chapter include those of Hardy and Wright [112, Chapter 20] and L. K. Hua [127, pp. 115–120, 204–210, 307–309]. The most extensive text on sums of squares is by Grosswald [101].

Chapter 4

Eisenstein Series

4.1. Bernoulli Numbers and Eisenstein Series

Appearing in the definition of the classical Eisenstein series are Bernoulli numbers, which we define first.

Definition 4.1.1. *The Bernoulli numbers $B_n, n \geq 0$, are defined by the generating function*

$$(4.1.1) \quad \frac{z}{e^z - 1} =: \sum_{n=0}^{\infty} B_n \frac{z^n}{n!}, \quad |z| < 2\pi.$$

(Note that the function $z/(e^z - 1)$ is analytic at $z = 0$, and its nearest singularities to 0 are $\pm 2\pi i$; thus, the series on the right side of (4.1.1) converges for $|z| < 2\pi$.) In particular,

$$B_0 = 1, \quad B_1 = -\frac{1}{2}, \quad B_2 = \frac{1}{6}, \quad B_3 = 0, \quad B_4 = -\frac{1}{30},$$

$$B_5 = 0, \quad B_6 = \frac{1}{42}, \quad B_7 = 0, \quad B_8 = -\frac{1}{30}, \quad B_9 = 0, \quad B_{10} = \frac{5}{66}.$$

It appears that $B_n = 0$ when n is odd and greater than 1. This is easy to prove.

Theorem 4.1.2. *For $n \geq 1$,*

$$B_{2n+1} = 0.$$

Proof. Define

$$(4.1.2) \quad f(z) := \frac{z(e^z + 1)}{2(e^z - 1)} = \frac{z}{e^z - 1} + \frac{1}{2}z = 1 + \sum_{n=2}^{\infty} B_n \frac{z^n}{n!}.$$

It is easily checked that

$$f(-z) = f(z),$$

i.e., $f(z)$ is an even function. Hence, from (4.1.2), we conclude that $B_n = 0$, when n is an odd integer at least equal to 3. $\square$

Theorem 4.1.3. For $|x| < 2\pi$,

$$\frac{x}{2} \cot \frac{x}{2} = \sum_{n=0}^{\infty} \frac{(-1)^n B_{2n}}{(2n)!} x^{2n}.$$

Proof. Using (4.1.1) with $z = -ix$, we find that

$$\begin{aligned} \frac{x}{2} \cot \frac{x}{2} &= -\frac{ix}{2} - \frac{ix}{e^{-ix} - 1} \\ &= -\frac{ix}{2} + \sum_{m=0}^{\infty} B_m \frac{(-ix)^m}{m!} \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n B_{2n}}{(2n)!} x^{2n}, \end{aligned}$$

because $B_1 = -\frac{1}{2}$ and $B_{2n+1} = 0, n \geq 1$, by Theorem 4.1.2. $\square$

Eisenstein series are special cases of Ramanujan's more general function

$$(4.1.3) \quad \Phi_{r,s}(q) := \sum_{k,n=1}^{\infty} k^r n^s q^{kn},$$

where r and s are nonnegative integers. Trivially, $\Phi_{r,s}(q) = \Phi_{s,r}(q)$. For our purposes, the most important special cases are when $r = 0$ and s is an odd positive integer, and when $r = 1$ and s is an even positive integer.

Definition 4.1.4. For each positive integer r , define

$$(4.1.4) \quad S_r := -\frac{B_{r+1}}{2(r+1)} + \Phi_{0,r}(q) = -\frac{B_{r+1}}{2(r+1)} + \sum_{k=1}^{\infty} \frac{k^r q^k}{1 - q^k}.$$

Definition 4.1.5. The classical Eisenstein series $E_{2n}(q)$, where n is a positive integer, are defined by

$$(4.1.5) \quad E_{2n}(q) := 1 - \frac{4n}{B_{2n}} \sum_{k=1}^{\infty} \frac{k^{2n-1} q^k}{1 - q^k}.$$

Thus,

$$(4.1.6) \quad E_{2n}(q) = -\frac{4n}{B_{2n}} S_{2n-1}, \quad n \geq 1.$$

(Our notation here is slightly different from the classical notation; normally, $E_{2n}(\tau)$ is defined by the right side of (4.1.5) when $q = e^{2\pi i \tau}$ and $\text{Im } \tau > 0$.) In Ramanujan's notation, the three most important special cases of (4.1.5) are

$$(4.1.7) \quad P = P(q) := E_2(q) = 1 - 24 \sum_{k=1}^{\infty} \frac{k q^k}{1 - q^k} = -24S_1,$$

$$(4.1.8) \quad Q = Q(q) := E_4(q) = 1 + 240 \sum_{k=1}^{\infty} \frac{k^3 q^k}{1 - q^k} = 240S_3,$$

$$(4.1.9) \quad R = R(q) := E_6(q) = 1 - 504 \sum_{k=1}^{\infty} \frac{k^5 q^k}{1 - q^k} = -504S_5.$$

Our first goal is to derive two recurrence relations for S_{2n+1} , the first of which also involves $\Phi_{1,2n}$.

4.2. Trigonometric Series

Our goal in this section is to present Ramanujan's elementary method [186], [192, pp. 136–162] for deriving recurrence relations for Eisenstein series.

We begin with an exercise.

Exercise 4.2.1. Verify (for example, by summing finite geometric series) the elementary identity

$$(4.2.1) \quad 1 + 2 \sum_{n=1}^{m-1} \cos(n\theta) + \cos(m\theta) = \cot\left(\frac{1}{2}\theta\right) \sin(m\theta),$$

valid for each positive integer m .

We use (4.2.1) when we square the left side of (4.2.2) below and collect terms. Note that the left side of (4.2.2) minus $(\frac{1}{4} \cot \frac{1}{2} \theta)^2$ is an even, continuously differentiable, periodic function of θ and so can be expressed as a Fourier cosine series. Accordingly, we write

$$(4.2.2) \quad \left(\frac{1}{4} \cot \frac{1}{2} \theta + \sum_{k=1}^{\infty} \frac{q^k \sin(k\theta)}{1 - q^k} \right)^2 = \left(\frac{1}{4} \cot \frac{1}{2} \theta \right)^2 + \sum_{n=0}^{\infty} C_n \cos(n\theta),$$

where the Fourier coefficients C_n depend upon q but not upon θ . Squaring the left side of (4.2.2), using (4.2.1) for the terms involving $\cot(\frac{1}{2}\theta) \sin(k\theta)$, and using the identity $\sin^2(k\theta) = \frac{1}{2}(1 - \cos(2k\theta))$, we find that

$$(4.2.3) \quad \begin{aligned} C_0 &= \frac{1}{2} \sum_{k=1}^{\infty} \frac{q^k}{1 - q^k} + \frac{1}{2} \sum_{k=1}^{\infty} \left(\frac{q^k}{1 - q^k} \right)^2 \\ &= \frac{1}{2} \sum_{k=1}^{\infty} \frac{q^k}{(1 - q^k)^2} \\ &= \frac{1}{2} \sum_{k=1}^{\infty} \sum_{m=1}^{\infty} m q^{mk} \\ &= \frac{1}{2} \sum_{m=1}^{\infty} \frac{m q^m}{1 - q^m}. \end{aligned}$$

To calculate C_n , $n \geq 1$, we need to be careful. First observe that we obtain contributions from

$$\begin{aligned} &\frac{q^j}{1 - q^j} \frac{q^k}{1 - q^k} \sin(j\theta) \sin(k\theta) \\ &= \frac{q^j}{1 - q^j} \frac{q^k}{1 - q^k} \frac{1}{2} (\cos(j - k)\theta - \cos(j + k)\theta), \end{aligned}$$

for $j, k \geq 1$. We also gain contributions from the terms $\cot(\frac{1}{2}\theta) \sin(m\theta)$, for $m \geq n$. Thus, we find that, for $n \geq 1$,

$$(4.2.4) \quad \begin{aligned} C_n &= \frac{1}{2} \frac{q^n}{1 - q^n} + \sum_{k=1}^{\infty} \frac{q^{n+k}}{1 - q^{n+k}} + \sum_{k=1}^{\infty} \frac{q^k}{1 - q^k} \frac{q^{n+k}}{1 - q^{n+k}} \\ &\quad - \frac{1}{2} \sum_{k=1}^{n-1} \frac{q^k}{1 - q^k} \frac{q^{n-k}}{1 - q^{n-k}}. \end{aligned}$$

Using the easily verified elementary identities

$$\frac{q^{n+k}}{1-q^{n+k}} \left(1 + \frac{q^k}{1-q^k}\right) = \frac{q^n}{1-q^n} \left(\frac{q^k}{1-q^k} - \frac{q^{n+k}}{1-q^{n+k}}\right)$$

and

$$\frac{q^n}{(1-q^k)(1-q^{n-k})} = \frac{q^n}{1-q^n} \left(1 + \frac{q^k}{1-q^k} + \frac{q^{n-k}}{1-q^{n-k}}\right)$$

in (4.2.4), we find that

$$\begin{aligned} C_n &= \frac{q^n}{2(1-q^n)} + \frac{q^n}{1-q^n} \sum_{k=1}^{\infty} \left(\frac{q^k}{1-q^k} - \frac{q^{n+k}}{1-q^{n+k}}\right) \\ &\quad - \frac{q^n}{2(1-q^n)} \sum_{k=1}^{n-1} \left(1 + \frac{q^k}{1-q^k} + \frac{q^{n-k}}{1-q^{n-k}}\right) \\ &= \frac{q^n}{2(1-q^n)} + \left(\frac{q^n}{1-q^n}\right)^2 - \frac{(n-1)q^n}{2(1-q^n)} \\ (4.2.5) \quad &= \frac{q^n}{1-q^n} \left(\frac{1}{1-q^n} - \frac{n}{2}\right). \end{aligned}$$

Substituting (4.2.3) and (4.2.5) in (4.2.2), we find that

$$\begin{aligned} \left(\frac{1}{4} \cot \frac{1}{2}\theta + \sum_{k=1}^{\infty} \frac{q^k \sin(k\theta)}{1-q^k}\right)^2 &= \left(\frac{1}{4} \cot \frac{1}{2}\theta\right)^2 + \sum_{k=1}^{\infty} \frac{q^k \cos(k\theta)}{(1-q^k)^2} \\ (4.2.6) \quad &\quad + \frac{1}{2} \sum_{k=1}^{\infty} \frac{kq^k}{1-q^k} (1 - \cos(k\theta)). \end{aligned}$$

This is the first of the two primary trigonometric series identities that we need. Using (4.2.6), we establish a recurrence formula for the functions S_r . Appearing in the recurrence relation are the functions from (4.1.3)

$$(4.2.7) \quad \Phi_{1,2n}(q) = \sum_{m,k=1}^{\infty} mk^{2n} q^{mk} = \sum_{k=1}^{\infty} \frac{k^{2n} q^k}{(1-q^k)^2}.$$

Theorem 4.2.2. For $n \geq 1$,

$$(4.2.8) \quad \frac{(2n+3)}{2(2n+1)} S_{2n+1} - \Phi_{1,2n} = \sum_{k=1}^n \binom{2n}{2k-1} S_{2k-1} S_{2n-2k+1}.$$

Proof. From the elementary identity

$$(4.2.9) \quad \cot^2 \theta = - \left(1 + \frac{d}{d\theta} \cot \theta \right)$$

and Theorem 4.1.3, we find that

$$(4.2.10) \quad \frac{1}{16} \cot^2 \frac{1}{2} \theta = \frac{1}{4\theta^2} - \frac{1}{24} + \frac{1}{4} \sum_{n=1}^{\infty} \frac{(-1)^n B_{2n+2} (2n+1)}{(2n+2)!} \theta^{2n}.$$

Using Theorem 4.1.3, (4.2.10), and the Maclaurin series for $\sin(k\theta)$ and $\cos(k\theta)$ in (4.2.6), we deduce that

$$(4.2.11) \quad \left(\frac{1}{2\theta} + \frac{1}{2} \sum_{n=1}^{\infty} \frac{(-1)^n B_{2n}}{(2n)!} \theta^{2n-1} + \sum_{k=1}^{\infty} \frac{q^k}{1-q^k} \sum_{n=0}^{\infty} \frac{(-1)^n k^{2n+1}}{(2n+1)!} \theta^{2n+1} \right)^2 \\ = \frac{1}{4\theta^2} - \frac{1}{24} + \frac{1}{4} \sum_{n=1}^{\infty} \frac{(-1)^n B_{2n+2}}{(2n)!(2n+2)} \theta^{2n} \\ + \sum_{k=1}^{\infty} \frac{q^k}{(1-q^k)^2} \sum_{n=0}^{\infty} \frac{(-1)^n k^{2n}}{(2n)!} \theta^{2n} \\ + \frac{1}{2} \sum_{k=1}^{\infty} \frac{kq^k}{1-q^k} \sum_{n=1}^{\infty} \frac{(-1)^{n-1} k^{2n}}{(2n)!} \theta^{2n}.$$

Collecting coefficients of like powers of θ in (4.2.11) and recalling the representations of S_r and $\Phi_{r,s}(q)$ given in (4.1.4) and (4.2.7), respectively, we find that

$$\left(\frac{1}{2\theta} + \frac{S_1}{1!} \theta - \frac{S_3}{3!} \theta^3 + \frac{S_5}{5!} \theta^5 - \dots \right)^2 \\ = \frac{1}{4\theta^2} + S_1 - \frac{\Phi_{1,2}}{2!} \theta^2 + \frac{\Phi_{1,4}}{4!} \theta^4 - \dots \\ + \frac{1}{2} \left(\frac{S_3}{2!} \theta^2 - \frac{S_5}{4!} \theta^4 + \frac{S_7}{6!} \theta^6 - \dots \right).$$

Equating coefficients of θ^{2n} , $n \geq 1$, on both sides above, we find that

$$\begin{aligned} & \frac{(-1)^{n-1} S_{2n+1}}{2} + \frac{(-1)^n \Phi_{1,2n}}{(2n)!} \\ &= (-1)^{n-1} \left(\frac{S_1}{1!} \frac{S_{2n-1}}{(2n-1)!} + \frac{S_3}{3!} \frac{S_{2n-3}}{(2n-3)!} + \cdots + \frac{S_{2n-1}}{(2n-1)!} \frac{S_1}{1!} \right) \\ & \quad + \frac{(-1)^n S_{2n+1}}{(2n+1)!}, \end{aligned}$$

or, upon simplification,

$$\frac{(2n+3)}{2(2n+1)} S_{2n+1} - \Phi_{1,2n} = \sum_{k=1}^n \binom{2n}{2k-1} S_{2k-1} S_{2n-2k+1},$$

which is (4.2.8). $\square$

Our next task is to use Theorem 4.2.2 to derive the three important first order differential equations satisfied by P , Q , and R . From (4.1.5),

$$\begin{aligned} q \frac{dE_{2n}(q)}{dq} &= q \frac{d}{dq} \left(1 - \frac{4n}{B_{2n}} \sum_{k=1}^{\infty} \frac{k^{2n-1} q^k}{1 - q^k} \right) \\ &= -\frac{4n}{B_{2n}} q \sum_{k=1}^{\infty} \left(\frac{k^{2n} q^{k-1}}{1 - q^k} + \frac{k^{2n} q^{2k-1}}{(1 - q^k)^2} \right) \\ &= -\frac{4n}{B_{2n}} \sum_{k=1}^{\infty} \frac{k^{2n} q^k}{(1 - q^k)^2} \\ (4.2.12) \quad &= -\frac{4n}{B_{2n}} \Phi_{1,2n}(q), \end{aligned}$$

by (4.2.7). In particular, from our table of Bernoulli numbers at the beginning of this chapter and the definitions (4.1.7)–(4.1.9),

$$(4.2.13) \quad q \frac{dP}{dq} = -24 \Phi_{1,2}(q),$$

$$(4.2.14) \quad q \frac{dQ}{dq} = 240 \Phi_{1,4}(q),$$

$$(4.2.15) \quad q \frac{dR}{dq} = -504 \Phi_{1,6}(q).$$

Putting $n = 1$ in (4.2.8), we find that

$$(4.2.16) \quad \frac{5}{6}S_3 - \Phi_{1,2} = 2S_1^2,$$

or

$$(4.2.17) \quad 288\Phi_{1,2} = Q - P^2;$$

putting $n = 2$ in (4.2.8), we find that

$$\frac{7}{10}S_5 - \Phi_{1,4} = 8S_1S_3,$$

or

$$(4.2.18) \quad 720\Phi_{1,4} = PQ - R;$$

and putting $n = 3$ in (4.2.8), we find that

$$\frac{9}{14}S_7 - \Phi_{1,6} = 12S_1S_5 + 20S_3^2,$$

or, with the use of the identity $Q^2 = 480S_7$, which we shall prove below in (4.2.38),

$$(4.2.19) \quad 1008\Phi_{1,6} = Q^2 - PR.$$

Hence, putting (4.2.17)–(4.2.19) in (4.2.13)–(4.2.15), respectively, we deduce the following important theorem.

Theorem 4.2.3. *We have*

$$(4.2.20) \quad q \frac{dP}{dq} = \frac{P^2 - Q}{12},$$

$$(4.2.21) \quad q \frac{dQ}{dq} = \frac{PQ - R}{3},$$

$$(4.2.22) \quad q \frac{dR}{dq} = \frac{PR - Q^2}{2}.$$

The next theorem is also of enormous importance.

Theorem 4.2.4. *We have*

$$(4.2.23) \quad Q^3(q) - R^2(q) = 1728q(q; q)_\infty^{24}.$$

Proof. By a straightforward use of logarithmic differentiation, we easily find that

$$q \frac{d}{dq} \log (q(q; q)_\infty^{24}) = P(q).$$

On the other hand, using straightforward differentiation along with Theorem 4.2.3, we find that

$$q \frac{d}{dq} \log (Q^3(q) - R^2(q)) = P(q).$$

The last two equalities imply that

$$\frac{d}{dq} \log (q(q; q)_{\infty}^{24}) = \frac{d}{dq} \log (Q^3(q) - R^2(q)),$$

or

$$(4.2.24) \quad q(q; q)_{\infty}^{24} = c (Q^3(q) - R^2(q)),$$

for some constant c . Equating coefficients of q on each side of (4.2.24), we deduce that

$$1 = c(3 \cdot 240 + 2 \cdot 504) = 1728c,$$

and so with the value $c = 1/1728$, Theorem 4.2.4 follows from (4.2.24). $\square$

To obtain a second recurrence relation for S_r , we begin with analogues of (4.2.1) and (4.2.6).

Exercise 4.2.5. *First, prove that*

$$(4.2.25) \quad \cot^2\left(\frac{1}{2}\theta\right) (1 - \cos(n\theta)) \\ = (2n - 1) + 4 \sum_{k=1}^{n-1} (n - k) \cos(k\theta) + \cos(n\theta).$$

Exercise 4.2.6. *(This is a very difficult exercise.) Second, using (4.2.25), prove that*

$$(4.2.26) \quad L(\theta) := \left(\frac{1}{8} \cot^2 \frac{1}{2} \theta + \frac{1}{12} + \sum_{k=1}^{\infty} \frac{kq^k}{1-q^k} (1 - \cos(k\theta)) \right)^2 \\ = \left(\frac{1}{8} \cot^2 \frac{1}{2} \theta + \frac{1}{12} \right)^2 + \frac{1}{12} \sum_{k=1}^{\infty} \frac{k^3 q^k}{1-q^k} (5 + \cos(k\theta)) =: R(\theta).$$

Theorem 4.2.7. *For each positive integer n ,*

$$(4.2.27) \quad \frac{(n-1)(2n+5)}{12(n+1)(2n+1)} S_{2n+3} = \sum_{k=1}^{n-1} \binom{2n}{2k} S_{2k+1} S_{2n-2k+1}.$$

Proof. Using (4.2.10), we find that

$$(4.2.28) \quad \frac{1}{8} \cot^2 \frac{1}{2} \theta + \frac{1}{12} = \frac{1}{2\theta^2} + \frac{1}{2} \sum_{n=1}^{\infty} \frac{(-1)^n B_{2n+2} (2n+1)}{(2n+2)!} \theta^{2n}.$$

Using (4.2.28) and the Maclaurin series for $\cos(k\theta)$, we find that the left side of (4.2.26) has the form

$$(4.2.29) \quad L(\theta) = \left(\frac{1}{2\theta^2} + \frac{1}{2} \sum_{n=1}^{\infty} \frac{(-1)^n B_{2n+2}}{(2n+2)(2n)!} \theta^{2n} + \sum_{k=1}^{\infty} \frac{kq^k}{1-q^k} \sum_{n=1}^{\infty} \frac{(-1)^{n+1} k^{2n}}{(2n)!} \theta^{2n} \right)^2.$$

For $n \geq 1$, the coefficient of θ^{2n} within parentheses above is equal to

$$\frac{(-1)^n B_{2n+2}}{2(2n+2)(2n)!} + \frac{(-1)^{n+1}}{(2n)!} \sum_{k=1}^{\infty} \frac{k^{2n+1} q^k}{1-q^k} = \frac{(-1)^{n+1}}{(2n)!} S_{2n+1},$$

by (4.1.4). Thus, the left side of (4.2.26), i.e., (4.2.29), is equal to

$$(4.2.30) \quad L(\theta) = \left(\frac{1}{2\theta^2} + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n)!} S_{2n+1} \theta^{2n} \right)^2.$$

Returning to (4.2.9), we find that

$$\begin{aligned} \frac{d^2}{d\theta^2} \cot \theta &= 2 \cot \theta \csc^2 \theta, \\ \frac{d^3}{d\theta^3} \cot \theta &= -2 - 8 \cot^2 \theta - 6 \cot^4 \theta \\ &= 6 + 8 \frac{d}{d\theta} \cot \theta - 6 \cot^4 \theta, \end{aligned}$$

or

$$(4.2.31) \quad \cot^4 \theta = 1 + \frac{4}{3} \frac{d}{d\theta} \cot \theta - \frac{1}{6} \frac{d^3}{d\theta^3} \cot \theta.$$

Thus, by (4.2.31), (4.2.9), and Theorem 4.1.3,

(4.2.32)

$$\begin{aligned}
 & \frac{1}{64} \cot^4 \theta + \frac{1}{48} \cot^2 \theta + \frac{1}{144} \\
 &= \frac{1}{64} \left(1 + \frac{4}{3} \frac{d}{d\theta} \cot \theta - \frac{1}{6} \frac{d^3}{d\theta^3} \cot \theta \right) + \frac{1}{48} \left(-1 - \frac{d}{d\theta} \cot \theta \right) + \frac{1}{144} \\
 &= \frac{1}{2^6 \cdot 3^2} - \frac{1}{2^7 \cdot 3} \frac{d^3}{d\theta^3} \cot \theta \\
 &= \frac{1}{2^6 \cdot 3^2} - \frac{1}{2^7 \cdot 3} \sum_{n=0}^{\infty} \frac{(-1)^n B_{2n} 2^{2n} (2n-1)(2n-2)(2n-3)}{(2n)!} \theta^{2n-4} \\
 &= \frac{1}{2^6 \cdot 3^2} - \frac{1}{2^7 \cdot 3} \left(-\frac{6}{\theta^4} - \frac{2}{15} + \sum_{n=1}^{\infty} \frac{(-1)^n B_{2n+4} 2^{2n+4}}{(2n+4)(2n)!} \theta^{2n} \right) \\
 &= \frac{1}{2^6 \theta^4} + \frac{1}{2^5 \cdot 3 \cdot 5} - \frac{1}{2^3 \cdot 3} \sum_{n=1}^{\infty} \frac{(-1)^n B_{2n+4} 2^{2n}}{(2n+4)(2n)!} \theta^{2n},
 \end{aligned}$$

where in the penultimate line, we used the fact that $B_4 = -\frac{1}{30}$. Using (4.2.32) with θ replaced by $\frac{1}{2}\theta$ and the Maclaurin series for $\cos(k\theta)$, we find that the right side of (4.2.26) is equal to

$$\begin{aligned}
 (4.2.33) \quad R(\theta) &= \frac{1}{4\theta^4} + \frac{1}{2^5 \cdot 3 \cdot 5} - \frac{1}{2^3 \cdot 3} \sum_{n=1}^{\infty} \frac{(-1)^n B_{2n+4} 2^{2n}}{(2n+4)(2n)!} \theta^{2n} \\
 &\quad + \frac{1}{12} \sum_{k=1}^{\infty} \frac{k^3 q^k}{1-q^k} \left(6 + \sum_{n=1}^{\infty} \frac{(-1)^n k^{2n}}{(2n)!} \theta^{2n} \right).
 \end{aligned}$$

The constant term in (4.2.33) equals

$$(4.2.34) \quad \frac{1}{2^5 \cdot 3 \cdot 5} + \frac{1}{2} \sum_{k=1}^{\infty} \frac{k^3 q^k}{1-q^k} = \frac{1}{2} S_3,$$

by (4.1.4). For $n \geq 1$, the coefficient of θ^{2n} in (4.2.33) equals

$$(4.2.35) \quad \frac{(-1)^{n+1} B_{2n+4}}{24(2n+4)(2n)!} + \frac{(-1)^n}{12(2n)!} \sum_{k=1}^{\infty} \frac{k^{2n+3} q^k}{1-q^k} = \frac{(-1)^n}{12(2n)!} S_{2n+3},$$

by (4.1.4). Hence, (4.2.33) can be rewritten in the shape

$$(4.2.36) \quad R(\theta) = \frac{1}{4\theta^4} + \frac{1}{2} S_3 + \frac{1}{12} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)!} S_{2n+3} \theta^{2n}.$$

Hence, by (4.2.30) and (4.2.36), we deduce that

$$\begin{aligned} \left(\frac{1}{2\theta^2} + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n)!} S_{2n+1} \theta^{2n} \right)^2 \\ = \frac{1}{4\theta^4} + \frac{1}{2} S_3 + \frac{1}{12} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)!} S_{2n+3} \theta^{2n}. \end{aligned}$$

Equating coefficients of θ^{2n} , $n \geq 1$, on both sides above, we find that

$$\begin{aligned} \frac{(-1)^n}{12(2n)!} S_{2n+3} &= \frac{(-1)^n}{(2n)!} \left\{ \binom{2n}{2} S_3 S_{2n-1} + \binom{2n}{4} S_5 S_{2n-3} + \cdots \right. \\ (4.2.37) \quad &\left. + \binom{2n}{2n-2} S_{2n-1} S_3 \right\} + \frac{(-1)^n}{(2n+2)!} S_{2n+3}, \end{aligned}$$

or

$$\frac{(n-1)(2n+5)}{12(n+1)(2n+1)} S_{2n+3} = \sum_{k=1}^{n-1} \binom{2n}{2k} S_{2k+1} S_{2n-2k+1},$$

which completes the proof of Theorem 4.2.7. $\square$

For example, putting $n = 2$ in (4.2.27) and using (4.1.6) and (4.1.8) yields

$$(4.2.38) \quad \frac{1}{20} S_7 = 6S_3^2 \quad \text{or} \quad E_8 = Q^2.$$

Putting $n = 3$ in (4.2.27) yields

$$(4.2.39) \quad E_{10} = E_4 E_6 = QR.$$

Using (4.2.27), the definitions (4.1.8) and (4.1.9), and induction on r , we show that, for each nonnegative integer r ,

$$(4.2.40) \quad S_{2r+1} = \sum_{\substack{m,n \geq 0 \\ 4m+6n=2r+2}} c_{m,n} Q^m R^n,$$

where the numbers $c_{m,n}$ are constants. It is clear that (4.2.40) is valid for $r = 1, 2$ by the definition (4.1.6) of S_{2r+1} , and for $r = 3, 4$ by (4.2.38) and (4.2.39), respectively. Assume that (4.2.40) holds. We prove (4.2.40) with r replaced by $r + 1$. By Theorem 4.2.7 and the

induction hypothesis,

$$\begin{aligned}
 S_{2r+3} &= \sum_{k=1}^{r-1} \sum_{\substack{m', n' \geq 0 \\ 4m' + 6n' = 2k+2}} c_{m', n'} Q^{m'} R^{n'} \\
 &\quad \times \sum_{\substack{m'', n'' \geq 0 \\ 4m'' + 6n'' = 2r - 2k + 2}} c_{m'', n''} Q^{m''} R^{n''} \\
 &= \sum_{\substack{m = m' + m'', n = n' + n'' \geq 0 \\ 4m + 6n = 2r + 4}} c_{m, n} Q^m R^n.
 \end{aligned}$$

Exercise 4.2.8. Similarly, using (4.2.8), along with the definitions (4.1.7)–(4.1.9), use induction on r to prove that, for each positive integer r ,

$$(4.2.41) \quad \Phi_{1,2r} = \sum_{\substack{\ell, m, n \geq 0, \ell \leq 2 \\ 2\ell + 4m + 6n = 2r + 2}} c_{\ell, m, n} P^\ell Q^m R^n,$$

where the numbers $c_{\ell, m, n}$ are constants. For the case $r = 1$, see (4.2.16).

4.3. A Class of Series from Ramanujan's Lost Notebook Expressible in Terms of P , Q , and R

On pages 188 and 369 of his lost notebook, in the pagination of [194], Ramanujan examines the series defined for $|q| < 1$ and each nonnegative integer k by

$$\begin{aligned}
 (4.3.1) \quad T_{2k} &:= T_{2k}(q) \\
 &:= 1 + \sum_{n=1}^{\infty} (-1)^n \left\{ (6n-1)^{2k} q^{n(3n-1)/2} + (6n+1)^{2k} q^{n(3n+1)/2} \right\}.
 \end{aligned}$$

Note that the exponents $n(3n \pm 1)/2$ are the generalized pentagonal numbers and that when $k = 0$, (4.3.1) reduces to the series in the pentagonal number theorem, Corollary 1.3.5. Ramanujan records formulas for T_{2k} , $k = 1, 2, \dots, 6$, in terms of the Eisenstein series P , Q , and R , defined by (4.1.7)–(4.1.9). Ramanujan's formulations of

these formulas are cryptic. The first is given by Ramanujan in the form

$$\frac{1 - 5^2q - 7^2q^2 + \cdots}{1 - q - q^2 + \cdots} = P.$$

In succeeding formulas, only the first two terms of the numerator are given, and in two instances the denominator is replaced by a dash —. At the bottom of the page, he gives the first five terms of a general formula for T_{2k} .

In this section, we indicate how to prove these seven formulas and one corollary. Keys to our proofs are the pentagonal number theorem (1.3.18),

$$(4.3.2) \quad (q; q)_{\infty} = 1 + \sum_{n=1}^{\infty} (-1)^n \left\{ q^{n(3n-1)/2} + q^{n(3n+1)/2} \right\} = T_0(q),$$

where $|q| < 1$, and Ramanujan's famous differential equations (4.2.20)–(4.2.22).

We now state Ramanujan's six formulas for T_{2k} followed by a corollary and his general formula.

Theorem 4.3.1. *If T_{2k} is defined by (4.3.1) and P, Q , and R are defined by (4.1.7)–(4.1.9), then*

- (i) $\frac{T_2(q)}{(q; q)_{\infty}} = P,$
- (ii) $\frac{T_4(q)}{(q; q)_{\infty}} = 3P^2 - 2Q,$
- (iii) $\frac{T_6(q)}{(q; q)_{\infty}} = 15P^3 - 30PQ + 16R,$
- (iv) $\frac{T_8(q)}{(q; q)_{\infty}} = 105P^4 - 420P^2Q + 448PR - 132Q^2,$
- (v) $\frac{T_{10}(q)}{(q; q)_{\infty}} = 945P^5 - 6300P^3Q + 10080P^2R - 5940PQ^2 + 1216QR,$

and

$$(vi) \quad \frac{T_{12}(q)}{(q; q)_{\infty}} = 10395P^6 - 103950P^4Q + 221760P^3R - 196020P^2Q^2 \\ + 80256PQR - 2712Q^3 - 9728R^2.$$

The first formula has an interesting arithmetical interpretation.

Corollary 4.3.2. For $n \geq 1$, let $\sigma(n) = \sum_{d|n} d$, and define $\sigma(0) = -\frac{1}{24}$. Let n denote a nonnegative integer. Then

$$(4.3.3) \quad -24 \sum_{\substack{j+k(3k \pm 1)/2=n \\ j,k \geq 0}} (-1)^k \sigma(j) = \begin{cases} (-1)^r (6r-1)^2, & \text{if } n=r(3r-1)/2, \\ (-1)^r (6r+1)^2, & \text{if } n=r(3r+1)/2, \\ 0, & \text{otherwise.} \end{cases}$$

Since $\sigma(j)$ is multiplicative, we note that $\sigma(j)$ is even except when j is a square or twice a square. Thus, from Corollary 4.3.2, we see that, unless $n = r(3r \pm 1)/2$, the number of representations of n as a sum of a square or twice a square and a generalized pentagonal number $k(3k \pm 1)/2$ is even. For example, if $n = 20$, then $20 = 8 + 12 = 18 + 2$; if $n = 5$, there is only one such representation: $5 = 4 + 1$.

In general, as Ramanujan indicated, for certain constants $c_{\ell,m,n}$,

$$T_{2k}(q) = (q)_\infty \sum_{\substack{\ell,m,n \geq 0 \\ 2\ell+4m+6n=2k}} c_{\ell,m,n} P^\ell Q^m R^n.$$

Important in our proofs are the simple identities

$$(4.3.4) \quad (6n \pm 1)^2 = 24 \frac{n(3n \pm 1)}{2} + 1.$$

We also use the notation

$$(4.3.5) \quad f_{2k}(P, Q, R) := \frac{T_{2k}(q)}{(q; q)_\infty}.$$

Proof of Theorem 4.3.1. Observe that

$$\begin{aligned} P(q) &= 1 + 24q \frac{d}{dq} \sum_{n=1}^{\infty} \log(1 - q^n) \\ &= 1 + 24q \frac{d}{dq} \log(q; q)_\infty \\ &= 1 + 24q \frac{\frac{d}{dq}(q; q)_\infty}{(q; q)_\infty}. \end{aligned}$$

Thus, using (4.3.2) and (4.3.4), we find that

(4.3.6)

$$\begin{aligned}
 & (q; q)_\infty P(q) - (q; q)_\infty \\
 &= 24q \frac{d}{dq} \left(1 + \sum_{n=1}^{\infty} (-1)^n \left\{ q^{n(3n-1)/2} + q^{n(3n+1)/2} \right\} \right) \\
 &= 24 \sum_{n=1}^{\infty} (-1)^n \left\{ \frac{n(3n-1)}{2} q^{n(3n-1)/2} + \frac{n(3n+1)}{2} q^{n(3n+1)/2} \right\} \\
 &= \sum_{n=1}^{\infty} (-1)^n \left\{ ((6n-1)^2 - 1) q^{n(3n-1)/2} + ((6n+1)^2 - 1) q^{n(3n+1)/2} \right\} \\
 &= \sum_{n=1}^{\infty} (-1)^n \left\{ (6n-1)^2 q^{n(3n-1)/2} + (6n+1)^2 q^{n(3n+1)/2} \right\} \\
 &\quad - (q; q)_\infty + 1 \\
 &= T_2(q) - (q; q)_\infty.
 \end{aligned}$$

This completes the proof of (i).

In the proofs of the remaining identities of Theorem 4.3.1, in each case, we apply the operator $24q \frac{d}{dq}$ to the preceding identity. In each proof we use the identities

$$(4.3.7) \quad 24q \frac{d}{dq} T_{2k}(q) = T_{2k+2}(q) - T_{2k}(q), \quad k \geq 0,$$

which follow from differentiation and the use of (4.3.4) in exactly the same manner that we employed above in (4.3.6) to prove the special case

$$(4.3.8) \quad 24q \frac{d}{dq} (q; q)_\infty = T_2(q) - (q; q)_\infty.$$

We now prove (ii). Applying the operator $24q \frac{d}{dq}$ to (4.3.6) and using (4.3.7) and (4.3.8), we deduce that

$$P(q) (T_2(q) - (q; q)_\infty) + (q; q)_\infty 24q \frac{d}{dq} P(q) = T_4(q) - T_2(q).$$

Employing (i) to simplify and using the differential equation in (4.2.20), we arrive at

$$P^2(q)(q; q)_\infty + 2(P^2(q) - Q(q))(q; q)_\infty = T_4(q),$$

or

$$(4.3.9) \quad T_4(q) = (3P^2(q) - 2Q(q))(q; q)_\infty,$$

as desired.

To prove (iii), we apply the operator $24q \frac{d}{dq}$ to (4.3.9) and use (4.3.7) and (4.3.8) to deduce that

$$\begin{aligned} T_6 - T_4 &= 24 \left(6Pq \frac{dP}{dq} - 2q \frac{dQ}{dq} \right) (q; q)_\infty + (3P^2 - 2Q)(T_2 - (q; q)_\infty) \\ &= (12P(P^2 - Q) - 16(PQ - R))(q; q)_\infty \\ &\quad + (3P^2 - 2Q)(P - 1)(q; q)_\infty, \end{aligned}$$

where we used (4.2.20), (4.2.21), and (i). If we now employ (4.3.9) and simplify, we conclude that

$$T_6 = (15P^3 - 30PQ + 16R)(q; q)_\infty.$$

In general, by applying the operator $24q \frac{d}{dq}$ to T_{2k} and using (4.3.5), (4.3.7), and (4.3.8), we find that

$$T_{2k+2} - T_{2k} = 24q \frac{d}{dq} f_{2k}(P, Q, R)(q; q)_\infty + f_{2k}(P, Q, R)(P - 1)(q; q)_\infty,$$

where we have used the notation (4.3.5). Then proceeding by induction while using the formula (4.3.5) for T_{2k} , we find that

$$\frac{T_{2k+2}}{(q; q)_\infty} = 24q \frac{d}{dq} f_{2k}(P, Q, R) + P f_{2k}(P, Q, R).$$

Thus, in the notation (4.3.5),

$$(4.3.10) \quad f_{2k+2}(P, Q, R) = 24q \frac{d}{dq} f_{2k}(P, Q, R) + P f_{2k}(P, Q, R).$$

Exercise 4.3.3. With the use of (4.3.10) and the differential equations (4.2.20)–(4.2.22), prove the remaining identities, (iv)–(vi).

□

Proof of Corollary 4.3.2. By expanding the summands of $P(q)$ in (4.1.7) in geometric series and collecting the coefficients of q^n for each positive integer n , we find that

$$P(q) = 1 - 24 \sum_{n=1}^{\infty} \sigma(n)q^n = -24 \sum_{n=0}^{\infty} \sigma(n)q^n,$$

upon using the definition $\sigma(0) = -\frac{1}{24}$. Thus, by (4.3.1) and (4.3.2), Theorem 4.3.1(i) can be written in the form

$$(4.3.11) \quad \begin{aligned} & -24 \sum_{j=0}^{\infty} \sigma(j) q^j \cdot \left(1 + \sum_{k=1}^{\infty} (-1)^k \left\{ q^{k(3k-1)/2} + q^{k(3k+1)/2} \right\} \right) \\ & = 1 + \sum_{n=1}^{\infty} (-1)^n \left\{ (6n-1)^2 q^{n(3n-1)/2} + (6n+1)^2 q^{n(3n+1)/2} \right\}. \end{aligned}$$

Equating coefficients of q^n , $n \geq 1$, on both sides of (4.3.11), we complete the proof. $\square$

4.4. Proofs of the Congruences

$$p(5n+4) \equiv 0 \pmod{5} \text{ and } p(7n+5) \equiv 0 \pmod{7}$$

In his manuscript [188], [192, pp. 210–213], in addition to proving the congruences (2.1.2)–(2.1.4), Ramanujan proves further congruences for $p(n)$. In particular, he establishes the identity

$$(4.4.1) \quad (q^{13}; q^{13})_{\infty} \sum_{n=1}^{\infty} p(13n-7) q^n = 11 \sum_{n=1}^{\infty} \tau(n) q^n + 13J.$$

He then writes, “It is not necessary to know all the details above in order to prove (4.4.1). The proof can be very much simplified as follows; using Theorem 4.2.3 and ... we can show that

$$(4.4.2) \quad (Q^3 - R^2)^7 = q \frac{dJ}{dq} + 3(Q^3 - R^2) + 13J,”$$

where $J = J(q)$ is a power series in q with integral coefficients. We use this notation of Ramanujan in what follows, and emphasize that J is not necessarily the same at each occurrence. Apparently, Ramanujan realized this simplification at precisely this juncture while writing his paper, for he did not return to his proofs of (2.1.2)–(2.1.4) to utilize this observation and thereby simplify them. Instead of using further (more complicated) Eisenstein series identities, the differential equations of Theorem 4.2.3 are employed. In his doctoral dissertation, J. M. Rushforth [201] used this idea to simplify Ramanujan’s proof of (2.1.4), as extracted by Hardy for [191]. In this section, we

use Ramanujan's observation above along with (4.2.23) and (4.2.20)–(4.2.22) to give simplified proofs of (2.1.2) and (2.1.3). The proof of (2.1.4) is more tedious, and we refer to Berndt's paper [40] for the proof of (2.1.4), which is precisely that of Rushforth [201].

Theorem 4.4.1. *For each nonnegative integer n ,*

$$(4.4.3) \quad p(5n + 4) \equiv 0 \pmod{5}.$$

Proof. From the definitions (4.1.8) and (4.1.9), respectively,

$$(4.4.4) \quad Q = 1 + 5J \quad \text{and} \quad R = P + 5J,$$

since $n^5 \equiv n \pmod{5}$ by Fermat's little theorem. It follows from (4.4.4) and (4.2.20) that

$$(4.4.5) \quad \begin{aligned} Q^3 - R^2 &= Q(1 + 5J)^2 - (P + 5J)^2 = Q - P^2 + 5J \\ &= -12q \frac{d}{dq} P + 5J = 3q \frac{d}{dq} P + 5J. \end{aligned}$$

But, by (4.2.23) and the binomial theorem,

$$(4.4.6) \quad Q^3 - R^2 = 1728q(q; q)_{\infty}^{24} = 3q \frac{(q; q)_{\infty}^{25}}{(q; q)_{\infty}} + 5J = 3q \frac{(q^5; q^5)_{\infty}^5}{(q; q)_{\infty}} + 5J.$$

Combining (4.4.5) and (4.4.6) and using the generating function for $p(n)$, we find that

$$(4.4.7) \quad q(q^5; q^5)_{\infty}^5 \sum_{n=0}^{\infty} p(n)q^n = q \frac{dP}{dq} + 5J.$$

We now equate those terms on both sides of (4.4.7) whose powers are of the form q^{5n} to find that

$$(4.4.8) \quad (q^5; q^5)_{\infty}^5 \sum_{n=0}^{\infty} p(5n + 4)q^{5n+5} = 5J.$$

Ramanujan's congruence (4.4.3) follows immediately from (4.4.8). $\square$

Theorem 4.4.2. *For each nonnegative integer n ,*

$$(4.4.9) \quad p(7n + 5) \equiv 0 \pmod{7}.$$

Proof. The first two steps of our proof are the same as those of Ramanujan and Hardy [191], [192, p. 235]. From the definition of R , it is obvious that

$$(4.4.10) \quad R = 1 + 7J.$$

Using one of the identities from (4.2.38), namely,

$$E_8(q) = 1 + 480 \sum_{n=1}^{\infty} \frac{n^7 q^n}{1 - q^n},$$

Fermat's little theorem, and the definition (4.1.7), we also find that

$$(4.4.11) \quad Q^2 = P + 7J.$$

Hence, from (4.4.10), (4.4.11), (4.2.20), and (4.2.21),

$$\begin{aligned} (Q^3 - R^2)^2 &= (PQ - 1 + 7J)^2 \\ &= P^2Q^2 - 2PQ + 1 + 7J \\ &= P(P^2 - Q) - PQ + R + 7J \\ &= 12Pq \frac{dP}{dq} - 3q \frac{dQ}{dq} + 7J \\ (4.4.12) \quad &= 6q \frac{d}{dq} P^2 - 3q \frac{dQ}{dq} + 7J = q \frac{dJ}{dq} + 7J. \end{aligned}$$

On the other hand, by (4.2.23) and the binomial theorem,

$$\begin{aligned} (Q^3 - R^2)^2 &= q^2 \frac{(q; q)_{\infty}^{49}}{(q; q)_{\infty}} + 7J = q^2 \frac{(q^7; q^7)_{\infty}^7}{(q; q)_{\infty}} + 7J \\ (4.4.13) \quad &= (q^7; q^7)_{\infty}^7 \sum_{n=0}^{\infty} p(n) q^{n+2} + 7J. \end{aligned}$$

We now equate the right sides of (4.4.12) and (4.4.13) and then extract those terms involving q^7 . Equating them, we find that

$$(q^7; q^7)_{\infty}^7 \sum_{n=0}^{\infty} p(7n+5) q^{7n+7} = 7J,$$

from whence (4.4.9) immediately follows. □

4.5. Notes

Eisenstein series were first introduced and studied extensively by G. Eisenstein [87], [89, pp. 357–478] in 1847. In contemporary notation, the Eisenstein series $G_{2j}(\tau)$ and $E_{2j}(\tau)$, where j is a positive integer exceeding one, are defined for $\text{Im } \tau > 0$ by

$$G_{2j}(\tau) := \sum_{\substack{m,n=0 \\ (m,n) \neq (0,0)}}^{\infty} (m\tau + n)^{-2j}$$

and

$$\begin{aligned} E_{2j}(\tau) &:= \frac{G_{2j}(\tau)}{2\zeta(2j)} = 1 - \frac{4j}{B_{2j}} \sum_{k=1}^{\infty} \frac{k^{2j-1} e^{2\pi i k \tau}}{1 - e^{2\pi i k \tau}} \\ (4.5.1) \qquad &= 1 - \frac{4j}{B_{2j}} \sum_{r=1}^{\infty} \sigma_{2j-1}(r) q^r, \quad q = e^{2\pi i \tau}, \end{aligned}$$

where $\zeta(z) = \sum_{n=1}^{\infty} n^{-z}$, $\text{Re } z > 1$, denotes the Riemann zeta function, B_n , $n \geq 0$, denotes the n th Bernoulli number, and $\sigma_{\nu}(n) = \sum_{d|n} d^{\nu}$. The series $G_{2j}(\tau)$ does not converge if $j = 1$, and so the first equality in (4.5.1) does not hold if $j = 1$, although the series on the right side converges for $j = 1$. We have defined Eisenstein series by the latter two representations in (4.5.1). These can be established from the former representations by using the Lipschitz summation formula or Fourier analysis. For example, see T. M. Apostol's text [25, pp. 18–20, 24] or B. Schoeneberg's book [205, pp. 51–53]. The Eisenstein series $E_{2j}(\tau)$ is a modular form of weight $2j$ on the full modular group. Along with the discriminant function $\Delta(\tau) := q(q; q)_{\infty}^{24}$, $q = e^{2\pi i \tau}$, these functions form the building blocks for all analytic modular forms on the full modular group. For basic properties of Eisenstein series from the point of view of modular forms, see, for example, the aforementioned texts as well as R. A. Rankin's text [196, Chap. 6].

Ramanujan made many original and fundamental contributions to the theory of Eisenstein series, most notably in his lost notebook and letters to Hardy [194]. In particular, in letters he wrote to Hardy from nursing homes near the end of his sojourn in England, he derived formulas for the coefficients of certain quotients of Eisenstein series.

These formulas are analogous to the asymptotic formula for $p(n)$ of Hardy and Ramanujan that was discussed in (1.1.6) and the notes for Chapter 1. However, their forms and proofs are much different. For accounts of this work, see papers of Hardy and Ramanujan [111], [192, pp. 310–321], Berndt and P. Bialek [42], and Berndt, Bialek, and A. J. Yee [43]. For an account of all of Ramanujan's theorems on Eisenstein series from his lost notebook, other unpublished papers, and letters, see the book [20] by G. E. Andrews and Berndt. For an expository survey of many of Ramanujan's theorems on Eisenstein series from his lost notebook, see the paper [53] by Berndt and Yee.

In his notebooks [193], Ramanujan made further claims about Eisenstein series. In particular, in Entry 14 of Chapter 15 of his second notebook, Ramanujan recorded another recurrence relation for Eisenstein series; see Berndt's book [33, p. 332]. Also, Chapters 17 and 21 in Ramanujan's second notebook contain many claims about Eisenstein series. See Berndt's book [34] for proofs of all these theorems.

An important function which we have not discussed in this book is the modular j -invariant defined by

$$j(\tau) := \frac{Q^3(q)}{q(q; q)_{\infty}^{24}}, \quad q = e^{2\pi i \tau}.$$

The values of $j(\tau)$ and its Fourier or power series coefficients are of great importance in the theory of modular forms.

At about the same time that Ramanujan discovered the differential equation (4.2.20) satisfied by $P(q)$, in 1911, J. Chazy [74] considered the third order differential equation

$$(4.5.2) \quad y''' = 2yy'' - 3(y')^2,$$

where $y = y(t)$ and showed that one solution is $\pi i P(q)$, where $q = e^{2\pi i t}$. If we eliminate Q and R from the system (4.2.20)–(4.2.22), we deduce the differential equation (4.5.2) for P . In fact, the system of equations (4.2.20)–(4.2.22) is equivalent to a system of three differential equations first studied by G. Halphen [104] in 1881. The equation (4.5.2) is of fundamental importance in the theory of integrable systems [1], [2], [3], [4].

The results in Sections 4.1 and 4.2 are taken from Ramanujan's paper [186], [192, pp. 136–162].

Exercise 4.5.1. *Show that Ramanujan's key identity (4.2.6) can be deduced from Venkatachaliengar's result (3.8.2).*

In her thesis [183], V. Ramamani extended Ramanujan's use of trigonometric series to study two analogues of $\Phi_{r,s}(q)$, namely,

$$\Psi_{r,s}(q) := \sum_{m,n=-\infty}^{\infty} (-1)^{n-1} m^r n^s q^{mn}$$

and

$$F_{r,s}(q) := \sum_{m,n=-\infty}^{\infty} (2m-1)^r n^s q^{(2m-1)n/2},$$

where $|q| < 1$. Ramamani employed the classical theory of elliptic functions in her work. In extending Ramamani's study of $\Psi_{r,s}(q)$, in her thesis [102], H. Hahn chiefly employed ideas from Ramanujan's viewpoint in the theory of elliptic functions.

H. H. Chan [68] derived an analogue of (4.2.2) with $\cot$ replaced by $\csc$ and used it to give new proofs of Ramanujan's famous formulas

$$\sum_{n=1}^{\infty} \left(\frac{n}{5}\right) \frac{q^n}{(1-q^n)^2} = q \frac{(q^5; q^5)_{\infty}^5}{(q; q)_{\infty}}$$

and

$$\sum_{n=1}^{\infty} \left(\frac{n}{7}\right) q^n \frac{1+q^n}{(1-q^n)^3} = q(q; q)_{\infty}^3 (q^7; q^7)_{\infty}^3 + 8q^2 \frac{(q^7; q^7)_{\infty}^7}{(q; q)_{\infty}},$$

where $\left(\frac{n}{p}\right)$ denotes the Legendre symbol. These formulas lead, respectively, to proofs of Ramanujan's congruences $p(5n+4) \equiv 0 \pmod{5}$ and $p(7n+5) \equiv 0 \pmod{7}$.

Z.-G. Liu [148] used the theory of elliptic functions and associated complex analysis to derive a trigonometric series identity involving theta functions that is analogous to Ramanujan's trigonometric series identities. Whereas Ramanujan used his results to study sums of squares, Liu used his identity to obtain representations for $t_{2k}(n)$.

The content of Section 4.3 is taken from a paper by Berndt and Yee [53]. K. Venkatachaliengar [214, pp. 31–32] has given a similar

but more abbreviated account of this work. On page 369 in his lost notebook [194], Ramanujan offers another infinite class of series

$$(4.5.3) \quad F_{2n}(q) := \sum_{k=0}^{\infty} (-1)^k (2k+1)^{2n+1} q^{k(k+1)/2}$$

that can be represented in terms of polynomials in P , Q , and R . Observe that when $n = 0$, $F_0(q) = (q; q)_{\infty}^3$, by Jacobi's identity, Theorem 1.3.9. Thus, $(q; q)_{\infty}^3$ and Jacobi's identity play the roles in the aforementioned identities that $(q; q)_{\infty}$ and the pentagonal number theorem play in the identities of Section 4.3. For proofs, see the paper by Berndt, S. H. Chan, Liu, and H. Yesilyurt [48]. These ideas of Ramanujan were developed further by H. H. Chan, S. Cooper, and P. C. Toh [70], who showed that further classes of series can be expressed as polynomials in P , Q , and R . For example, in one class, the powers $(q; q)_{\infty}$ and $(q; q)_{\infty}^3$ above are replaced by $(q; q)_{\infty}^{10}$.

The proofs in Section 4.4 can be found in Berndt's paper [40]. Rushforth's proof of the congruence $p(11n + 6) \equiv 0 \pmod{11}$ can be found in his thesis [201] and was reproduced in Berndt's paper [40]. As indicated earlier, the key idea behind these proofs arises from Ramanujan's previously unpublished manuscript on the partition and tau functions [194], for which Berndt and Ono [50] gave additional details and references for several proofs. The latter paper was extended and annotated still further by Andrews and Berndt in their book [21].

Chapter 5

The Connection Between Hypergeometric Functions and Theta Functions

In this chapter, we prove one of the central theorems in the theory of elliptic functions and, without a doubt, the primary theorem for Ramanujan in the development of his highly original approach to the theory of elliptic functions.

5.1. Definitions of Hypergeometric Series and Elliptic Integrals

First, we review the notation for a *rising factorial* or a *shifted factorial* given in (1.1.3). Define $(a)_0 = 1$ and, for $n \geq 1$,

$$(5.1.1) \quad (a)_n := a(a+1)(a+2) \cdots (a+n-1).$$

Unfortunately, the notation (5.1.1) is the same as that in (1.1.1). However, the notations (1.1.1) and (5.1.1) are rarely used in the same context, and so no confusion should arise. Note that $(1)_n = n!$. The notation (5.1.1) is also called the Pochhammer symbol in especially

the older literature, but the notation is misnamed, not suggestive, and should be laid to rest.

Definition 5.1.1. Let a , b , and c be arbitrary complex numbers, except that c cannot be a nonpositive integer. Then, for $|z| < 1$, the Gaussian or ordinary hypergeometric function ${}_2F_1(a, b; c; z)$ is defined by

$$(5.1.2) \quad {}_2F_1(a, b; c; z) := \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n n!} z^n.$$

Readers can easily apply the ratio test to show that the series in (5.1.2) indeed does converge for $|z| < 1$.

Certain special cases of hypergeometric functions are connected with elliptic integrals.

Definition 5.1.2. The complete elliptic integral of the first kind is defined for $|k| < 1$ by

$$(5.1.3) \quad K(k) := \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}}.$$

The number k is called the modulus, and the number $k' = \sqrt{1 - k^2}$ is called the complementary modulus.

Lemma 5.1.3. For $|k| < 1$,

$$(5.1.4) \quad K(k) = \frac{\pi}{2} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; k^2\right).$$

Proof. In our proof, we express the value of an integral in terms of the classical Gamma function $\Gamma(z)$, and use the well-known value $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$. Readers unfamiliar with the Gamma function might want to use induction on n to evaluate the integral below. Using the binomial series in the integrand of $K(k)$, inverting the order of summation and integration by absolute convergence, and using the aforementioned value of the integral of $\sin^{2n} \phi$ over $[0, \pi/2]$, we find

that

$$\begin{aligned}
 \int_0^{\pi/2} \frac{d\phi}{\sqrt{1-k^2 \sin^2 \phi}} &= \sum_{n=0}^{\infty} \frac{(\frac{1}{2})_n}{n!} k^{2n} \int_0^{\pi/2} \sin^{2n} \phi \, d\phi \\
 &= \sum_{n=0}^{\infty} \frac{(\frac{1}{2})_n}{n!} k^{2n} \frac{\Gamma(n + \frac{1}{2}) \Gamma(\frac{1}{2})}{2\Gamma(n+1)} \\
 &= \frac{1}{2} \sum_{n=0}^{\infty} \frac{(\frac{1}{2})_n}{n!} k^{2n} \frac{(\frac{1}{2})_n \pi}{n!} \\
 &= \frac{\pi}{2} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; k^2\right).
 \end{aligned}$$

□

Although not necessary for the development of the theory in this chapter, we next define elliptic integrals of the second kind.

Definition 5.1.4. *The complete elliptic integral of the second kind is defined for $|k| < 1$ by*

$$(5.1.5) \quad E(k) := \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \phi} \, d\phi.$$

Readers will be familiar with complete integrals of the second kind from elementary calculus. Recall that if an ellipse is given by the parametric equations $x = a \cos \phi$, $y = b \sin \phi$, then the length L of the perimeter of this ellipse is given by

$$L = 4a \int_0^{\pi/2} \sqrt{1 - e^2 \cos^2 \phi} \, d\phi,$$

where e denotes the eccentricity of the ellipse. (Note that L can be expressed by an elliptic integral of the second kind.) An analogue of Lemma 5.1.3 holds for elliptic integrals of the second kind.

Exercise 5.1.5. *Prove that*

$$(5.1.6) \quad E(k) = \frac{\pi}{2} {}_2F_1\left(-\frac{1}{2}, \frac{1}{2}; 1; k^2\right).$$

We next prove one of the most important formulas in the theory of elliptic integrals and hypergeometric functions.

Theorem 5.1.6 (Landen's Transformation). For $0 < x < 1$,

$$K\left(\frac{2\sqrt{x}}{1+x}\right) = (1+x)K(x),$$

or, equivalently, by Lemma 5.1.3,

$$(5.1.7) \quad {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; \frac{4x}{(1+x)^2}\right) = (1+x) {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x^2\right).$$

Proof. Using the binomial series below and inverting the order of summation and integration by absolute convergence, we find that

$$\begin{aligned} K\left(\frac{2\sqrt{x}}{1+x}\right) &= \frac{1}{2} \int_0^\pi \left(1 - \frac{4x}{(1+x)^2} \sin^2 \phi\right)^{-1/2} d\phi \\ &= \frac{1+x}{2} \int_0^\pi (1 + x^2 + 2x(1 - 2\sin^2 \phi))^{-1/2} d\phi \\ &= \frac{1+x}{2} \int_0^\pi (1 + x^2 + 2x \cos 2\phi)^{-1/2} d\phi \\ &= \frac{1+x}{2} \int_0^\pi (1 + xe^{2i\phi})^{-1/2} (1 + xe^{-2i\phi})^{-1/2} d\phi \\ &= \frac{1+x}{2} \sum_{m=0}^{\infty} \frac{\left(\frac{1}{2}\right)_m}{m!} (-x)^m \sum_{n=0}^{\infty} \frac{\left(\frac{1}{2}\right)_n}{n!} (-x)^n \int_0^\pi e^{2i(m-n)\phi} d\phi \\ &= \frac{1+x}{2} \sum_{n=0}^{\infty} \frac{\left(\frac{1}{2}\right)_n^2}{(n!)^2} x^{2n} \pi \\ &= (1+x) \frac{\pi}{2} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x^2\right) \\ &= (1+x)K(x), \end{aligned}$$

by Lemma 5.1.3. □

Corollary 5.1.7. For $0 < x < 1$,

$$(5.1.8) \quad {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1 - \left(\frac{1-x}{1+x}\right)^2\right) = (1+x) {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x^2\right).$$

Proof. Observing that

$$\frac{4x}{(1+x)^2} = 1 - \left(\frac{1-x}{1+x}\right)^2,$$

we see that (5.1.8) follows immediately from (5.1.7). □

The version of Landen's transformation that we have given is actually a special case of a more general Landen transformation, namely, (5.1.9)

$${}_2F_1\left(a, b; 2b; \frac{4x}{(1+x)^2}\right) = (1+x)^{2a} {}_2F_1\left(a, a-b+\frac{1}{2}; b+\frac{1}{2}; x^2\right).$$

By Lemma 5.1.3 and Theorem 5.1.6, the special case of (5.1.9) that we have proved is for $a = b = \frac{1}{2}$.

There is still one further version of Landen's transformation. To describe this transformation, we need to define incomplete elliptic integrals.

Definition 5.1.8. If $0 < \alpha \leq \pi/2$, the incomplete elliptic integral of the first kind is defined by

$$\int_0^\alpha \frac{d\phi}{\sqrt{1-k^2 \sin^2 \phi}}.$$

We now state Landen's transformation for incomplete elliptic integrals of the first kind.

Exercise 5.1.9. If $|x| < 1$ and

$$x \sin \alpha = \sin(2\beta - \alpha), \quad 0 \leq \alpha \leq \pi, \quad 0 \leq \beta \leq \pi/2,$$

prove that

$$(5.1.10) \quad (1+x) \int_0^\alpha \frac{d\phi}{\sqrt{1-x^2 \sin^2 \phi}} = 2 \int_0^\beta \frac{d\phi}{\sqrt{1-\frac{4x}{(1+x)^2} \sin^2 \phi}}.$$

When $\alpha = \pi$ and $\beta = \pi/2$, (5.1.10) reduces to Theorem 5.1.6. We highly recommend that readers provide a rigorous proof of (5.1.10).

Examining closely the definition of ${}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; x)$ as $x \rightarrow 1$, we conjecture that a partial sum of ${}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; x)$ behaves like a partial sum of the harmonic series. We make this observation more precise in the following lemma, which is used in the next section.

Lemma 5.1.10. As $x \rightarrow 0^+$,

$$(5.1.11) \quad \pi {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1-x\right) \sim -\log x + C,$$

where C is a constant. Copyrighted Material

Proof. By Lemma 5.1.3, it suffices to prove that

$$(5.1.12) \quad 2K(\sqrt{x}) \sim -\log(1-x) + C,$$

as $x \rightarrow 1^-$.

In the representation (5.1.3), let $t = \sin \phi$, so that $\cos \phi = \sqrt{1-t^2}$. Thus,

$$(5.1.13) \quad 2K(\sqrt{x}) = 2 \int_0^1 \frac{dt}{\sqrt{1-t^2}\sqrt{1-xt^2}}.$$

Note that, for $0 < x < 1$,

$$(5.1.14) \quad \int_0^1 \frac{x dt}{1-xt} = -\log(1-x).$$

Now examine, for $0 < x < 1$,

$$R(x) := \int_0^1 \left\{ \frac{2}{\sqrt{1-t^2}\sqrt{1-xt^2}} - \frac{x}{1-xt} \right\} dt.$$

Observe that $R(x)$ is continuous at $x = 1^-$. In fact,

$$(5.1.15) \quad R(1) = \int_0^1 \left\{ \frac{2}{1-t^2} - \frac{1}{1-t} \right\} dt = \int_0^1 \frac{dt}{1+t} = \log 2.$$

Thus, as $x \rightarrow 1^-$, the two integrals in (5.1.13) and (5.1.14) are asymptotic to each other, since their difference approaches a constant, namely, $\log 2$, as $x \rightarrow 1^-$. Hence, (5.1.12) follows from (5.1.14) and (5.1.15). $\square$

5.2. The Main Theorem

Briefly, our primary goal is to prove that

$$\varphi^2(q) = {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x\right),$$

where q is a certain function of x . We shall establish this relation after a series of ancillary lemmas.

First define, for $0 < x < 1$,

$$(5.2.1) \quad F(x) := \exp\left(-\pi \frac{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1-x\right)}{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x\right)}\right).$$

By continuity, since ${}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 0\right) = 1$ and ${}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x\right) \rightarrow \infty$, as $x \rightarrow 1^-$, by Lemma 5.1.10, we can extend the definition of $F(x)$ to $x = 0$ and $x = 1$ by setting

$$F(0) = 0 \quad \text{and} \quad F(1) = 1.$$

Lemma 5.2.1. For $0 < x < 1$,

$$(5.2.2) \quad F(x^2) = F^2\left(\frac{4x}{(1+x)^2}\right).$$

Proof. In (5.1.7) replace x by $(1-x)/(1+x)$. Observe that

$$\begin{aligned} x^2 &\rightarrow 1 - \frac{4x}{(1+x)^2}, \\ \frac{4x}{(1+x)^2} &\rightarrow 1 - x^2, \\ 1+x &\rightarrow \frac{2}{1+x}. \end{aligned}$$

Hence, we arrive at

$$(5.2.3) \quad \frac{2}{1+x} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1 - \frac{4x}{(1+x)^2}\right) = {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1 - x^2\right).$$

Now divide (5.2.3) by (5.1.7) to deduce that

$$(5.2.4) \quad 2 \frac{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1 - \frac{4x}{(1+x)^2}\right)}{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; \frac{4x}{(1+x)^2}\right)} = \frac{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1 - x^2\right)}{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x^2\right)}.$$

Multiplying both sides of (5.2.4) by $-\pi$, exponentiating both sides, and invoking the definition of $F(x)$ from (5.2.1), we complete the proof of (5.2.2). $\square$

Lemma 5.2.2. If x , $0 < x < 1$, is defined by

$$(5.2.5) \quad \frac{1-x}{1+x} = \frac{\varphi^2(-q)}{\varphi^2(q)} =: \lambda = \lambda(q),$$

then

$$(5.2.6) \quad 1 - x^2 = \frac{\varphi^4(-q^2)}{\varphi^4(q^2)} = \lambda^2(q^2).$$

Proof. By (1.3.32), (3.6.7), and the definition of λ in (5.2.5),

$$\begin{aligned}\lambda^2(q^2) &= \frac{\varphi^4(-q^2)}{\varphi^4(q^2)} = \left\{ \frac{2\varphi(q)\varphi(-q)}{\varphi^2(q) + \varphi^2(-q)} \right\}^2 \\ &= \frac{4\lambda}{(1+\lambda)^2} = \frac{4\frac{1-x}{1+x}}{\left(1 + \frac{1-x}{1+x}\right)^2} = 1 - x^2.\end{aligned}$$

This concludes the proof. $\square$

Lemma 5.2.3. *If $n = 2^m$, where m is any nonnegative integer,*

$$(5.2.7) \quad F\left(\frac{\varphi^4(-q)}{\varphi^4(q)}\right) = F^n\left(\frac{\varphi^4(-q^n)}{\varphi^4(q^n)}\right).$$

Proof. Replacing x by $(1-x)/(1+x)$ in Lemma 5.2.1, we readily find that

$$(5.2.8) \quad F\left(\left(\frac{1-x}{1+x}\right)^2\right) = F^2(1-x^2).$$

Applying Lemma 5.2.2 to (5.2.8), we find that

$$(5.2.9) \quad F(\lambda^2(q)) = F^2(\lambda^2(q^2)).$$

Iterate (5.2.9) to deduce that

$$F(\lambda^2(q)) = F^2(\lambda^2(q^2)) = F^{2^2}(\lambda^2(q^{2^2})) = \cdots = F^{2^m}(\lambda^2(q^{2^m})),$$

that is to say,

$$F\left(\frac{\varphi^4(-q)}{\varphi^4(q)}\right) = F^n\left(\frac{\varphi^4(-q^n)}{\varphi^4(q^n)}\right).$$

$\square$

Lemma 5.2.4. *If $n = 2^m$, where m is any nonnegative integer, then*

$$(5.2.10) \quad F^n\left(1 - \frac{\varphi^4(-q)}{\varphi^4(q)}\right) = F\left(1 - \frac{\varphi^4(-q^n)}{\varphi^4(q^n)}\right).$$

Proof. From the definition of $F(x)$ in (5.2.1) and Lemma 5.2.3, we see that if

$$x_n := \frac{\varphi^4(-q^n)}{\varphi^4(q^n)},$$

then

$$\frac{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1-x_1\right)}{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x_1\right)} = n \frac{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1-x_n\right)}{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x_n\right)}.$$

We rewrite the last equality in the form

$$n \frac{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x_1\right)}{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1-x_1\right)} = \frac{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x_n\right)}{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1-x_n\right)}.$$

But this implies that $F^n(1-x_1) = F(1-x_n)$, which is the same as (5.2.10). $\square$

Theorem 5.2.5. *We have*

$$(5.2.11) \quad F\left(1 - \frac{\varphi^4(-q)}{\varphi^4(q)}\right) = q.$$

Proof. Write Lemma 5.1.10 in the form

$$(5.2.12) \quad \pi {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1-x\right) = -\log x + C + o(1),$$

where $o(1)$ denotes a function which tends to 0 as $x \rightarrow 0^+$, not necessarily the same with each appearance below. Using (5.2.12) and the definition (5.2.1), we see that

$$(5.2.13) \quad F(x) = e^{\log x - C + o(1)} = A x e^{o(1)} = A x (1 + o(1)),$$

as $x \rightarrow 0^+$, where $A = e^{-C}$.

Let

$$(5.2.14) \quad x_n := \frac{\varphi^4(-q^n)}{\varphi^4(q^n)}.$$

Note that x_n tends to 1 as $n \rightarrow \infty$. We now take the n th root of both sides of (5.2.10) and let n tend to ∞ . We need to take care on the right side when letting $n \rightarrow \infty$. Thus, in the notation (5.2.14) and with the use of (5.2.13), we find that

$$\begin{aligned} F\left(1 - \frac{\varphi^4(-q)}{\varphi^4(q)}\right) &= \lim_{n \rightarrow \infty} \sqrt[n]{F(1-x_n)} \\ &= \lim_{n \rightarrow \infty} \sqrt[n]{A(1-x_n)(1+o(1))} \\ &= \lim_{n \rightarrow \infty} \sqrt[n]{1-x_n} \\ &= \lim_{n \rightarrow \infty} \sqrt[n]{\frac{\varphi^4(q^n) - \varphi^4(-q^n)}{\varphi^4(q^n)}} \end{aligned}$$

$$\begin{aligned}
 &= \lim_{n \rightarrow \infty} \sqrt[n]{\frac{16q^n \psi^4(q^{2n})}{\varphi^4(q^n)}} \\
 (5.2.15) \quad &= q,
 \end{aligned}$$

where in the penultimate line we used (3.6.8). This completes the proof of Theorem 5.2.5. $\square$

Lemma 5.2.6. *We have*

$$(5.2.16) \quad {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1 - \frac{\varphi^4(-q)}{\varphi^4(q)}\right) = \frac{\varphi^2(q)}{\varphi^2(q^2)} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1 - \frac{\varphi^4(-q^2)}{\varphi^4(q^2)}\right).$$

Proof. From (5.2.5) and (5.2.6),

$$\begin{aligned}
 (1+x)^2 &= \frac{1+x}{1-x}(1-x^2) = \frac{\varphi^2(q)}{\varphi^2(-q)} \frac{\varphi^4(-q^2)}{\varphi^4(q^2)} \\
 (5.2.17) \quad &= \frac{\varphi^2(q)}{\varphi^2(-q)} \frac{\varphi^2(q)\varphi^2(-q)}{\varphi^4(q^2)} = \frac{\varphi^4(q)}{\varphi^4(q^2)},
 \end{aligned}$$

by (1.3.32). Now substitute from (5.2.5), (5.2.6), and (5.2.17) into Corollary 5.1.7, and the result follows. $\square$

Lemma 5.2.7. *We have*

$$(5.2.18) \quad \varphi^2(q) = {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1 - \frac{\varphi^4(-q)}{\varphi^4(q)}\right).$$

Proof. Iterate (5.2.16) m times to obtain the equalities

$$\begin{aligned}
 {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1 - \frac{\varphi^4(-q)}{\varphi^4(q)}\right) &= \frac{\varphi^2(q)}{\varphi^2(q^4)} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1 - \frac{\varphi^4(-q^4)}{\varphi^4(q^4)}\right) \\
 (5.2.19) \quad &= \cdots = \frac{\varphi^2(q)}{\varphi^2(q^{2^m})} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1 - \frac{\varphi^4(-q^{2^m})}{\varphi^4(q^{2^m})}\right).
 \end{aligned}$$

Now let $m \rightarrow \infty$ in (5.2.19). Observing that $\varphi^2(\pm q^{2^m}) \rightarrow \varphi(0) = 1$, as m tends to ∞ , and remembering that ${}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 0\right) = 1$, we see that the right side of (5.2.19) tends to $\varphi^2(q)$ as m tends to ∞ , which is what we wanted to prove. $\square$

We are now ready to view the *Main Event*.

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Theorem 5.2.8. Recall that $F(x)$ is defined by (5.2.1). For $0 < x < 1$,

$$(5.2.20) \quad \varphi^2(F(x)) = {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x\right) =: z,$$

or, in other words, if

$$(5.2.21) \quad q = F(x) := \exp\left(-\pi \frac{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1-x\right)}{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x\right)}\right),$$

then

$$(5.2.22) \quad \varphi^2(q) = z.$$

Proof. We begin by summarizing first Lemma 5.2.7 and second Theorem 5.2.5. If

$$u := \frac{\varphi^4(-F(x))}{\varphi^4(F(x))},$$

then

$$(5.2.23) \quad \varphi^2(F(x)) = {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1-u\right)$$

and

$$(5.2.24) \quad F(x) = F(1-u).$$

The last equality (5.2.24) implies that

$$(5.2.25) \quad \frac{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1-x\right)}{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x\right)} = \frac{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; u\right)}{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1-u\right)}.$$

Suppose that we can show that the denominators in (5.2.25) are equal, i.e., that

$$(5.2.26) \quad {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x\right) = {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1-u\right).$$

Then it will follow from (5.2.23) that

$$\varphi^2(F(x)) = {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x\right),$$

i.e., (5.2.20) holds. We show that (5.2.26) easily follows from the monotonicity of ${}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x\right)$ on $(0, 1)$. Suppose there is some point x_0 such that (5.2.26) does not hold. Let $u_0 := u(x_0)$. Without loss of generality, assume that $1 - u_0 < x_0$, so that, by the monotonicity of ${}_2F_1$,

$${}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1-u_0\right) < {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x_0\right).$$

It follows from (5.2.25) that

$${}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; u_0\right) < {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1 - x_0\right),$$

which implies, by monotonicity, that $u_0 < 1 - x_0$, which is incompatible with the previous condition $1 - u_0 < x_0$. Hence, our assumption that there exists a point x_0 such that (5.2.26) does not hold has produced a contradiction. $\square$

We place Theorem 5.2.8 in the context of the classical theory of elliptic functions and summarize some of our principal results in this chapter. Set $x = k^2$, where, as above, k is the modulus. Also set

$$(5.2.27) \quad k^2 = x = 1 - \frac{\varphi^4(-q)}{\varphi^4(q)},$$

so that the complementary modulus k' is given by

$$(5.2.28) \quad k' = \sqrt{1 - k^2} = \frac{\varphi^2(-q)}{\varphi^2(q)}.$$

Then, by Theorem 5.2.8,

$$(5.2.29) \quad \varphi^2(q) = {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; k^2\right) =: z,$$

where

$$(5.2.30) \quad q = e^{-y} := \exp\left(-\pi \frac{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1 - k^2\right)}{{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; k^2\right)}\right) = \exp\left(-\pi \frac{K'}{K}\right),$$

where $K = K(k)$ and $K' = K(k')$.

5.3. Principles of Duplication and Dimidiation

In this section we show how a formula involving x (or k^2), y (defined by (5.2.30)), and z (defined by (5.2.29)) can be converted to an equation involving $2y$ or $y/2$.

Define, for $0 < x' < 1$,

$$(5.3.1) \quad x' = \left(\frac{1 - \sqrt{1 - x}}{1 + \sqrt{1 - x}}\right)^2,$$

from which it follows that

$$(5.3.2) \quad x := \frac{4\sqrt{x'}}{(1 + \sqrt{x'})^2}.$$

Furthermore, define

$$(5.3.3) \quad e^{-y'} := F(x') \quad \text{and} \quad z' = {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x'\right).$$

By (5.2.1), (5.3.2), Lemma 5.2.1, and (5.3.3),

$$(5.3.4) \quad e^{-y} = F(x) = F\left(\frac{4\sqrt{x'}}{(1+\sqrt{x'})^2}\right) = \sqrt{F(x')} = e^{-y'/2}$$

Hence,

$$(5.3.5) \quad y = \frac{1}{2}y'.$$

Furthermore, by (5.2.29), (5.3.2), (5.1.7), and (5.3.3),

$$\begin{aligned} z &= {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x\right) = {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; \frac{4\sqrt{x'}}{(1+\sqrt{x'})^2}\right) \\ &= (1+\sqrt{x'}) {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x'\right) \\ (5.3.6) \quad &= (1+\sqrt{x'})z'. \end{aligned}$$

Thus, solving (5.3.6) for z' and using (5.3.1), we find that

$$(5.3.7) \quad z' = \frac{z}{1+\sqrt{x'}} = \frac{z}{1+\frac{1-\sqrt{1-x}}{1+\sqrt{1-x}}} = \frac{1}{2}(1+\sqrt{1-x})z.$$

Theorem 5.3.1 (Principle of Duplication). *Suppose that two sets of parameters, x , y , and z and x' , y' , and z' , are related by the equations (5.2.27), (5.2.29), and (5.2.30) with x , y , and z replaced by x' , y' , and z' , respectively. Suppose they satisfy an equation of the form*

$$\Omega(x', y', z') = 0,$$

and x is related to x' by (5.3.2). Then, by (5.3.1), (5.3.5), and (5.3.7), we obtain an equation of the form

$$(5.3.8) \quad \Omega\left(\left(\frac{1-\sqrt{1-x}}{1+\sqrt{1-x}}\right)^2, 2y, \frac{1}{2}(1+\sqrt{1-x})z\right) = 0.$$

Theorem 5.3.2 (Principle of Dimidiation). *As in the previous theorem, suppose that two sets of parameters, x , y , and z and x' , y' , and z' , are related by the equations (5.2.27), (5.2.29), and (5.2.30) with x ,*

y , and z replaced by x' , y' , and z' , respectively. Suppose they satisfy an equation of the form

$$\Omega(x', y', z') = 0,$$

and we reverse the roles of x , y , and z with those of x' , y' , and z' , respectively. Then, by (5.3.2), (5.3.5), and (5.3.6), we obtain an equation of the form

$$(5.3.9) \quad \Omega\left(\frac{4\sqrt{x}}{(1+\sqrt{x})^2}, \frac{1}{2}y, (1+\sqrt{x})z\right) = 0.$$

5.4. A Catalogue of Formulas for Theta Functions and Eisenstein Series

Using (5.2.29), (5.2.27), the principles of duplication and dimidiation from the previous section, and elementary theta function identities, such as those given in Theorem 1.3.10, we can derive a plethora of evaluations of the functions φ , ψ , f , and χ at different powers of the argument q in terms of z , x , and q . Using the tools mentioned above, readers should be able to easily derive each of the formulas below. Proofs of all the results in Theorems 5.4.1–5.4.4 can be found in [33, pp. 122–125, Entries 10–12]. After we state the formulas, we offer a few proofs in illustration.

Theorem 5.4.1. *If x , q , and z are related by (5.2.27), (5.2.29), and (5.2.30), then*

- (i) $\varphi(q) = \sqrt{z},$
- (ii) $\varphi(-q) = \sqrt{z}(1-x)^{1/4},$
- (iii) $\varphi(-q^2) = \sqrt{z}(1-x)^{1/8},$
- (iv) $\varphi(q^2) = \sqrt{z}\sqrt{\frac{1}{2}(1+\sqrt{1-x})},$
- (v) $\varphi(q^4) = \frac{1}{2}\sqrt{z}\left(1+(1-x)^{1/4}\right),$
- (vi) $\varphi(\sqrt{q}) = \sqrt{z}(1+\sqrt{x})^{1/2},$
- (vii) $\varphi(-\sqrt{q}) = \sqrt{z}(1-\sqrt{x})^{1/2}.$

Proof. We offer proofs of (ii)–(iv). To prove (ii)–(iv), employ (i) and use, respectively, (5.2.27), (1.3.32), and (3.6.7). $\square$

Theorem 5.4.2. *If x , q , and z are related by (5.2.27), (5.2.29), and (5.2.30), then*

- (i) $\psi(q) = \sqrt{\frac{1}{2}z(x/q)^{1/8}},$
- (ii) $\psi(-q) = \sqrt{\frac{1}{2}z(x(1-x)/q)^{1/8}},$
- (iii) $\psi(q^2) = \frac{1}{2}\sqrt{z}(x/q)^{1/4},$
- (iv) $\psi(q^4) = \frac{1}{2}\sqrt{\frac{1}{2}z}\{(1-\sqrt{1-x})/q\}^{1/2},$
- (v) $\psi(q^8) = \frac{1}{4}\sqrt{z}\{1-(1-x)^{1/4}\}/q,$
- (vi) $\psi(\sqrt{q}) = \sqrt{z}\{\frac{1}{2}(1+\sqrt{x})\}^{1/4}(x/q)^{1/16},$
- (vii) $\psi(-\sqrt{q}) = \sqrt{z}\{\frac{1}{2}(1-\sqrt{x})\}^{1/4}(x/q)^{1/16}.$

Proof. We prove only (i)–(iii). To prove (iii), use (3.6.8) along with Theorem 5.4.1(i), (ii). To prove (i), use (iii) and the principle of dimidiation. Part (ii) follows from (1.3.30) and Theorems 5.4.1(i), (ii) and 5.4.2(i). $\square$

Theorem 5.4.3. *If x , q , and z are related by (5.2.27), (5.2.29), and (5.2.30), then*

- (i) $f(q) = \sqrt{z}2^{-1/6}\{x(1-x)/q\}^{1/24},$
- (ii) $f(-q) = \sqrt{z}2^{-1/6}(1-x)^{1/6}(x/q)^{1/24},$
- (iii) $f(-q^2) = \sqrt{z}2^{-1/3}\{x(1-x)/q\}^{1/12},$
- (iv) $f(-q^4) = \sqrt{z}4^{-1/3}(1-x)^{1/24}(x/q)^{1/6}.$

Proof. We prove (i)–(iii). Both (ii) and (iii) follow from (1.3.34) and previously proved results. Next employ (1.3.30) to prove (i). $\square$

Theorem 5.4.4. *If x , q , and z are related by (5.2.27), (5.2.29), and (5.2.30), then*

- (i) $\chi(q) = 2^{1/6}\{x(1-x)/q\}^{-1/24},$
- (ii) $\chi(-q) = 2^{1/6}(1-x)^{1/12}(x/q)^{-1/24},$
- (iii) $\chi(-q^2) = 2^{1/3}(1-x)^{1/24}(x/q)^{-1/12}.$

Exercise 5.4.5. Prove some further results in Theorems 5.4.1–5.4.4.

We conclude this chapter by deriving representations for $P(q)$, $P(q^2)$, $Q(q)$, $Q(q^2)$, $R(q)$, and $R(q^2)$ in terms of x and z . However, first we need the following two important theorems, the first of which readers can easily prove.

Theorem 5.4.6. The function $z = {}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; x)$ satisfies the differential equation

$$(5.4.1) \quad x(1-x)\frac{d^2z}{dx^2} + (1-2x)\frac{dz}{dx} - \frac{1}{4}z = 0.$$

Exercise 5.4.7. The differential equation (5.4.1) has a regular singular point at $x = 0$. Solve (5.4.1) by the method of Frobenius to find that one of the two linearly independent solutions is ${}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; x)$.

Those readers unfamiliar with solving ordinary differential equations with regular singular points may use the definition of $z = {}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; x)$ to easily check that $z = {}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; x)$ is indeed a solution of (5.4.1).

Theorem 5.4.8. If y is defined by (5.2.30) and z is defined by (5.2.29), then

$$(5.4.2) \quad \frac{dy}{dx} = -\frac{1}{x(1-x)z^2}.$$

Proof. Using (5.2.27), (5.2.29), (1.2.4), and Theorem 5.4.3(ii), (iv), we can readily establish the identity

$$(5.4.3) \quad \frac{x}{1-x} = 16q \frac{f^8(-q^4)}{f^8(-q)} = 16q \frac{(q^4; q^4)_\infty^8}{(q; q)_\infty^8}.$$

Take the logarithm of both sides of (5.4.3), differentiate with respect to q , and multiply both sides by q to deduce that

$$(5.4.4) \quad \begin{aligned} \frac{q}{x(1-x)} \frac{dx}{dq} &= 1 - 8 \sum_{n=1}^{\infty} \frac{4nq^{4n}}{1-q^{4n}} + 8 \sum_{n=1}^{\infty} \frac{nq^n}{1-q^n} \\ &= 1 + 8 \sum_{n=1}^{\infty} \frac{nq^n}{1-q^n} = \varphi^4(q) = z^2, \end{aligned}$$

where we have employed the fifth line in the display (3.3.7), and (5.2.29). Using the trivial fact

$$\frac{dq}{dx} = -q \frac{dy}{dx}$$

in (5.4.4), we complete the proof. $\square$

Theorem 5.4.9. *With z defined by (5.2.29),*

$$(5.4.5) \quad P(q) = (1 - 5x)z^2 + 12x(1 - x)z \frac{dz}{dx},$$

$$(5.4.6) \quad P(q^2) = (1 - 2x)z^2 + 6x(1 - x)z \frac{dz}{dx}.$$

Proof. We prove only (5.4.6); the proof of (5.4.5) is similar. Applying the definition of $P(q)$, Theorem 5.4.3(iii), the definition of $f(-q)$ from (1.2.4), and lastly the chain rule along with Theorem 5.4.8, we find that

$$\begin{aligned} P(q^2) &= 1 - 24 \sum_{n=1}^{\infty} \frac{ne^{-2ny}}{1 - e^{-2ny}} \\ &= 1 - 12 \frac{d}{dy} \sum_{n=1}^{\infty} \log(1 - e^{-2ny}) \\ &= 1 - 12 \frac{d}{dy} \log(e^{-2y}; e^{-2y})_{\infty} \\ &= -\frac{d}{dy} \log\{e^{-y} f^{12}(-e^{-2y})\} \\ &= -\frac{d}{dy} \log\{e^{-y} z^6 2^{-4} x(1 - x) e^y\} \\ &= -\frac{d}{dy} \log\{z^6 x(1 - x)\} \\ &= x(1 - x) z^2 \frac{d}{dx} \log\{z^6 x(1 - x)\} \\ &= (1 - x) z^2 - x z^2 + 6x(1 - x) z \frac{dz}{dx} \\ &= (1 - 2x) z^2 + 6x(1 - x) z \frac{dz}{dx}. \end{aligned}$$

Exercise 5.4.10. *Alternatively, apply the process of dimidiation to (5.4.6) to derive (5.4.5).*

Theorem 5.4.11. *We have*

$$(5.4.7) \quad Q(q) = z^4(1 + 14x + x^2),$$

$$(5.4.8) \quad Q(q^2) = z^4(1 - x + x^2).$$

Proof. We first prove (5.4.8). Secondly, we establish (5.4.7) by applying the process of dimidiation to (5.4.8).

From the differential equation (4.2.20) and the chain rule, we find that

$$(5.4.9) \quad q \frac{dP(q^2)}{dq} = \frac{P^2(q^2) - Q(q^2)}{6}.$$

From the definition of q , i.e., (5.2.30), Theorem 5.4.8, and the chain rule, we next find that

$$(5.4.10) \quad \frac{dq}{dx} = -\frac{dy}{dx}q = \frac{q}{x(1-x)z^2}.$$

It follows from (5.4.9) and (5.4.10) that

$$(5.4.11) \quad x(1-x)z^2 \frac{dP(q^2)}{dx} = \frac{P^2(q^2) - Q(q^2)}{6}.$$

We next use (5.4.11) and Theorem 5.4.9 to establish (5.4.8).

Differentiating (5.4.6) with respect to x , applying the product rule several times, and simplifying, we deduce that

$$\begin{aligned} (5.4.12) \quad & \frac{dP(q^2)}{dx} \\ &= -2z^2 + 8z(1-2x)\frac{dz}{dx} + 6x(1-x)\left(\frac{dz}{dx}\right)^2 + 6x(1-x)z\frac{d^2z}{dx^2} \\ &= -2z\left(z - 4(1-2x)\frac{dz}{dx}\right) + 6x(1-x)\left(\frac{dz}{dx}\right)^2 + 6x(1-x)z\frac{d^2z}{dx^2} \\ &= -2z \cdot 4x(1-x)\frac{d^2z}{dx^2} + 6x(1-x)\left(\frac{dz}{dx}\right)^2 + 6x(1-x)z\frac{d^2z}{dx^2} \\ &= -2zx(1-x)\frac{d^2z}{dx^2} + 6x(1-x)\left(\frac{dz}{dx}\right)^2 \end{aligned}$$

$$= -2x(1-x) \left(z \frac{d^2 z}{dx^2} - 3 \left(\frac{dz}{dx} \right)^2 \right),$$

where in the antepenultimate step we employed the differential equation (5.4.1). Using (5.4.12) and (5.4.6) in (5.4.11), we find that

$$\begin{aligned} Q(q^2) &= \left((1-2x)z^2 + 6x(1-x)z \frac{dz}{dx} \right)^2 \\ &\quad - 6x(1-x)z^2 \left(-2x(1-x) \left(z \frac{d^2 z}{dx^2} - 3 \left(\frac{dz}{dx} \right)^2 \right) \right) \\ &= (1-2x)^2 z^4 + 12x(1-x)(1-2x)z^3 \frac{dz}{dx} + 12x^2(1-x)^2 z^3 \frac{d^2 z}{dx^2} \\ &= (1-2x)^2 z^4 + 12x(1-x)(1-2x)z^3 \frac{dz}{dx} \\ &\quad + 12x(1-x)z^3 \left(-(1-2x) \frac{dz}{dx} + \frac{1}{4}z \right) \\ &= (1-x+x^2)z^4, \end{aligned}$$

where in the penultimate line we utilized the differential equation (5.4.1). This completes the proof of (5.4.8).

We now use the Principle of Dimidiation to establish (5.4.7). Accordingly, from (5.4.8),

$$\begin{aligned} Q(q) &= \left(1 - \frac{4\sqrt{x}}{(1+\sqrt{x})^2} + \frac{16x}{(1+\sqrt{x})^4} \right) (1+\sqrt{x})^4 z^4 \\ &= (1+14x+x^2)z^4, \end{aligned}$$

after simplification. □

Theorem 5.4.12. *We have*

$$(5.4.13) \quad R(q) = z^6(1+x)(1-34x+x^2),$$

$$(5.4.14) \quad R(q^2) = z^6(1+x)(1-\tfrac{1}{2}x)(1-2x).$$

Proof. We begin by proving (5.4.14) and proceed as we did in the derivation of (5.4.11) of the preceding proof. Using (4.2.21), (5.2.30), (5.4.10), Theorem 5.4.8, and the chain rule, we deduce that

$$(5.4.15) \quad 3x(1-x)z^2 \frac{dQ(q^2)}{dx} = 2P(q^2)Q(q^2) - 2R(q^2).$$

Solving (5.4.15) for $R(q^2)$ and employing (5.4.6) and (5.4.8) (twice), we find that

$$\begin{aligned}
 R(q^2) &= P(q^2)Q(q^2) - \frac{3}{2}x(1-x)z^2 \frac{dQ(q^2)}{dx} \\
 &= \left\{ (1-2x)z^2 + 6x(1-x)z \frac{dz}{dx} \right\} z^4(1-x+x^2) \\
 &\quad - \frac{3}{2}x(1-x)z^2 \left\{ 4z^3 \frac{dz}{dx}(1-x+x^2) + z^4(-1+2x) \right\} \\
 &= z^6 \left\{ (1-2x)(1-x+x^2) + \frac{3}{2}x(1-x)(1-2x) \right\} \\
 &= z^6(1-2x)(1+x) \left(1 - \frac{1}{2}x \right),
 \end{aligned}$$

which completes the proof of (5.4.14).

Applying the Principle of Dimidiation to (5.4.14), we find that

$$\begin{aligned}
 R(q) &= z^6(1+\sqrt{x})^6 \left(1 + \frac{4\sqrt{x}}{(1+\sqrt{x})^2} \right) \left(1 - \frac{8\sqrt{x}}{(1+\sqrt{x})^2} \right) \\
 &\quad \times \left(1 - \frac{2\sqrt{x}}{(1+\sqrt{x})^2} \right) \\
 &= z^6(1+6\sqrt{x}+x)(1-6\sqrt{x}+x)(1+x) \\
 &= z^6(1+x)(1-34x+x^2),
 \end{aligned}$$

which completes the proof of (5.4.13). □

Exercise 5.4.13. *Prove that*

$$\begin{aligned}
 P(q^4) &= z^2 \left(1 - \frac{5}{4}x \right) + 3x(1-x)z \frac{dz}{dx}, \\
 Q(q^4) &= z^4 \left(1 - x + \frac{1}{16}x^2 \right), \\
 R(q^4) &= z^6 \left(1 - \frac{1}{2}x \right) \left(1 - x - \frac{1}{32}x^2 \right).
 \end{aligned}$$

5.5. Notes

We have followed Ramanujan's presentation in Chapter 17 of his second notebook [193], as presented by the author in [34, pp. 91–102, Sections 2–6], in our development of Theorem 5.2.8. However, the clever proof of Theorem 5.4.8 that we have given was communicated to us by H. H. Chan and is much simpler than the proof in [34, p. 120];

see his paper with Y. L. Ong [71]. The Principles of Duplication and Dimidiation are due to Jacobi [131]. Ramanujan states the Principle of Duplication in Entry 13(vii) of Chapter 17 in his second notebook [193], [34, p. 127]. It is clear that Ramanujan frequently used the Principles of Duplication and Dimidiation in his work on elliptic functions. Our presentation of these principles is similar to that given in the author's book [34, pp. 125–126].

For Ramanujan, it was natural to ask whether there were other theories of elliptic functions in which the hypergeometric function ${}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; x)$ is replaced by other hypergeometric functions. In his famous paper, *Modular equations and approximations to π* [184], [192, pp. 23–39], Ramanujan wrote, “There are corresponding theories in which q is replaced by one or other of the functions

$$q_1 = e^{-\pi K'_1 \sqrt{2}/K_1}, \quad q_2 = e^{-2\pi K'_2/(K_2 \sqrt{3})}, \quad q_3 = e^{-2\pi K'_3/K_3},$$

where

$$\begin{aligned} K_1 &= {}_2F_1\left(\frac{1}{4}, \frac{3}{4}; 1; x\right), \\ K_2 &= {}_2F_1\left(\frac{1}{3}, \frac{2}{3}; 1; x\right), \\ K_3 &= {}_2F_1\left(\frac{1}{6}, \frac{5}{6}; 1; x\right). \end{aligned}$$

Pages 257–262 in Ramanujan's second notebook are devoted to developing these theories. The many claims on these six pages were first proved by Berndt, S. Bhargava, and F. G. Garvan [41]. See also Chapter 33 in Berndt's book [38]. However, Ramanujan's theories are by no means complete, and since the appearance of [41], many other papers have been written on Ramanujan's alternative theories, with still much remaining to be accomplished.

The formula (5.1.6) is due to the Scottish mathematician C. Maclaurin [153] in 1742. Our proof of Landen's transformation, Theorem 5.1.6, is almost identical to the proof that the English astronomer J. Ivory [129] gave in 1796 for (5.1.6). When Ivory submitted his paper to editor John Playfair, he naturally included a cover letter, but, rather surprisingly (at least to us), the cover letter was published along with the paper. In this letter, he relates how he discovered his formula. “Having, as you know, bestowed a good deal of time and attention on the study of that part of physical astronomy

which relates to the mutual disturbances of the planets, I have, naturally, been led to consider the various methods of resolving the formula $(a^2 + b^2 - 2ab \cos \phi)^n$ into infinite series of the form $A + B \cos \phi + C \cos 2\phi + \&c$. In the course of these investigations, a series for the rectification of the ellipsis occurred to me, remarkable for its simplicity, as well as its rapid convergency. As I believe it to be new, I send it to you”

Landen’s transformation was introduced by J. Landen in a paper written in 1771 [137] but developed more completely in his paper [138] published in 1775. This transformation was crucial in our proof of the fundamental Theorem 5.2.8. The importance of Landen’s transformation is conveyed by G. Mittag-Leffler, who, in his survey article [160] on elliptic functions written in 1923, emphasizes, “Euler’s addition theorem and the transformation theorem of Landen and Lagrange were the two fundamental ideas of which the theory of elliptic functions was in possession when this theory was brought up for renewed consideration by Legendre in 1786.”

Born in 1719, Landen was appointed as the land-agent to the Earl Fitzwilliam, a post he held until his retirement two years before his death in 1790. According to an edition of *Encyclopedia Britannica* published in 1882, “He [Landen] lived a very retired life, and saw little or nothing of society; when he did mingle in it, his dogmatism and pugnacity caused him to be generally shunned.” Landen made several contributions to the *Ladies Diary*, which was published in England from 1704 to 1816 and “designed principally for the amusement and instruction of the fair sex.” As was common with other contributors, Landen frequently used pseudonyms, such as Sir Stately Stiff, Peter Walton, Waltonienses, C. Bumpkin, and Peter Puzzlem, for problems he proposed and solved. The largest portion of each issue was devoted to the presentation of mathematical problems and their solutions. Despite its name, of the 913 contributors of mathematical problems and solutions over the years of its publication, only 32 were women. For additional information about Landen and the *Ladies Diary*, see a paper by G. Almkvist and the author [11]. Readers are also recommended to read G. N. Watson’s article, *The marquis and the land-agent; a tale of the eighteenth century* [217]. (You know the

identity of the land-agent; to augment your curiosity, we refrain from telling you the identity of the marquis.)

Exercise 5.1.9 is just one of many beautiful transformation formulas for elliptic integrals, many of which are due to Jacobi and/or Ramanujan [34, pp. 104–113]. We offer a few of these transformation formulas as exercises.

Exercise 5.5.1. If $0 < \alpha, \beta < \frac{1}{2}\pi$ and $\tan \alpha = \sqrt{1-x} \tan \beta$, show that

$$\int_0^\alpha \frac{d\phi}{\sqrt{1-x \cos^2 \phi}} = \int_0^\beta \frac{d\phi}{\sqrt{1-x \sin^2 \phi}}.$$

The next exercise is a form of the *addition theorem* in the theory of elliptic functions.

Exercise 5.5.2. If $0 < \alpha, \beta < \frac{1}{2}\pi$ and $\cot \alpha \cot \beta = \sqrt{1-x}$, prove that

$$\int_0^\alpha \frac{d\phi}{\sqrt{1-x \sin^2 \phi}} + \int_0^\beta \frac{d\phi}{\sqrt{1-x \sin^2 \phi}} = \frac{\pi}{2} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; x\right).$$

The following exercise gives the classical *duplication formula* for elliptic integrals.

Exercise 5.5.3. If $0 < \alpha, \beta < \frac{1}{2}\pi$ and $\cot \alpha \tan(\beta/2) = \sqrt{1-x \sin^2 \alpha}$, prove that

$$2 \int_0^\alpha \frac{d\phi}{\sqrt{1-x \sin^2 \phi}} = \int_0^\beta \frac{d\phi}{\sqrt{1-x \sin^2 \phi}}.$$

Exercise 5.5.4. If $0 < \alpha, \beta < \frac{1}{2}\pi$ and $(1+x \sin^2 \alpha) \sin \beta = (1+x) \sin \alpha$, prove that

$$(1+x) \int_0^\alpha \frac{d\phi}{\sqrt{1-x^2 \sin^2 \phi}} = \int_0^\beta \frac{d\phi}{\sqrt{1-\frac{4x}{(1+x)^2} \sin^2 \phi}}.$$

The transformation above is called Gauss's transformation and is similar in form to Landen's transformation given in Exercise 5.1.9.

Historically, the theory of elliptic functions arose from the problem of inverting incomplete elliptic integrals, such as the incomplete

elliptic integral of the first kind given in Definition 5.1.8. These inversion problems were motivated by the well-known inversion of the trigonometric integral

$$\arcsin x = \int_0^x \frac{d\phi}{\sqrt{1-\phi^2}}, \quad 0 < x < 1.$$

Landen's transformation for incomplete elliptic integrals is only one example of many beautiful relations that exist between elliptic integrals, many of which are due to Jacobi [131] and Ramanujan [193, Chapter 17, Section 8], [34, pp. 104–113]. The text [180, Chapters 2–4] by V. Prasolov and Y. Solov'yev is a superb introduction to elliptic integrals describing the fundamental theorems of Legendre, Jacobi, and N. H. Abel.

Lemma 5.1.10 is a primitive, special case of a much more general asymptotic formula for Gaussian **hypergeometric** series; for example, see [33, pp. 77–78].

The catalogue of formulas for theta functions given in Section 5.4 is taken from Sections 10–12 of Chapter 17 in Ramanujan's second notebook [193], [33, pp. 122–124]. The formulas for $Q(q)$ and $R(q)$ in Theorems 5.4.11 and 5.4.12 can be found in Section 13 of Chapter 17 in Ramanujan's second notebook [193], [34, pp. 126–127]. Although the formulas for $P(q)$ in Theorem 5.4.9 are not found in Section 13, from the appearances of other formulas for $P(q)$ in that section, it is clear that Ramanujan knew Theorem 5.4.9. Ramanujan used his formulas for $P(q)$, $Q(q)$, and $R(q)$ to derive a multitude of elegant evaluations of infinite series in Sections 13–17 of Chapter 17 [34, pp. 126–139]. For example,

$$1 + 2 \sum_{k=1}^{\infty} \frac{1}{\cosh(ky)} = z$$

and

$$\sum_{k=0}^{\infty} \frac{(-1)^k (2k+1)}{\cosh\{\frac{1}{2}(2k+1)y\}} = \frac{1}{2} z^2 \sqrt{x(1-x)}.$$

See also Corollary 6.2.10.

Chapter 6

Applications of the Primary Theorem of Chapter 5

6.1. Introduction

Our goal in this chapter is to provide some applications of Theorem 5.2.8 and the several representations of theta functions and Eisenstein series in terms of z and x that arose from Theorem 5.2.8. We first demonstrate in Section 6.2 how the formulas we derived for Eisenstein series in Chapter 5 can be applied to give simple proofs of the formulas for $r_4(n)$ and $r_8(n)$ that we proved in Chapter 3. We also derive an elegant formula for $t_8(n)$.

In Section 6.3 we define one of the most important concepts in the theory of elliptic functions, namely, a *modular equation*. Landen's transformation, Theorem 5.1.6, can be thought of as a modular equation of degree 2. However, the theory of modular equations more properly began in 1825 with Legendre's discovery of a modular equation of degree 3. The construction of modular equations is not easy. There is no single method that one can use to produce modular equations. In Section 6.3, we shall derive some modular equations of degree 3.

6.2. Sums of Squares and Triangular Numbers

We first offer some corollaries of Theorem 5.4.9. Recall that $0 < x < 1$ and q and z are defined by (5.2.30) and (5.2.29), respectively.

Corollary 6.2.1. *We have*

$$(6.2.1) \quad 2P(q^2) - P(q) = 1 + 24 \sum_{n=1}^{\infty} \frac{nq^n}{1+q^n} = z^2(1+x).$$

Proof. Applying Theorem 5.4.9, we immediately deduce that

$$(6.2.2) \quad 2P(q^2) - P(q) = z^2(1+x).$$

On the other hand,

$$(6.2.3) \quad 2P(q^2) - P(q) = 1 + 24 \sum_{n=1}^{\infty} \left(-\frac{2nq^{2n}}{1-q^{2n}} + \frac{nq^n}{1-q^n} \right) = 1 + 24 \sum_{n=1}^{\infty} \frac{nq^n}{1+q^n}.$$

Combining (6.2.2) and (6.2.3), we deduce (6.2.1). $\square$

Corollary 6.2.2. *We have*

$$2P(q^4) - P(q^2) = 1 + 24 \sum_{n=1}^{\infty} \frac{nq^{2n}}{1+q^{2n}} = z^2 \left(1 - \frac{1}{2}x \right).$$

Proof. Using the process of duplication along with Corollary 6.2.1, we find that

$$\begin{aligned} 2P(q^4) - P(q^2) &= 1 + 24 \sum_{n=1}^{\infty} \frac{nq^{2n}}{1+q^{2n}} \\ &= \frac{1}{4}z^2 (1 + \sqrt{1-x})^2 \left(1 + \left(\frac{1 - \sqrt{1-x}}{1 + \sqrt{1-x}} \right)^2 \right) \\ &= z^2 \left(1 - \frac{1}{2}x \right). \end{aligned}$$

$\square$

Corollary 6.2.3. *We have*

$$1 + 8 \sum_{n=1}^{\infty} \frac{(-1)^n nq^n}{1+q^n} = z^2(1-x).$$

Proof. Using Corollaries 6.2.2 and 6.2.1, we find that

$$3 \left(1 + 8 \sum_{n=1}^{\infty} \frac{(-1)^n n q^n}{1 + q^n} \right) = 4 \left(1 + 24 \sum_{n=1}^{\infty} \frac{n q^{2n}}{1 + q^{2n}} \right) \\ - \left(1 + 24 \sum_{n=1}^{\infty} \frac{n q^n}{1 + q^n} \right) = 4z^2 \left(1 - \frac{1}{2}x \right) - z^2(1 + x) = 3z^2(1 - x),$$

from which Corollary 6.2.3 is immediate. $\square$

Theorem 6.2.4. *For each positive integer n ,*

$$(6.2.4) \quad r_4(n) = 8 \sum_{\substack{d|n \\ 4 \nmid d}} d.$$

First Proof of Theorem 6.2.4. From Theorem 5.4.1(ii) and Corollary 6.2.3,

$$\varphi^4(-q) = z^2(1 - x) = 1 + 8 \sum_{n=1}^{\infty} \frac{(-1)^n n q^n}{1 + q^n},$$

that is, with q replaced by $-q$,

$$\varphi^4(q) = 1 + 8 \sum_{n=1}^{\infty} \frac{n q^n}{1 + (-q)^n},$$

which arises from (3.3.12) and (3.3.13) in our second proof of Theorem 3.3.1, i.e., Theorem 6.2.4. Thus, the remainder of the proof of Theorem 6.2.4 is identical to that of the aforementioned proof. $\square$

We now give a second proof of Theorem 6.2.4, which is a variation of our first proof above.

Second Proof of Theorem 6.2.4. Recall that $\sigma(n) = \sum_{d|n} d$. We extend the definition of $\sigma(n)$ by setting $\sigma(n) = 0$, if n is not an integer. With this in mind, applying Theorem 5.4.1, Theorem 5.4.9,

and Exercise 5.4.13, we find that

$$\begin{aligned}\sum_{n=0}^{\infty} r_4(n)q^n &= \varphi^4(q) = z^2 = \frac{4}{3}P(q^4) - \frac{1}{3}P(q) \\ &= \frac{4}{3} \left(1 - 24 \sum_{n=1}^{\infty} \sigma(n/4)q^n \right) - \frac{1}{3} \left(1 - 24 \sum_{n=1}^{\infty} \sigma(n)q^n \right) \\ &= 1 + \sum_{n=1}^{\infty} (8\sigma(n) - 32\sigma(n/4))q^n.\end{aligned}$$

Equating coefficients of q^n , $n \geq 1$, on the extremal sides above, we find that

$$r_4(n) = 8\sigma(n) - 32\sigma(n/4) = 8 \sum_{\substack{d|n \\ 4 \nmid d}} d,$$

which completes our second proof of Theorem 6.2.4. $\square$

To re-derive Jacobi's formula for $r_8(n)$, we need the following consequence of Theorem 5.4.11.

Corollary 6.2.5. *We have*

$$(6.2.5) \quad 16Q(q^2) - Q(q) = 15 \left(1 + 16 \sum_{n=1}^{\infty} \frac{(-1)^n n^3 q^n}{1 - q^n} \right) = 15z^4(1 - x)^2.$$

Proof. By a direct calculation,

$$\begin{aligned}16Q(q^2) - Q(q) &= 15 + 16 \cdot 240 \sum_{n=1}^{\infty} \frac{n^3 q^{2n}}{1 - q^{2n}} - 240 \sum_{n=1}^{\infty} \frac{n^3 q^n}{1 - q^n} \\ &= 15 \left(1 + 16 \sum_{n=1}^{\infty} \frac{(-1)^n n^3 q^n}{1 - q^n} \right).\end{aligned}$$

On the other hand, using Theorem 5.4.11, we find that

$$16Q(q^2) - Q(q) = 16z^4(1 - x + x^2) - z^4(1 + 14x + x^2) = 15z^4(1 - x)^2.$$

The desired result now follows from the last two identities. $\square$

Theorem 6.2.6. *For each positive integer n ,*

$$(6.2.6) \quad r_8(n) = 16(-1)^n \sum_{d|n} (-1)^d d^3.$$

Proof. From Theorem 5.4.1(ii) and Corollary 6.2.5,

$$\varphi^8(-q) = z^4(1-x)^2 = 1 + 16 \sum_{n=1}^{\infty} \frac{(-1)^n n^3 q^n}{1-q^n}.$$

Thus, replacing q by $-q$, we deduce that

$$\varphi^8(q) = 1 + 16 \sum_{n=1}^{\infty} \frac{n^3 q^n}{1-(-q)^n},$$

which is precisely (3.5.14) in our previous proof of Theorem 3.5.4, or Theorem 6.2.6. The remainder of the proof is then the same as before. $\square$

Exercise 6.2.7. Give another proof of Theorem 6.2.6 in the following manner. First, using formulas for $Q(q)$, $Q(q^2)$, and $Q(q^4)$ from Theorem 5.4.11 and Exercise 5.4.13, respectively, prove that

$$\sum_{n=0}^{\infty} r_8(n) q^n = z^4 = 1 + \sum_{n=1}^{\infty} (16\sigma_3(n) - 32\sigma_3(n/2) + 256\sigma_3(n/4)) q^n,$$

where $\sigma_3(m) = 0$, if m is not an integer. Second, by considering the cases n even and n odd separately, show that

$$16\sigma_3(n) - 32\sigma_3(n/2) + 256\sigma_3(n/4) = 16(-1)^n \sum_{d|n} (-1)^d d^3.$$

We now turn our attention to sums of triangular numbers. First, we give a proof of Theorem 3.6.3 along the lines of the previous proofs in this section. The following simple exercise will be used in our proof below.

Exercise 6.2.8. If m is an even positive integer, then

$$\sigma(m) - 3\sigma(m/2) + 2\sigma(m/4) = 0.$$

Theorem 6.2.9. For each nonnegative integer n ,

$$t_4(n) = \sigma(2n+1).$$

Proof. By Theorem 5.4.2(iii),

$$(6.2.7) \quad \sum_{n=0}^{\infty} t_4(n) q^{2n+1} = q\psi^4(q^2) = \frac{1}{16}xz^2.$$

Then, using (6.2.7) in conjunction with Corollaries 6.2.1 and 6.2.2 and Exercise 6.2.8, we deduce that

$$\begin{aligned}\sum_{n=0}^{\infty} t_4(n) q^{2n+1} &= -\frac{1}{24} P(q) + \frac{1}{8} P(q^2) - \frac{1}{12} P(q^4) \\ &= \sum_{m=1}^{\infty} (\sigma(m) - 3\sigma(m/2) + 2\sigma(m/4)) q^m \\ &= \sum_{n=0}^{\infty} \sigma(2n+1) q^{2n+1}.\end{aligned}$$

Equating the coefficients of q^{2n+1} on the extremal sides above, we complete the proof. $\square$

In the notes to Chapter 3, we mentioned a formula for $t_8(n)$, which we now prove. We first need another corollary.

Corollary 6.2.10. *We have*

$$\sum_{k=1}^{\infty} \frac{k^3}{\sinh(ky)} = \frac{1}{8} z^4 x.$$

Proof. Using the definition (5.2.30) and Theorem 5.4.11, we find that

$$\begin{aligned}(6.2.8) \quad \sum_{k=1}^{\infty} \frac{k^3}{\sinh(ky)} &= 2 \sum_{k=1}^{\infty} \frac{k^3}{e^{ky} - e^{-ky}} = 2 \sum_{k=1}^{\infty} \frac{k^3 e^{-ky}}{1 - e^{-2ky}} \\ &= 2 \sum_{k=1}^{\infty} k^3 \left(\frac{q^k}{1 - q^k} - \frac{q^{2k}}{1 - q^{2k}} \right) \\ &= \frac{1}{120} (Q(q) - Q(q^2)) \\ &= \frac{1}{120} z^4 (1 + 14x + x^2 - 1 + x - x^2) \\ &= \frac{1}{8} z^4 x.\end{aligned}$$

$\square$

Corollary 6.2.11. *We have*

$$q\psi^8(q) = \sum_{k=1}^{\infty} \frac{k^3 q^k}{1 - q^{2k}}.$$

Proof. From Theorem 5.4.2(i),

$$q\psi^8(q) = \frac{1}{16}z^4x.$$

Employing Corollary 6.2.10 and (6.2.8), we complete the proof. $\square$

Theorem 6.2.12. *For each positive integer n ,*

$$t_8(n) = \sum_{\substack{d|(n+1) \\ (n+1)/d \text{ odd}}} d^3.$$

Proof. By Corollary 6.2.11,

$$q\psi^8(q) = \sum_{k=1}^{\infty} k^3 q^k \sum_{m=0}^{\infty} q^{2km} = \sum_{n=1}^{\infty} \sum_{\substack{d|n \\ n/d \text{ odd}}} d^3 q^n.$$

Equating coefficients of q^{n+1} on both sides above, we complete the proof. $\square$

Recall that in Theorem 4.2.4 we established the following fundamental theorem.

Theorem 6.2.13. *We have*

$$(6.2.9) \quad Q^3(q) - R^2(q) = 1728q(q; q)^{24}.$$

Our representations from Theorems 5.4.11 and 5.4.12 enable us to give a very simple proof of Theorem 6.2.13.

Proof. It will be simpler to use the argument q^2 instead of q . Thus, from Theorems 5.4.11 and 5.4.12, we find that

$$(6.2.10) \quad \begin{aligned} Q^3(q^2) - R^2(q^2) &= z^{12}(1-x+x^2)^3 - z^{12}(1+x)^2(1-\tfrac{1}{2}x)^2(1-2x)^2 \\ &= \tfrac{1}{4}3^3 z^{12} x^2 (1-x)^2, \end{aligned}$$

after a dosage of elementary algebra. On the other hand, by Theorem 5.4.3(iii),

$$(6.2.11) \quad \begin{aligned} 1728q^2(q^2; q^2)_{\infty}^{24} &= 1728q^2 f^{24}(-q^2) = 2^6 \cdot 3^3 q^2 z^{12} 2^{-8} x^2 (1-x)^2 / q^2 \\ &= \tfrac{1}{4}3^3 z^{12} x^2 (1-x)^2. \end{aligned}$$

Combining (6.2.10) and (6.2.11) and replacing q^2 by q , we complete the proof. $\square$

6.3. Modular Equations

Definition 6.3.1. Let K , K' , L , and L' denote complete elliptic integrals of the first kind associated with the moduli k , $k' := \sqrt{1 - k^2}$, ℓ , and $\ell' := \sqrt{1 - \ell^2}$, respectively, where $0 < k, \ell < 1$. Suppose that

$$(6.3.1) \quad n \frac{K'}{K} = \frac{L'}{L}$$

for some positive integer n . A relation between k and ℓ induced by (6.3.1) is called a modular equation of degree n .

Following Ramanujan, set

$$\alpha = k^2 \quad \text{and} \quad \beta = \ell^2.$$

We often say that β has degree n over α .

Using Lemma 5.1.3, we may replace the defining relation (6.3.1) for a modular equation by the equivalent relation

$$(6.3.2) \quad n \frac{{}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; 1 - \alpha)}{{}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; \alpha)} = \frac{{}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; 1 - \beta)}{{}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; \beta)}.$$

Using (5.2.29) and the formulas from Section 5.4, we see that a modular equation can be considered as an identity amongst theta functions with arguments q and theta functions with arguments q^n . In fact, most often one establishes a modular equation of degree n by first proving the requisite theta function identity. Then we use the formulas from Section 5.4 to express theta functions with argument q in terms of α , $z = z_1$, and (possibly) q , and the theta functions with argument q^n in terms of β , z_n , and (possibly) q^n , where

$$(6.3.3) \quad z_n := \varphi^2(q^n).$$

The multiplier m of degree n is defined by

$$(6.3.4) \quad m = \frac{\varphi^2(q)}{\varphi^2(q^n)} = \frac{z_1}{z_n}.$$

We note that the method of establishing modular equations briefly described above always yields an algebraic relation between α and β . Although by no means obvious, every modular equation arising from (6.3.1) is an algebraic relation between α and β . As mentioned in the Introduction to this chapter, there is no single method one can use to

discover or construct modular equations. One needs to be resourceful and use a variety of tools. Generally, as the degree of the modular equation increases, the difficulty of establishing modular equations rises sharply. The following simple, but enormously useful, device enables us to produce new modular equations from previously derived modular equations.

Theorem 6.3.2 (Method of Reciprocation). *If we replace α by $1 - \beta$, β by $1 - \alpha$, and m by n/m in a modular equation of degree n , then we obtain a new modular equation of the same degree.*

Proof. If we make the indicated substitutions in the definition (6.3.4) of the multiplier, we find that

$$(6.3.5) \quad \frac{n}{m} = \frac{{}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; 1 - \beta)}{{}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; 1 - \alpha)}.$$

Rearranging (6.3.5) and using (6.3.4), we deduce that

$$n \frac{{}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; 1 - \alpha)}{{}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; \alpha)} = \frac{{}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; 1 - \beta)}{{}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; \beta)},$$

i.e., we obtain the defining relation (6.3.2) for a modular equation. $\square$

We now discuss modular equations of degree 3. We need several identities for Lambert series.

Theorem 6.3.3. *If $(\frac{n}{3})$ denotes the Legendre symbol, then*

- (i) $q\psi(q^2)\psi(q^6) = \sum_{n=0}^{\infty} \left(\frac{q^{6n+1}}{1 - q^{12n+2}} - \frac{q^{6n+5}}{1 - q^{12n+10}} \right),$
- (ii) $\varphi(q)\varphi(q^3) = 1 + 2 \sum_{n=1}^{\infty} \left(\frac{n}{3} \right) \frac{q^n}{1 + (-q)^n},$
- (iii) $\frac{\varphi^3(q)}{\varphi(q^3)} = 1 + 6 \sum_{n=1}^{\infty} \left(\frac{(-1)^n q^{3n+1}}{1 + (-q)^{3n+1}} + \frac{(-1)^n q^{3n+2}}{1 + (-q)^{3n+2}} \right),$
- (iv) $\frac{\psi^3(q)}{\psi(q^3)} = 1 + 3 \left(\sum_{n=0}^{\infty} \frac{q^{6n+1}}{1 - q^{6n+1}} - \sum_{n=0}^{\infty} \frac{q^{6n+5}}{1 - q^{6n+5}} \right)$

Part (i) is identical to Theorem 3.7.7, and part (ii) can be found in (3.7.8). Copyrighted Material

Proof of (iii). Replacing q^2 by q in (1.3.60) and employing the Jacobi triple product identity, equation (1.3.11), we find that

(6.3.6)

$$\begin{aligned}
 \varphi^2(-q)f(-q) &= \sum_{n=-\infty}^{\infty} (6n+1)q^{(3n^2+n)/2} \\
 &= \frac{d}{dz} \sum_{n=-\infty}^{\infty} q^{(3n^2+n)/2} z^{6n+1} \Big|_{z=1} \\
 &= \frac{d}{dz} \{zf(q^2z^6, q/z^6)\} \Big|_{z=1} \\
 &= f(q, q^2) \frac{d}{dz} (\log \{zf(q^2z^6, q/z^6)\}) \Big|_{z=1} \\
 &= f(q, q^2) \left(1 + \frac{d}{dz} (\log(-q/z^6; q^3)_{\infty} (-q^2z^6; q^3)_{\infty} (q^3; q^3)_{\infty}) \Big|_{z=1} \right).
 \end{aligned}$$

Using the Maclaurin series for $\log(1+z)$, inverting the order of summation, and then summing the resulting geometric series, we find that, for $|a| < 1$,

$$\begin{aligned}
 (6.3.7) \quad \log(-a; q^3)_{\infty} &= \sum_{n=0}^{\infty} \log(1 + aq^{3n}) \\
 &= - \sum_{n=0}^{\infty} \sum_{m=1}^{\infty} \frac{(-1)^m (aq^{3n})^m}{m} \\
 &= - \sum_{m=1}^{\infty} \frac{(-a)^m}{m(1 - q^{3m})}.
 \end{aligned}$$

Using (6.3.7) in (6.3.6), we find that

$$\begin{aligned}
 \frac{\varphi^2(-q)f(-q)}{f(q, q^2)} &= \left(1 - \frac{d}{dz} \sum_{n=1}^{\infty} \frac{(-qz^{-6})^n}{n(1 - q^{3n})} - \frac{d}{dz} \sum_{n=1}^{\infty} \frac{(-q^2z^6)^n}{n(1 - q^{3n})} \right) \Big|_{z=1} \\
 &= 1 + 6 \sum_{n=1}^{\infty} \frac{(-1)^n q^n}{1 - q^{3n}} - 6 \sum_{n=1}^{\infty} \frac{(-1)^n q^{2n}}{1 - q^{3n}} \\
 (6.3.8) \quad &= 1 - 6 \sum_{n=0}^{\infty} \frac{q^{3n+1}}{1 + q^{3n+1}} + 6 \sum_{n=0}^{\infty} \frac{q^{3n+2}}{1 + q^{3n+2}},
 \end{aligned}$$

where we expanded the summands in geometric series and then inverted the order of summation. Now, by Theorem 1.3.9, (1.3.15), and

(3.2.7),

$$\begin{aligned}
 \frac{\varphi^2(q)f(q)}{f(-q, q^2)} &= \frac{\varphi^2(q)(-q; -q)_\infty}{(q; -q^3)_\infty(-q^2; -q^3)_\infty(-q^3; -q^3)_\infty} \\
 &= \varphi^2(q) \frac{(-q; -q)_\infty}{(q; -q)_\infty} \frac{(q; -q)_\infty}{(q; -q^3)_\infty(-q^2; -q^3)_\infty} \frac{1}{(-q^3; -q^3)_\infty} \\
 &= \varphi^3(q) \frac{(q^3; -q^3)_\infty}{(-q^3; -q^3)_\infty} \\
 (6.3.9) \quad &= \frac{\varphi^3(q)}{\varphi(q^3)}.
 \end{aligned}$$

Replacing q by $-q$ in (6.3.8) and then using (6.3.9), we complete the proof of (iii). $\square$

Proof of (iv). The proof is similar to the previous proof. Using (1.3.61), the Jacobi triple product identity in equation (1.3.11), and (6.3.7), we find that

$$\begin{aligned}
 (6.3.10) \quad \psi(q^2)f^2(-q) &= \sum_{n=-\infty}^{\infty} (3n+1)q^{3n^2+2n} \\
 &= \frac{d}{dz} \sum_{n=-\infty}^{\infty} q^{3n^2+2n} z^{3n+1} \Big|_{z=1} \\
 &= \frac{d}{dz} \{zf(q^5z^3, q/z^3)\} \Big|_{z=1} \\
 &= f(q, q^5) \frac{d}{dz} (\log \{zf(q^5z^3, q/z^3)\}) \Big|_{z=1} \\
 &= f(q, q^5) \frac{d}{dz} (\log \{z(-q^5z^3; q^6)_\infty(-q/z^3; q^6)_\infty(q^6; q^6)_\infty\}) \Big|_{z=1} \\
 &= f(q, q^5) \left(1 - 3 \sum_{n=0}^{\infty} \left\{ \frac{q^{6n+1}}{1+q^{6n+1}} - \frac{q^{6n+5}}{1+q^{6n+5}} \right\} \right).
 \end{aligned}$$

Using (1.3.11), (1.3.14), and (1.3.15), we find that

$$\begin{aligned}
 \frac{\psi(q^2)f^2(q)}{f(-q, -q^5)} &= \frac{(q^4; q^4)_\infty(-q; q^2)_\infty^2(q^2; q^2)_\infty^2}{(q^2; q^4)_\infty(q; q^6)_\infty(q^5; q^6)_\infty(q^6; q^6)_\infty} \\
 &= \frac{(q^4; q^4)_\infty(q^2; q^4)_\infty^2(q^2; q^2)_\infty^2(q^3; q^6)_\infty}{(q^2; q^4)_\infty(q; q^2)_\infty^3(q^6; q^6)_\infty}
 \end{aligned}$$

$$= \frac{(q^2; q^2)_\infty^3 (q^3; q^6)_\infty}{(q; q^2)_\infty^3 (q^6; q^6)_\infty} = \frac{\psi^3(q)}{\psi(q^3)}.$$

Replacing q by $-q$ in (6.3.10) and then using the calculation above, we finish the proof. $\square$

In the theory of modular equations of degree 3, it is advantageous to introduce the parameter p defined by

$$(6.3.11) \quad m = 1 + 2p.$$

Theorem 6.3.4 (Modular Equations of Degree 3). *Let β have degree 3 over α , and let m denote the multiplier of degree 3. Then*

$$(i) \quad \left(\frac{\alpha^3}{\beta}\right)^{1/8} - \left(\frac{(1-\alpha)^3}{1-\beta}\right)^{1/8} = 1 = \left(\frac{(1-\beta)^3}{1-\alpha}\right)^{1/8} - \left(\frac{\beta^3}{\alpha}\right)^{1/8},$$

$$(ii) \quad (\alpha\beta)^{1/4} + \{(1-\alpha)(1-\beta)\}^{1/4} = 1,$$

$$(iii) \quad m = 1 + 2\left(\frac{\beta^3}{\alpha}\right)^{1/8}, \quad \frac{3}{m} = 1 + 2\left(\frac{(1-\alpha)^3}{1-\beta}\right)^{1/8},$$

$$(iv) \quad m^2 \left\{ \left(\frac{\alpha^3}{\beta}\right)^{1/8} - \alpha \right\} = \left(\frac{\alpha^3}{\beta}\right)^{1/8} - \beta,$$

$$(v) \quad \alpha = p \left(\frac{2+p}{1+2p}\right)^3, \quad \beta = p^3 \left(\frac{2+p}{1+2p}\right),$$

$$1 - \alpha = (1+p) \left(\frac{1-p}{1+2p}\right)^3, \quad 1 - \beta = (1+p)^3 \left(\frac{1-p}{1+2p}\right),$$

$$(vi) \quad m^2 = \left(\frac{\beta}{\alpha}\right)^{1/2} + \left(\frac{1-\beta}{1-\alpha}\right)^{1/2} - \left(\frac{\beta(1-\beta)}{\alpha(1-\alpha)}\right)^{1/2},$$

$$\frac{9}{m^2} = \left(\frac{\alpha}{\beta}\right)^{1/2} + \left(\frac{1-\alpha}{1-\beta}\right)^{1/2} - \left(\frac{\alpha(1-\alpha)}{\beta(1-\beta)}\right)^{1/2},$$

$$(\alpha\beta^5)^{1/8} + \{(1-\alpha)(1-\beta)^5\}^{1/8} = 1 - \left(\frac{\beta^3(1-\alpha)^3}{\alpha(1-\beta)}\right)^{1/8}$$

$$(vii) \quad \begin{aligned} &= (\alpha^5\beta)^{1/8} + \{(1-\alpha)^5(1-\beta)\}^{1/8} \\ &= \left\{ \frac{1}{2} (1 + (\alpha\beta)^{1/2} + \{(1-\alpha)(1-\beta)\}^{1/2}) \right\}^{1/2}. \end{aligned}$$

Proof of (i). Using (iii) of Theorem 6.3.3 twice, once with q replaced by $-q^2$, we find that

$$\begin{aligned}
 \frac{\varphi^3(q)}{\varphi(q^3)} + \frac{\varphi^3(-q^2)}{\varphi(-q^6)} &= 2 + 6 \sum_{n=1}^{\infty} \left(\frac{q^{6n+1}}{1 - q^{6n+1}} + \frac{q^{6n+2}}{1 + q^{6n+2}} \right) \\
 &\quad - 6 \sum_{n=0}^{\infty} \left(\frac{q^{6n+4}}{1 + q^{6n+4}} + \frac{q^{6n+5}}{1 - q^{6n+5}} \right) \\
 &\quad - 6 \sum_{n=1}^{\infty} \left(\frac{q^{6n+2}}{1 + q^{6n+2}} - \frac{q^{6n+4}}{1 + q^{6n+4}} \right) \\
 &= 2 + 6 \sum_{n=1}^{\infty} \left(\frac{q^{6n+1}}{1 - q^{6n+1}} - \frac{q^{6n+5}}{1 - q^{6n+5}} \right) \\
 &= 2 \frac{\psi^3(q)}{\psi(q^3)},
 \end{aligned}
 \tag{6.3.12}$$

by Theorem 6.3.3(iv). Employing Theorems 5.4.1(i), (iii) and 5.4.2(i), we transcribe (6.3.12) into the form

$$\frac{z_1^{3/2}}{z_3^{1/2}} + \frac{z_1^{3/2}(1-\alpha)^{3/8}}{z_3^{1/2}(1-\beta)^{1/8}} = 2 \frac{2^{-3/2} z_1^{3/2} (\alpha/q)^{3/8}}{2^{-1/2} z_3^{1/2} (\beta/q)^{1/8}},$$

which upon simplification yields the first equality of (i).

The second equality of (i) is the reciprocal of the first. $\square$

Proof of (ii). From Theorem 6.3.3(ii),

$$\begin{aligned}
 \varphi(q)\varphi(q^3) - \varphi(-q)\varphi(-q^3) &= 2 \sum_{n=1}^{\infty} \binom{n}{3} \left(\frac{q^n}{1 + (-q)^n} - \frac{(-q)^n}{1 + q^n} \right) \\
 &= 2 \sum_{n=1}^{\infty} \left(\frac{q^{6n+1}}{1 - q^{6n+1}} + \frac{q^{6n+1}}{1 + q^{6n+1}} \right. \\
 &\quad \left. - \frac{q^{6n+5}}{1 - q^{6n+5}} - \frac{q^{6n+5}}{1 + q^{6n+5}} \right) \\
 &= 4 \sum_{n=0}^{\infty} \left(\frac{q^{6n+1}}{1 - q^{12n+2}} - \frac{q^{6n+5}}{1 - q^{12n+10}} \right) \\
 &= 4q\psi(q^2)\psi(q^6),
 \end{aligned}
 \tag{6.3.13}$$

by Theorem 6.3.3(i). Converting (6.3.13) into a modular equation by applications of Theorems 5.4.1(i), (ii) and 5.4.2(iii), we deduce that

$$\sqrt{z_1 z_3} - \sqrt{z_1 z_3} \{(1 - \alpha)(1 - \beta)\}^{1/4} = \sqrt{z_1 z_3} q(\alpha/q)^{1/4} (\beta/q^3)^{1/4},$$

from which (ii) follows. $\square$

Proof of (iii). Using Theorem 6.3.3(iii), we find that

$$\begin{aligned} \frac{\varphi^3(q)}{\varphi(q^3)} + 2 \frac{\varphi^3(-q^2)}{\varphi(-q^6)} &= 3 + 6 \sum_{n=1}^{\infty} \left(\frac{(-1)^n q^{3n+1}}{1 + (-q)^{3n+1}} + \frac{(-1)^n q^{3n+2}}{1 + (-q)^{3n+2}} \right) \\ &\quad + 12 \sum_{n=1}^{\infty} \left(-\frac{q^{6n+2}}{1 + q^{6n+2}} + \frac{q^{6n+4}}{1 + q^{6n+4}} \right) \\ &= 3 \left(1 + 2 \sum_{n=1}^{\infty} \binom{n}{3} \frac{q^n}{1 + (-q)^n} \right) \\ (6.3.14) \quad &= 3\varphi(q)\varphi(q^3), \end{aligned}$$

by Theorem 6.3.3(ii). Using Theorem 5.4.1(i), (iii), we may rewrite (6.3.14) in the form

$$\frac{z_1^{3/2}}{z_3^{1/2}} + 2 \frac{z_1^{3/2}(1 - \alpha)^{3/8}}{z_3^{1/2}(1 - \beta)^{1/8}} = 3\sqrt{z_1 z_3}.$$

Simplifying, we obtain the second equality of (iii). The first equality of (iii) is the reciprocal of the second. $\square$

Before embarking on the proofs of the remaining modular equations, we provide some useful parameterizations that greatly facilitate the derivation of modular equations of degree 3. From Theorem 6.3.4(iii),

$$(6.3.15) \quad \left(\frac{\beta^3}{\alpha} \right)^{1/8} = \frac{m-1}{2}.$$

Thus, from Theorem 6.3.4(i) and (6.3.15),

$$(6.3.16) \quad \left(\frac{(1 - \beta)^3}{1 - \alpha} \right)^{1/8} = 1 + \left(\frac{\beta^3}{\alpha} \right)^{1/8} = \frac{m+1}{2}.$$

Next, from Theorem 6.3.4(iii),

$$(6.3.17) \quad \left(\frac{(1 - \alpha)^3}{\beta} \right)^{1/8} = \frac{3 - m}{2^n}.$$

Hence, from Theorem 6.3.4(i) and (6.3.17),

$$(6.3.18) \quad \left(\frac{\alpha^3}{\beta}\right)^{1/8} = 1 + \left(\frac{(1-\alpha)^3}{1-\beta}\right)^{1/8} = \frac{3+m}{2m}.$$

Taking the product of the cube of (6.3.18) and (6.3.15), we find that

$$(6.3.19) \quad \alpha = \frac{(m-1)(3+m)^3}{16m^3},$$

while taking the product of the cube of (6.3.15) and (6.3.18), we find that

$$(6.3.20) \quad \beta = \frac{(m-1)^3(3+m)}{16m}.$$

Next multiply the cube of (6.3.17) by (6.3.16) to deduce that

$$(6.3.21) \quad 1 - \alpha = \frac{(m+1)(3-m)^3}{16m^3},$$

and lastly multiply the cube of (6.3.16) by (6.3.17) to find that

$$(6.3.22) \quad 1 - \beta = \frac{(m+1)^3(3-m)}{16m}.$$

Proof of (iv). By (6.3.19), (6.3.20), and (6.3.18),

$$m^2\alpha - \beta = (m^2 - 1)\frac{3+m}{2m} = (m^2 - 1)\left(\frac{\alpha^3}{\beta}\right)^{1/8},$$

from which (iv) is immediate. $\square$

Proof of (v). Substituting (6.3.11) into (6.3.19), (6.3.20), (6.3.21), and (6.3.22), we readily deduce the four formulas in (v). $\square$

Proof of (vi). Using first (6.3.19) and (6.3.20) and secondly (6.3.21) and (6.3.22), we find, respectively, that

$$(6.3.23) \quad \left(\frac{\beta}{\alpha}\right)^{1/2} = \frac{m(m-1)}{3+m}$$

and

$$(6.3.24) \quad \left(\frac{1-\beta}{1-\alpha}\right)^{1/2} = \frac{m(m+1)}{3-m}.$$

If we substitute (6.3.23) and (6.3.24) in the right side of the first equality of (vi), we easily verify its truth. The second formula follows from the first by the method of reciprocation. $\square$

Proof of (vii). From (6.3.19) and (6.3.20),

$$(\alpha\beta^5)^{1/8} = \frac{(m-1)^2(3+m)}{8m},$$

and from (6.3.21) and (6.3.22),

$$\{(1-\alpha)(1-\beta)^5\}^{1/8} = \frac{(m+1)^2(3-m)}{8m}.$$

Hence,

$$\begin{aligned} (\alpha\beta^5)^{1/8} + \{(1-\alpha)(1-\beta)^5\}^{1/8} &= \frac{3+m^2}{4m} \\ &= 1 - \frac{(3-m)(m-1)}{4m} \\ (6.3.25) \quad &= 1 - \left(\frac{\beta^3(1-\alpha)^3}{\alpha(1-\beta)} \right)^{1/8}, \end{aligned}$$

by (6.3.19)–(6.3.22). This proves the first equality in (vii).

By the principle of reciprocation, the second equality follows from the first.

Lastly, from (6.3.19)–(6.3.22),

$$\begin{aligned} (6.3.26) \quad 1 + (\alpha\beta)^{1/2} + \{(1-\alpha)(1-\beta)\}^{1/2} \\ = 1 + \frac{(m-1)^2(3+m)^2}{16m^2} + \frac{(m+1)^2(3-m)^2}{16m^2} = \frac{(m^2+3)^2}{8m^2}. \end{aligned}$$

If we multiply both sides of (6.3.26) by $\frac{1}{2}$, take the square root of both sides, and appeal to the first equality in (6.3.25), we complete the proof of the third equality of (vii). $\square$

The next theorem gives two further modular equations of degree 3 that are prototypes of modular equations of further orders.

Theorem 6.3.5. *If*

$$P := \{16\alpha\beta(1-\alpha)(1-\beta)\}^{1/8} \quad \text{and} \quad Q := \left(\frac{\beta(1-\beta)}{\alpha(1-\alpha)} \right)^{1/4},$$

then

$$(6.3.27) \quad Q + \frac{1}{Q} + 2\sqrt{2} \left(P - \frac{1}{P} \right) = 0.$$

If

$$P := (\alpha\beta)^{1/8} \quad \text{and} \quad Q := (\beta/\alpha)^{1/4},$$

then

$$(6.3.28) \quad Q - \frac{1}{Q} = 2 \left(P - \frac{1}{P} \right).$$

Proof. From (6.3.19)–(6.3.22), we easily find that

$$P^2 = \frac{(m^2 - 1)(9 - m^2)}{8m^2} \quad \text{and} \quad Q^2 = \frac{(m^2 - 1)m^2}{9 - m^2}.$$

Thus,

$$PQ = \frac{m^2 - 1}{2\sqrt{2}} \quad \text{and} \quad \frac{P}{Q} = \frac{9 - m^2}{2m^2\sqrt{2}}.$$

Eliminating m from this last pair of equations yields

$$\frac{P}{Q} = \frac{2\sqrt{2} - PQ}{2PQ\sqrt{2} + 1}.$$

Rearranging this last equality, we readily deduce (6.3.27).

The proof of (6.3.28) is similar. From (6.3.19) and (6.3.20),

$$P^2 = \frac{(m - 1)(3 + m)}{4m} \quad \text{and} \quad Q^2 = \frac{m(m - 1)}{3 + m}.$$

It follows that

$$PQ = \frac{m - 1}{2} \quad \text{and} \quad \frac{P}{Q} = \frac{3 + m}{2m}.$$

Eliminating m from this last pair of equations, we find that

$$\frac{P}{Q} = \frac{2 + PQ}{2PQ + 1},$$

which upon rearrangement yields (6.3.28). □

With the use of (6.3.11) in conjunction with (6.3.15) and (6.3.17), it follows that $p > 0$ and $p < 1$, respectively, or equivalently that $1 < m < 3$. From the formulas for α and β in Theorem 6.3.4(v), we readily find that, for $0 \leq p \leq 1$,

$$\frac{d\alpha}{dp} = \frac{2(1 - p)^2(2 + p)^2}{(1 + 2p)^4} \geq 0$$

and

$$\frac{d\beta}{dp} = \frac{6p^2(1+p)^2}{(1+2p)^2} \geq 0.$$

There is consequently a one-to-one correspondence between α and p and also between β and p when $0 \leq p \leq 1$.

Theorem 6.3.6. *Let p be defined by (6.3.11). Then*

(6.3.29)

$${}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; p\left(\frac{2+p}{1+2p}\right)^3\right) = (1+2p) {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; p^3\left(\frac{2+p}{1+2p}\right)\right).$$

Proof. Recall the definition (6.3.4) of the multiplier m and use Theorem 5.2.8 and (6.3.3) to write this definition in the form

$${}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; \alpha\right) = m {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; \beta\right).$$

Now use the representations for m , α , and β given in (6.3.11) and Theorem 6.3.4(v), and (6.3.29) follows immediately. $\square$

6.4. Notes

We are grateful to K. S. Williams for providing the second proof of Theorem 6.2.4, the proof of Theorem 6.2.9, and Exercise 6.2.7.

The theory of modular equations begins with Legendre's [139, vol. 1, p. 229] modular equation of degree 3 in 1825, namely,

$$(6.4.1) \quad (\alpha\beta)^{1/4} + \{(1-\alpha)(1-\beta)\}^{1/4} = 1.$$

In the century that followed, several mathematicians, including A. Berry, A. Cayley, A. Enneper, E. Fiedler, R. Fricke, C. Guetzlaff, M. Hanna, C. G. J. Jacobi, F. Klein, R. Russell, L. Schläfli, H. Schröter, L. A. Sohncke, G. N. Watson, and H. Weber, contributed to the growing list of modular equations. However, the mathematician who discovered far more modular equations than any of these mathematicians was Ramanujan, who constructed over 200 modular equations.

As indicated immediately above, the form of the modular equation given in Theorem 6.3.4(ii) is due to Legendre [139, vol. 1, p. 229], and can also be found in both A. Cayley's text on elliptic functions

[67, p. 196] and in Jacobi's epic work [131, p. 68]. This type of modular equation has been established for several other degrees. In particular, formulas due to Schröter in his dissertation [206] are useful in establishing such formulas; see also [34, pp. 66–72]. These modular equations are also called “of Russell-type,” after the English mathematician R. Russell, who derived several modular equations of this sort [202], [203]. For example, the modular equations of degrees 5 and 7 of this type are given by, respectively [34, pp. 280, 314],

$$(\alpha\beta)^{1/2} + \{(1-\alpha)(1-\beta)\}^{1/2} + 2\{16\alpha\beta(1-\alpha)(1-\beta)\}^{1/6} = 1$$

and

$$(\alpha\beta)^{1/8} + \{(1-\alpha)(1-\beta)\}^{1/8} = 1.$$

The parameterizations for α and β given in (6.3.19) and (6.3.20), respectively, were first discovered by Legendre [139, vol. 1, p. 223] and rediscovered by Jacobi [131, p. 25]. As we have seen, the parameterizations of α and β in terms of the multiplier m are extremely useful in deriving modular equations of degree 3. Similar parameterizations exist for α and β in the theory of modular equations of degree 5 [34, pp. 280–288], but for higher degrees we do not know of any further parameterizations, which reflects the fact that with increasing n , the shapes of modular equations become increasingly more complex, and finding them becomes increasingly more difficult.

The paper [105] by M. Hanna provides a summary of much of the work accomplished on modular equations up until 1928.

Ramanujan summarized much of his work on modular equations in fragments that were published with his lost notebook [194] on pages 55 and 350–352. For accounts of these fragments, see Chapter 17 in the book by Andrews and Berndt [19]. In particular, the formulas for m in Theorem 6.3.4(vi) are examples of a large class of similar formulas for m . Although we have provided a simple proof based on parameterizations for α and β , in general, we do not know Ramanujan's methods for deriving them. In our book [34], we used the theory of modular forms to verify several of them; Ramanujan most likely did not use such ideas.

The modular equation in (6.3.27) is of “Schläfli-type” [204], and Ramanujan recorded many modular equations of this sort in his notebooks and lost notebook [193], [194]. See [19, Chap. 17], [34, Chaps. 19–20], and [37, Chap. 25].

In this short monograph providing an introduction to Ramanujan’s theory of theta functions, we do not have the space to give applications of modular equations. The proofs of Ramanujan’s congruences for $p(n)$ modulo arbitrary powers of 5, 7, and 11 depend upon modular equations [194], [50], [218], [28]. Many of the results in the following Chapter 7 crucially utilize modular equations of degree 5.

The Rogers–Ramanujan Continued Fraction

7.1. Definition and Historical Background

A continued fraction is an expression of the sort

$$(7.1.1) \quad b_0 + \frac{a_1}{b_1 + \frac{a_2}{b_2 + \frac{a_3}{b_3 + \frac{a_4}{b_4 + \dots}}}},$$

which is commonly written in the more compact form

$$(7.1.2) \quad b_0 + \frac{a_1}{b_1} + \frac{a_2}{b_2} + \frac{a_3}{b_3} + \frac{a_4}{b_4} + \dots.$$

The continued fraction (7.1.1) or (7.1.2) may terminate, i.e., the fractions $\frac{a_k}{b_k}$ do not continue indefinitely. For example,

$$b_0 + \frac{a_1}{b_1} + \frac{a_2}{b_2} + \frac{a_3}{b_3} + \frac{a_4}{b_4}$$

is a terminating continued fraction. More interesting are those continued fractions that do not terminate, i.e., infinite continued fractions.

Suppose that we define the sequences P_n and Q_n , $n \geq -1$, by

$$\begin{aligned} P_n &= b_n P_{n-1} + a_n P_{n-2}, & n \geq 0, \\ Q_n &= b_n Q_{n-1} + a_n Q_{n-2}, & n \geq 0, \\ P_{-1} &= 1, \quad Q_{-1} = 0, \quad P_0 = b_0, \quad Q_0 = 1. \end{aligned}$$

Exercise 7.1.1. *Prove that*

$$(7.1.3) \quad \frac{P_n}{Q_n} := b_0 + \frac{a_1}{b_1} + \frac{a_2}{b_2} + \cdots + \frac{a_n}{b_n}.$$

Then if

$$\lim_{n \rightarrow \infty} \frac{P_n}{Q_n}$$

exists, we say that the continued fraction (7.1.2) converges; otherwise it diverges. Clearly, the first task in developing a theory of continued fractions is to derive criteria for convergence and divergence. In particular, when the numerators a_n and denominators b_n are functions of a complex variable, an extensive theory has been developed, and it continues to evolve. It is not the purpose of this monograph to develop such a theory, and so we refer readers to the excellent text by L. Lorentzen and H. Waadeland [151, Chapter 1] for many criteria for convergence and divergence. In particular, see [151, p. 35, Theorem 3].

Most mathematics students first encounter continued fractions in a course in elementary number theory. The first infinite continued fractions that students may be asked to evaluate are those in the following exercise.

Exercise 7.1.2. *Prove that*

$$(7.1.4) \quad 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \cdots}}} = \frac{\sqrt{5} + 1}{2}$$

and

$$(7.1.5) \quad 1 - \frac{1}{1 + \frac{1}{1 - \frac{1}{1 + \cdots}}} = \frac{\sqrt{5} - 1}{2}.$$

These are, in fact, special cases of perhaps the most interesting continued fraction in mathematics, the Rogers–Ramanujan continued fraction, which first appeared in a paper by L. J. Rogers [198] in 1894.

Definition 7.1.3. The Rogers–Ramanujan continued fraction $R(q)$ is defined by

$$(7.1.6) \quad R(q) := \frac{q^{1/5}}{1 + \frac{q}{1 + \frac{q^2}{1 + \frac{q^3}{1 + \dots}}}},$$

provided that it converges. Furthermore, set

$$(7.1.7) \quad F(q) := q^{-1/5}R(q)$$

and

$$(7.1.8) \quad T(q) := \frac{1}{F(q)} = 1 + \frac{q}{1 + \frac{q^2}{1 + \frac{q^3}{1 + \dots}}}.$$

Readers will immediately ask, “Why does $q^{1/5}$ appear in the definition of $R(q)$?” The reason is that $R(q)$ belongs to the world of theta functions, and $R(q)$ ’s modular properties are more symmetric and elegant with $q^{1/5}$ appearing in its definition than if $q^{1/5}$ were absent. However, there are occasions when the factor $q^{1/5}$ is not helpful, and so we then use one of the representations (7.1.7) or (7.1.8).

7.2. The Convergence, Divergence, and Values of $R(q)$

In the definition above, we have not indicated where $R(q)$ converges or where it diverges. By standard theorems [151, pp. 35, 94], $R(q)$ converges in the complex q -plane for $|q| < 1$, and it diverges for $|q| > 1$. What happens if $|q| = 1$? Taking the reciprocals of (7.1.4) and (7.1.5), we find, respectively, that

$$(7.2.1) \quad R(1) = \frac{\sqrt{5}-1}{2} \quad \text{and} \quad -R(-1) = \frac{\sqrt{5}+1}{2},$$

where we have taken the root $(-1)^{1/5} = -1$ in the definition of $R(q)$. Furthermore, I. Schur [207, pp. 319–321], [208, pp. 117–136] and Ramanujan in his notebooks [193] established the following theorem [38, p. 35].

Theorem 7.2.1. Recall that $T(q)$ is defined by (7.1.8). Let q be a primitive m th root of unity. If m is a multiple of 5, $T(q)$ diverges. Otherwise, $T(q)$ converges and

$$(7.2.2) \quad T(q) = \alpha T(\alpha) q^{(1-\alpha\rho m)/5},$$

where α denotes the Legendre symbol $\left(\frac{m}{5}\right)$ and ρ is the least positive residue of m modulo 5.

Thus, for a root of unity, we know when $R(q)$ converges and when it diverges. What about other points on $|q| = 1$? We do not know the answer in general, but D. Bowman and J. McLaughlin [61] have found an uncountable set of measure 0 on the unit circle (not including 5 n th roots of unity) where $R(q)$ diverges. It is conjectured that $R(q)$ diverges on the unit circle except for those points of convergence described in Theorem 7.2.1.

As mentioned above, the Rogers–Ramanujan continued fraction was first defined by Rogers, who proved a few of its properties. However, most of the results that we know about this continued fraction are due to Ramanujan. Several theorems about $R(q)$ appear in his notebooks [193], but his lost notebook [194] contains considerably more material on $R(q)$; see [19, Chapters 1–5]. In his first two letters to G. H. Hardy [192, pp. xxvii, xxviii], [51, pp. 29, 57], Ramanujan communicated several theorems on $R(q)$. In particular, in his first letter [192, p. xxvii], [51, p. 29], Ramanujan gave the first nonelementary evaluations of $R(q)$, namely,

$$(7.2.3) \quad R(e^{-2\pi}) = \sqrt{\frac{5 + \sqrt{5}}{2}} - \frac{\sqrt{5} + 1}{2}$$

and

$$(7.2.4) \quad S(e^{-\pi}) = \sqrt{\frac{5 - \sqrt{5}}{2}} - \frac{\sqrt{5} - 1}{2},$$

where we have set

$$(7.2.5) \quad S(q) = -R(-q).$$

In his second letter to Hardy [192, p. xxviii], [51, p. 57], Ramanujan further asserted that

$$R(e^{-2\pi\sqrt{5}}) = \frac{\sqrt{5}}{1 + \sqrt[5]{5^{3/4} \left(\frac{\sqrt{5}-1}{2}\right)^{5/2}} - 1} - \frac{\sqrt{5} + 1}{2}.$$

In both letters, Ramanujan claimed that [192, p. xxvii], [51, pp. 29, 57], “It is always possible to find exactly the value of $R(e^{-\pi\sqrt{n}})$.”

The meaning of this last statement was not clarified until 1996 when Berndt, H. H. Chan, and L.-C. Zhang [46] demonstrated that if n is a positive integer and if the requisite *class invariants* could be determined, then $R(e^{-\pi\sqrt{n}})$ could be explicitly evaluated. (Class invariants are certain multiples of Ramanujan's function $\chi(\pm e^{-\pi\sqrt{n}})$. Historically, they were first brought into prominence by H. Weber [220], who explicitly calculated many class invariants and who used them to generate Hilbert class fields. Ramanujan explicitly determined over 100 class invariants and used them to calculate values of certain quotients of theta functions and the Rogers–Ramanujan continued fraction. Ramanujan's work on class invariants is described in [38, Chapter 34]. For an introduction to class invariants and Ramanujan's applications of them, see [47].) Both in his notebooks [193] and his lost notebook [194], Ramanujan recorded many values for the Rogers–Ramanujan continued fraction. For example, on pages 204 and 210 of his lost notebook [194], Ramanujan offered the value

$$S(e^{-\pi/\sqrt{35}}) = \left(5\sqrt{5} - 7 + \sqrt{35(5 - 2\sqrt{5})}\right)^{1/5},$$

where $S(q)$ is defined by (7.2.5). For proofs of these claims, see [44], [46], and [19, Chapter 2]. Those readers with some background in algebraic number theory and who have been exceptionally observant will have observed that each of the values for $R(q)$ that we have quoted is a unit in some algebraic number field. This is true for all values of $R(e^{-\pi\sqrt{n}})$, when n is a positive integer; see [46] for a proof.

We record one further result on $R(q)$ communicated from India by Ramanujan to Hardy. Let α and β be positive numbers such that $\alpha\beta = \pi^2$. In his second letter to Hardy, Ramanujan [192, p. xxviii], [51, p. 57] claimed that

$$(7.2.6) \quad \left(\frac{1+\sqrt{5}}{2} + R(e^{-2\alpha})\right) \left(\frac{1+\sqrt{5}}{2} + R(e^{-2\beta})\right) = \frac{5+\sqrt{5}}{2},$$

which was first proved in print by Watson [216]. This result is also recorded as Entry 39(i) in Chapter 16 of Ramanujan's second notebook; see [34, pp. 84–85] for another proof and further references, and also see [19, p. 94]. Note that if one knows the value of $R(e^{-2\alpha})$

for a certain α , then (7.2.6) enables one to immediately calculate another value $R(e^{-2\beta})$. Thus, besides being a beautiful formula, (7.2.6) enables one to obtain two values from one.

7.3. The Rogers–Ramanujan Functions

The Rogers–Ramanujan continued fraction is intimately connected with the famous Rogers–Ramanujan functions $G(q)$ and $H(q)$, which are defined by

$$(7.3.1) \quad G(q) := \sum_{n=0}^{\infty} \frac{q^{n^2}}{(q; q)_n} \quad \text{and} \quad H(q) := \sum_{n=0}^{\infty} \frac{q^{n(n+1)}}{(q; q)_n}.$$

Both [Rogers](#) [198] and Ramanujan [193, Vol. II, Chapter 16, Sect. 15], [34, p. 30] proved that

$$(7.3.2) \quad R(q) = q^{1/5} \frac{H(q)}{G(q)},$$

which we now prove. In fact, we first prove a *finite form* of (7.3.2), which can be found as Entry 16 in Chapter 16 of Ramanujan's second notebook [193], [34, p. 31], and from which (7.3.2) follows as an immediate corollary.

Theorem 7.3.1. *For each nonnegative integer n , let*

$$(7.3.3) \quad \mu := \mu_n(a, q) := \sum_{k=0}^{[(n+1)/2]} \frac{(q)_{n-k+1} a^k q^{k^2}}{(q)_k (q)_{n-2k+1}},$$

$$(7.3.4) \quad \nu := \nu_n(a, q) := \sum_{k=0}^{[n/2]} \frac{(q)_{n-k} a^k q^{k(k+1)}}{(q)_k (q)_{n-2k}},$$

where $[x]$ denotes the greatest integer less than or equal to x . Then, for $n \geq 1$,

$$(7.3.5) \quad \frac{\mu}{\nu} = 1 + \frac{aq}{1} + \frac{aq^2}{1} + \cdots + \frac{aq^n}{1}.$$

Proof. For each nonnegative integer r , define

$$F_r := F_r(a, q) := \sum_{k=0}^{[(n-r+1)/2]} \frac{(q)_{n-r-k+1} a^k q^{k(r+k)}}{(q)_k (q)_{n-r-2k+1}}.$$

Observe that

$$(7.3.6) \quad F_0 = \mu \quad \text{and} \quad F_1 = \nu.$$

Also note that

$$(7.3.7) \quad F_n = 1 \quad \text{and} \quad F_{n-1} = 1 + aq^n.$$

We now develop a recurrence relation for F_r . When we combine the two sums in the first step below, we use the fact that $1/(q)_{-1} = 0$ by (1.3.35). To that end,

$$\begin{aligned}
 F_r - F_{r+1} &= \sum_{k=0}^{[(n-r+1)/2]} \frac{(q)_{n-r-k+1} a^k q^{k(r+k)}}{(q)_k (q)_{n-r-2k+1}} \\
 &\quad - \sum_{k=0}^{[(n-r)/2]} \frac{(q)_{n-r-k} a^k q^{k(r+1+k)}}{(q)_k (q)_{n-r-2k}} \\
 &= \sum_{k=1}^{[(n-r+1)/2]} \frac{(q)_{n-r-k} a^k q^{k(r+k)}}{(q)_k (q)_{n-r-2k}} \left(\frac{1 - q^{n-r-k+1}}{1 - q^{n-r-2k+1}} - q^k \right) \\
 &= \sum_{k=1}^{[(n-r+1)/2]} \frac{(q)_{n-r-k} a^k q^{k(r+k)}}{(q)_k (q)_{n-r-2k}} \frac{(1 - q^k)}{(1 - q^{n-r-2k+1})} \\
 &= \sum_{k=1}^{[(n-r+1)/2]} \frac{(q)_{n-r-k} a^k q^{k(r+k)}}{(q)_{k-1} (q)_{n-r-2k+1}} \\
 &= \sum_{j=0}^{[(n-r-1)/2]} \frac{(q)_{n-r-j-1} a^{j+1} q^{(j+1)(r+j+1)}}{(q)_j (q)_{n-r-2j-1}} \\
 &= aq^{r+1} \sum_{j=0}^{[(n-r-1)/2]} \frac{(q)_{n-(r+2)-j+1} a^j q^{j(r+2+j)}}{(q)_j (q)_{n-(r+2)-2j+1}} \\
 (7.3.8) \quad &= aq^{r+1} F_{r+2}.
 \end{aligned}$$

Using (7.3.6), (7.3.8) repeatedly, and lastly (7.3.7), we conclude that

$$\begin{aligned}
 \frac{\mu}{\nu} &= \frac{F_0}{F_1} = \frac{F_1 + aqF_2}{F_1} = 1 + \frac{aq}{F_1/F_2} \\
 &= 1 + \frac{aq}{(F_2 + aq^2F_3)/F_2} = 1 + \frac{aq}{1} + \frac{aq^2}{F_2/F_3}
 \end{aligned}$$

$$\begin{aligned}
&= 1 + \frac{aq}{1} + \frac{aq^2}{1} + \cdots + \frac{aq^{n-1}}{F_{n-1}/F_n} \\
&= 1 + \frac{aq}{1} + \frac{aq^2}{1} + \cdots + \frac{aq^{n-1}}{1} + \frac{aq^n}{1}.
\end{aligned}$$

□

Corollary 7.3.2. For any complex number a and $|q| < 1$,

$$(7.3.9) \quad \frac{\sum_{k=0}^{\infty} \frac{a^k q^{k^2}}{(q)_k}}{\sum_{k=0}^{\infty} \frac{a^k q^{k(k+1)}}{(q)_k}} = 1 + \frac{aq}{1} + \frac{aq^2}{1} + \cdots + \frac{aq^n}{1} + \cdots.$$

Proof. Let $n \rightarrow \infty$ in (7.3.5). □

The continued fraction in (7.3.9) is called the *Generalized Rogers–Ramanujan Continued Fraction*.

The renowned Rogers–Ramanujan identities, which we do not prove in this book, are given by

$$(7.3.10) \quad G(q) = \frac{1}{(q; q^5)_{\infty} (q^4; q^5)_{\infty}} \quad \text{and} \quad H(q) = \frac{1}{(q^2; q^5)_{\infty} (q^3; q^5)_{\infty}}$$

Using (7.3.10), we immediately deduce the elegant representation for $R(q)$ in the next theorem.

Theorem 7.3.3. We have

$$(7.3.11) \quad R(q) = q^{1/5} \frac{(q; q^5)_{\infty} (q^4; q^5)_{\infty}}{(q^2; q^5)_{\infty} (q^3; q^5)_{\infty}}.$$

Proof. Set $a = 1$ in Corollary 7.3.2, take the reciprocal of both sides, use the definitions (7.3.1), and lastly use (7.3.10). □

The results in this section can be greatly generalized in that many more general continued fractions can be represented as quotients of two q -series. See, in particular, [19, Chapter 6] and [34, pp. 30–31].

7.4. Identities for $R(q)$

Ramanujan discovered two beautiful identities connecting $R(q)$ and $1/R(q)$, which we prove in this section. The first is given below.

Theorem 7.4.1. *Recall that $T(q)$ is defined in (7.1.8). Then*

$$(7.4.1) \quad T(q^5) - q - \frac{q^2}{T(q^5)} = \frac{(q; q)_\infty}{(q^{25}; q^{25})_\infty}.$$

Comparing (7.4.1) with (2.3.13) in the proof of Theorem 2.3.4, we see that the unidentified series $J_1(q)$ and $J_2(q)$ are equal to $T(q)$ and $-1/T(q)$, respectively.

We now prove a more general theorem from which Theorem 7.4.1 follows by specialization. It will be convenient to introduce the notation

$$(a_1, a_2, \dots, a_m; q)_\infty = (a_1; q)_\infty (a_2; q)_\infty \cdots (a_m; q)_\infty.$$

Theorem 7.4.2. *For any complex number a ,*

$$(7.4.2) \quad \begin{aligned} & (a, a^2, q/a, q/a^2, q; q)_\infty \\ &= (q^5; q^5)_\infty \left(\frac{(a^5 q; q^5)_\infty (a^{-5} q^4; q^5)_\infty}{(q; q^5)_\infty (q^4; q^5)_\infty} - a \frac{(a^5 q^2; q^5)_\infty (a^{-5} q^3; q^5)_\infty}{(q^2; q^5)_\infty (q^3; q^5)_\infty} \right. \\ & \quad \left. - a^2 \frac{(a^5 q^3; q^5)_\infty (a^{-5} q^2; q^5)_\infty}{(q^2; q^5)_\infty (q^3; q^5)_\infty} + a^3 \frac{(a^5 q^4; q^5)_\infty (a^{-5} q; q^5)_\infty}{(q; q^5)_\infty (q^4; q^5)_\infty} \right). \end{aligned}$$

Before proving Theorem 7.4.2, we show that Theorem 7.4.1 follows immediately from Theorem 7.4.2.

Proof of Theorem 7.4.1. If we replace q by q^5 in (7.4.2), then set $a = q$, realize that $(1; q^{25})_\infty = 0$, and recall (7.3.11), we deduce (7.4.1) forthwith. $\square$

The decompositions in Theorems 7.4.1 and 7.4.2 are called *5-dissections*, because in the former theorem, the series terms of $(q; q)_\infty$ are separated out in powers of q according to their residue classes modulo 5, and in the latter theorem, the terms are separated out in powers of a according to their residue classes modulo 5.

Proof of Theorem 7.4.2. Using the Jacobi triple product identity (1.3.11) twice, we find that

$$\begin{aligned}
 (a, a^2, q/a, q/a^2, q; q)_\infty &= \frac{(a, q/a, q; q)_\infty (a^2, q/a^2, q; q)_\infty}{(q; q)_\infty} \\
 &= \frac{1}{(q; q)_\infty} \sum_{r=-\infty}^{\infty} (-1)^r a^r q^{(r^2-r)/2} \sum_{s=-\infty}^{\infty} (-1)^s a^{2s} q^{(s^2-s)/2} \\
 &= \frac{1}{(q; q)_\infty} \sum_{r,s=-\infty}^{\infty} (-1)^{r+s} a^{r+2s} q^{(r^2-r+s^2-s)/2} \\
 (7.4.3) \quad &= \sum_{n=-\infty}^{\infty} a^n c_n(q),
 \end{aligned}$$

where, for $-\infty < n < \infty$,

$$c_n(q) := \frac{1}{(q; q)_\infty} \sum_{\substack{r,s=-\infty \\ r+2s=n}}^{\infty} (-1)^{r+s} q^{(r^2-r+s^2-s)/2}.$$

We now determine $c_n(q)$ according to the residue class of n modulo 5.

First, consider the residue class 0 (mod 5). Replace n by $5n$ and make the change of variables $r = n - 2t$ and $s = 2n + t$. Note that $r + 2s = 5n$. Then, simplifying and applying the Jacobi triple product identity (1.3.11), we find that

$$\begin{aligned}
 c_{5n}(q) &= \frac{1}{(q; q)_\infty} \sum_{t=-\infty}^{\infty} (-1)^{n+t} q^{((n-2t)^2 - (n-2t) + (2n+t)^2 - (2n+t))/2} \\
 &= \frac{(-1)^n q^{(5n^2-3n)/2}}{(q; q)_\infty} \sum_{t=-\infty}^{\infty} (-1)^t q^{(5t^2+t)/2} \\
 &= \frac{(-1)^n q^{(5n^2-3n)/2}}{(q; q)_\infty} f(-q^3, -q^2) \\
 (7.4.4) \quad &= \frac{(-1)^n q^{(5n^2-3n)/2}}{(q; q^5)_\infty (q^4; q^5)_\infty}.
 \end{aligned}$$

Second, consider the residue class 1 (mod 5). Set $r = n - 2t + 1$ and $s = 2n + t$, so that $r + 2s = 5n + 1$. Then, upon simplification

and the use of the Jacobi triple product identity (1.3.11), we see that

$$\begin{aligned}
 c_{5n+1}(q) &= \frac{1}{(q; q)_{\infty}} \sum_{t=-\infty}^{\infty} (-1)^{n+t+1} \\
 &\quad \times q^{((n-2t+1)^2 - (n-2t+1) + (2n+t)^2 - (2n+t))/2} \\
 &= \frac{(-1)^{n+1} q^{(5n^2-n)/2}}{(q; q)_{\infty}} \sum_{t=-\infty}^{\infty} (-1)^t q^{(5t^2-3t)/2} \\
 &= \frac{(-1)^{n+1} q^{(5n^2-n)/2}}{(q; q)_{\infty}} f(-q, -q^4) \\
 (7.4.5) \quad &= \frac{(-1)^{n+1} q^{(5n^2-n)/2}}{(q^2; q^5)_{\infty} (q^3; q^5)_{\infty}}.
 \end{aligned}$$

It should now be clear how to calculate the three remaining cases.

Exercise 7.4.3. *Prove that*

$$(7.4.6) \quad c_{5n+2}(q) = \frac{(-1)^{n+1} q^{(5n^2+n)/2}}{(q^2; q^5)_{\infty} (q^3; q^5)_{\infty}},$$

$$(7.4.7) \quad c_{5n+3}(q) = \frac{(-1)^n q^{(5n^2+3n)/2}}{(q; q^5)_{\infty} (q^4; q^5)_{\infty}},$$

$$(7.4.8) \quad c_{5n+4}(q) = 0.$$

Substitute (7.4.4)–(7.4.8) in (7.4.3) and use the Jacobi triple product identity (1.3.11) four times to conclude that

$$\begin{aligned}
 &(a, a^2, q/a, q/a^2, q; q)_{\infty} \\
 &= \frac{1}{(q; q^5)_{\infty} (q^4; q^5)_{\infty}} \sum_{n=-\infty}^{\infty} (-1)^n a^{5n} q^{(5n^2-3n)/2} \\
 &\quad - \frac{1}{(q^2; q^5)_{\infty} (q^3; q^5)_{\infty}} \sum_{n=-\infty}^{\infty} (-1)^n a^{5n+1} q^{(5n^2-n)/2} \\
 &\quad - \frac{1}{(q^2; q^5)_{\infty} (q^3; q^5)_{\infty}} \sum_{n=-\infty}^{\infty} (-1)^n a^{5n+2} q^{(5n^2+n)/2} \\
 &\quad + \frac{1}{(q; q^5)_{\infty} (q^4; q^5)_{\infty}} \sum_{n=-\infty}^{\infty} (-1)^n a^{5n+3} q^{(5n^2+3n)/2}
 \end{aligned}$$

$$= (q^5; q^5)_\infty \left(\frac{(a^5 q; q^5)_\infty (a^{-5} q^4; q^5)_\infty}{(q; q^5)_\infty (q^4; q^5)_\infty} - a \frac{(a^5 q^2; q^5)_\infty (a^{-5} q^3; q^5)_\infty}{(q^2; q^5)_\infty (q^3; q^5)_\infty} \right. \\ \left. - a^2 \frac{(a^5 q^3; q^5)_\infty (a^{-5} q^2; q^5)_\infty}{(q^2; q^5)_\infty (q^3; q^5)_\infty} + a^3 \frac{(a^5 q^4; q^5)_\infty (a^{-5} q; q^5)_\infty}{(q; q^5)_\infty (q^4; q^5)_\infty} \right).$$

Thus, the proof of Theorem 7.4.2 is complete. $\square$

From Theorem 7.4.2, we derive the second major identity involving $R(q)$ or $T(q)$.

Theorem 7.4.4. *We have*

$$(7.4.9) \quad T^5(q^5) - 11q^5 - \frac{q^{10}}{T^5(q^5)} = \frac{(q^5; q^5)_\infty^6}{(q^{25}; q^{25})_\infty^6}.$$

Proof. The proof is almost identical to a portion of the proof of Theorem 2.3.1. We therefore forego most of the details. Write (7.4.1) in the form

$$(7.4.10) \quad \frac{(q, q^2, q^3, q^4, q^5; q^5)_\infty}{(q^{25}; q^{25})_\infty} = T(q^5) - q - \frac{q^2}{T(q^5)}.$$

Let ω be any fifth root of unity, and replace q by ωq in (7.4.10) to find that

$$(7.4.11) \quad \frac{(\omega q, \omega^2 q^2, \omega^3 q^3, \omega^4 q^4, q^5; q^5)_\infty}{(q^{25}; q^{25})_\infty} = T(q^5) - \omega q - \frac{\omega^2 q^2}{T(q^5)}.$$

Now multiply all five equalities (7.4.11) together. On the left side of our product, use precisely the same argument that was used in (2.3.18)–(2.3.21) to arrive at the right-hand side of (7.4.9). For the right-hand side of our product of the expressions in (7.4.11), employ the same argument that was used in deducing (2.3.23) from (2.3.22). We then obtain the left-hand side of (7.4.9) to complete the proof. $\square$

Although the proofs are essentially the same, the use of Theorem 7.4.4 leads to a somewhat cleaner and more satisfying proof of Theorem 2.3.1 than the proof employing the pentagonal number theorem, Jacobi's identity, and the division of power series in Chapter 2. We complete our work on the Rogers–Ramanujan continued fraction in this section by giving this alternative proof of Theorem 2.3.1.

Theorem 7.4.5. *For each nonnegative integer n ,*

$$(7.4.12) \quad p(5n+4) \equiv 0 \pmod{5}.$$

Proof. Write (7.4.9) in the form

$$(7.4.13) \quad 1 = \frac{(q^{25}; q^{25})_{\infty}^6}{(q^5; q^5)_{\infty}^6} \left(T^5(q^5) - 11q^5 - \frac{q^{10}}{T^5(q^5)} \right).$$

Divide both sides of (7.4.13) by (7.4.1) in the form

$$(q; q)_{\infty} = (q^{25}; q^{25})_{\infty} \left(T(q^5) - q - \frac{q^2}{T(q^5)} \right).$$

Then, using long division and the abbreviated notation $x = T(q^5)$, we find that

$$(7.4.14) \quad \begin{aligned} \sum_{n=0}^{\infty} p(n)q^n &= \frac{(q^{25}; q^{25})_{\infty}^5}{(q^5; q^5)_{\infty}^6} \frac{T^5(q^5) - 11q^5 - q^{10}/T^5(q^5)}{T(q^5) - q - q^2/T(q^5)} \\ &= \frac{(q^{25}; q^{25})_{\infty}^5}{(q^5; q^5)_{\infty}^6} \left(x^4 + qx^3 + 2q^2x^2 + 3q^3x + 5q^4 \right. \\ &\quad \left. - \frac{3q^5}{x} + \frac{2q^6}{x^2} - \frac{q^7}{x^3} + \frac{q^8}{x^4} \right). \end{aligned}$$

We now extract those terms from both sides of (7.4.14) that involve only the powers q^{5n+4} , $n \geq 0$, to deduce that

$$(7.4.15) \quad \sum_{n=0}^{\infty} p(5n+4)q^{5n+4} = 5q^4 \frac{(q^{25}; q^{25})_{\infty}^5}{(q^5; q^5)_{\infty}^6}.$$

Dividing both sides of (7.4.15) by q^4 and replacing q^5 by q , we complete the proof. $\square$

Theorems 7.4.1 and 7.4.4 were found by G. N. Watson [215], [216] in Ramanujan's notebooks and proved by him [215] in order to establish (7.2.3) and (7.2.4). Observe that if we can evaluate the right side of either (7.4.1) or (7.4.9) for a certain value of q , then we can find the value of $R(q)$ by simply solving a quadratic equation.

Exercise 7.4.6. The shifted Fibonacci numbers f_n , $n \geq 0$, are defined by $f_0 = f_1 = 1$ and

$$(7.4.16) \quad f_n = f_{n-1} + f_{n-2}, \quad n \geq 2.$$

Establish the following generating function for the shifted Fibonacci numbers:

$$\sum_{n=0}^{\infty} f_n q^n = \frac{1}{1 - q - q^2}, \quad |q| < 1.$$

Next prove that

$$\sum_{n=0}^{\infty} f_{5n+4} q^n = \frac{5}{1-11q-q^2}.$$

Conclude that $f_{5n+4} \equiv 0 \pmod{5}$. In particular, $f_4 = 5$, $f_9 = 55$, $f_{14} = 610$, $f_{19} = 6765$, and $f_{24} = 75025$. Lastly, prove that $f_{25n+24} \equiv 0 \pmod{25}$. (To prove this last congruence, you will probably need to use further properties of Fibonacci numbers.)

The previous exercise relating $p(5n+4)$ to Fibonacci numbers was suggested by M. D. Hirschhorn [125]. In [124], he used the fact $5|f_{5n+4}$ and a variation of the argument we gave above in our proof of Theorem 7.4.5, to give a similar proof of Theorem 7.4.5. See also (2.3.25). It is remarkable that f_{5n+4} and f_{25n+24} obey the same congruences as $p(5n+4)$ and $p(25n+24)$, respectively. Ramanujan's original conjecture (2.1.5) in the case that $\delta = 5^3$ is $p(125n+99) \equiv 0 \pmod{125}$. Unfortunately, in general, it is not true that $f_{125n+99}$ is divisible by 125, and so the analogy fails for 5^3 .

7.5. Modular Equations for $R(q)$

Recall that a modular equation of degree n can be thought of as a relation among theta functions with argument q and theta functions with argument q^n . We close this chapter by offering some of Ramanujan's modular equations for $R(q)$.

First, on page 326 in his second notebook [193], [38, p. 12], Ramanujan recorded a relation between $R(q)$ and $R(q^2)$ in different notation, which was also proved by Rogers [199, eq. (5.4)].

Theorem 7.5.1. *Let $u := R(q)$ and $v := R(q^2)$. Then*

$$uv^2 = \frac{v-u^2}{v+u^2}.$$

Also on page 326, Ramanujan adroitly defines the parameter

$$(7.5.1) \quad k := R(q)R^2(q^2),$$

and states the following two elegant and symmetric relations [38, p. 13].

Theorem 7.5.2. *With k defined by (7.5.1),*

$$R^5(q) = k \left(\frac{1-k}{1+k} \right)^2 \quad \text{and} \quad R^5(q^2) = k^2 \left(\frac{1+k}{1-k} \right).$$

In his lost notebook [194], Ramanujan recorded several exquisite identities for theta functions in the argument k of (7.5.1); see [19, Sects. 1.8, 1.9].

The next beautiful modular equation of degree 3 is found on page 321 in Ramanujan's second notebook [193], [38, p. 17], and was also established by Rogers [38, p. 392].

Theorem 7.5.3. *Let $u := R(q)$ and $v := R(q^3)$. Then*

$$(v - u^3)(1 + uv^3) = 3u^2v^2.$$

Lastly, we conclude with a modular equation of degree 5 for $R(q)$ that Ramanujan communicated in his first letter to Hardy [107, p. xxvii], [51, p. 29], and that is found on page 289 of Ramanujan's second notebook [193], [38, pp. 19–20]. Again, this modular equation was first established by Rogers [199, p. 392]. For references to further proofs, see [38, p. 20] or [51, p. 43].

Theorem 7.5.4. *Let $u := R(q)$ and $v := R(q^5)$. Then*

$$u^5 = v \frac{1 - 2v + 4v^2 - 3v^3 + v^4}{1 + 3v + 4v^2 + 2v^3 + v^4}.$$

7.6. Notes

The Rogers–Ramanujan continued fraction was one of Ramanujan's favorite functions. We have related to readers some of its most fascinating properties, but Ramanujan recorded many further results, especially in his lost notebook [194]. We hope readers will be stimulated to read about these discoveries in Chapters 1–6 of [19], which is the most complete source of theorems and references on the Rogers–Ramanujan continued fraction. An expository account of several theorems on the Rogers–Ramanujan continued fraction can be found in [45]. A subset of the results proved in [19] are established in [49].

Hardy was intrigued by Ramanujan's claims about the Rogers–Ramanujan continued fraction. One day Hardy wrote to Ramanujan

urging him to write a paper about it. In a letter probably written on 24 December 1913, Hardy exhorted [51, p. 87]

If you will send me your proof written out carefully (so that it is easy to follow), I will (assuming that I agree with it—of which I have very little doubt) try to get it published for you in England. Write it in the form of a paper “On the continued fraction

$$\frac{x}{1} + \frac{x^2}{1} + \frac{x^3}{1} + \cdots,”$$

giving a full proof of the principal and most remarkable theorem, viz. that the fraction can be expressed in finite terms when $x = e^{-\pi\sqrt{n}}$, when n is rational.

However, Ramanujan never followed Hardy’s advice.

The history of the famous Rogers–Ramanujan identities (7.3.10) is now well known. They were originally discovered by Rogers [198] and rediscovered by Ramanujan, who at first did not have proofs of them. One day while at Cambridge, Ramanujan was perusing back issues of the *Proceedings of the London Mathematical Society* and found Rogers’s paper [198] giving proofs of (7.3.10). Ramanujan soon found his own proofs and published them in [189], [192, pp. 214–215]. For further historical accounts, see Hardy’s book [107, pp. 90–99], Andrews’s text [14, Chapter 7], or Berndt’s book [34, pp. 77–79]. Many proofs of the identities now exist; a description and classification of all known proofs up to 1989 can be found in Andrews’s paper [17].

The identities (7.3.10) have beautiful combinatorial interpretations. In the definition of $G(q)$ given in (7.3.1), write $n^2 = 1 + 3 + \cdots + (2n - 1)$. The first identity in (7.3.10) is equivalent to the assertion that the number of partitions of a positive integer N into distinct parts with differences at least 2 equals the number of partitions of N into parts congruent to either 1 or 4 modulo 5. For example, there are five partitions of 9 into distinct parts with differences between parts at least 2, namely, 9, 8+1, 7+2, 6+3, and 5+3+1. The five partitions of 9 into parts congruent to either 1 or 4 modulo 5 are 9, 6+1+1+1, 4+4+1, 4+1+1+1+1+1, and 1+1+1+1+1+1+1. For the second identity,

in the definition of $H(q)$, write $n(n+1) = 2 + 4 + \cdots + 2n$. The second Rogers–Ramanujan identity in (7.3.10) is an analytic statement of the fact that the number of partitions of N into distinct parts with differences at least 2 and with no 1's is equal to the number of partitions of N into parts congruent to either 2 or 3 modulo 5. For example, the three partitions of 8 into distinct parts with parts differing by at least 2 and with no 1's are 8, $6 + 2$, and $5 + 3$, while the three partitions of 8 into parts congruent to either 2 or 3 modulo 5 are 8, $3 + 3 + 2$, and $2 + 2 + 2 + 2$.

Exercise 7.6.1. *Prove that the Rogers–Ramanujan identities have the combinatorial interpretations described in the preceding paragraph.*

Theorem 7.4.2 is due to Hirschhorn [123], and the proofs of Theorems 7.4.1 and 7.4.2 that we have given are also due to Hirschhorn [123]. The only other proof of Theorem 7.4.1 known to us is by Watson [215], who employed the quintuple product identity. Hirschhorn's proof is somewhat simpler. The proof of Theorem 7.4.1 given by Berndt [34, p. 267] is similar to that of Watson. Our deduction of Theorem 7.4.4 from Theorem 7.4.1 is the same as that given in the aforementioned works of Watson, Berndt, and Hirschhorn.

Exercise 7.4.6 arose from the combined efforts of Hirschhorn, P. Crutcher, O. Yeat Chan, and the author.

An approach to the Rogers–Ramanujan continued fraction via modular forms has been written by W. Duke [85].

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